

APPENDIX H

APPLICATION NO. 12/390,509

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number			
	Filing Date			
	First Named Inventor	Junwen LIU		
	Art Unit			
	Examiner Name	Not Assigned Yet		
	Attorney Docket Number	CSM P 2009 NAT 03 US 0		

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	Examiner Name	Not Assigned Yet	
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	1	LEONARD RUBIN ET AL., Narrow n+/p+ isolation in retrograde well implants, Solid State Technology Magazine, 01 July 2003, pages 119-125, Volume 46 Issue 7, Solid State Technology, USA.	☐
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**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	
Filing Date	
First Named Inventor	Junwen LIU
Art Unit	
Examiner Name	Not Assigned Yet
Attorney Docket Number	CSM P 2009 NAT 03 US 0

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

☐ That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

☐ See attached certification statement.

☐ Fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☒ None

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/dexter chin/	Date (YYYY-MM-DD)	2009-02-23
Name/Print	Dexter CHIN	Registration Number	38,842

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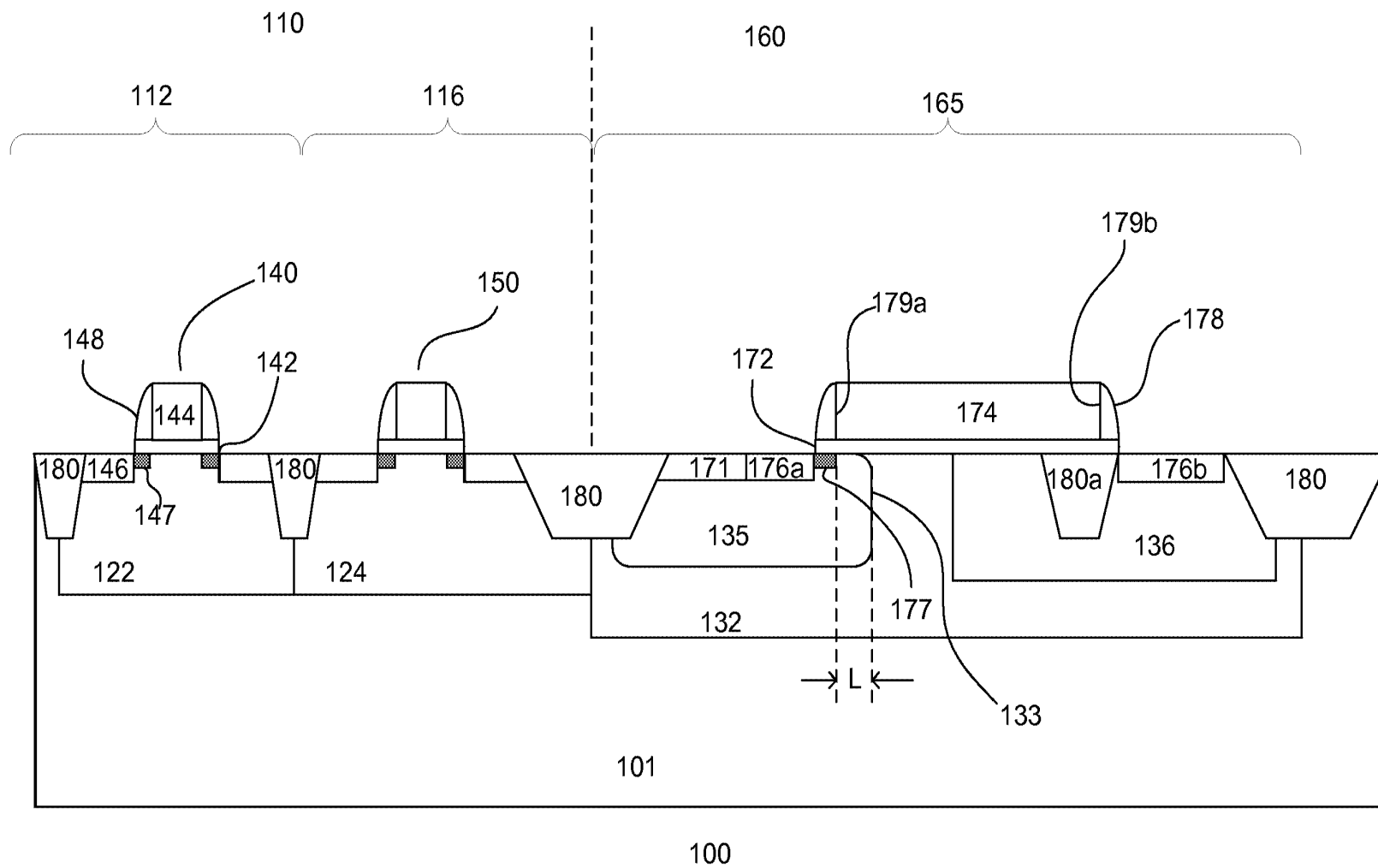


Fig. 1

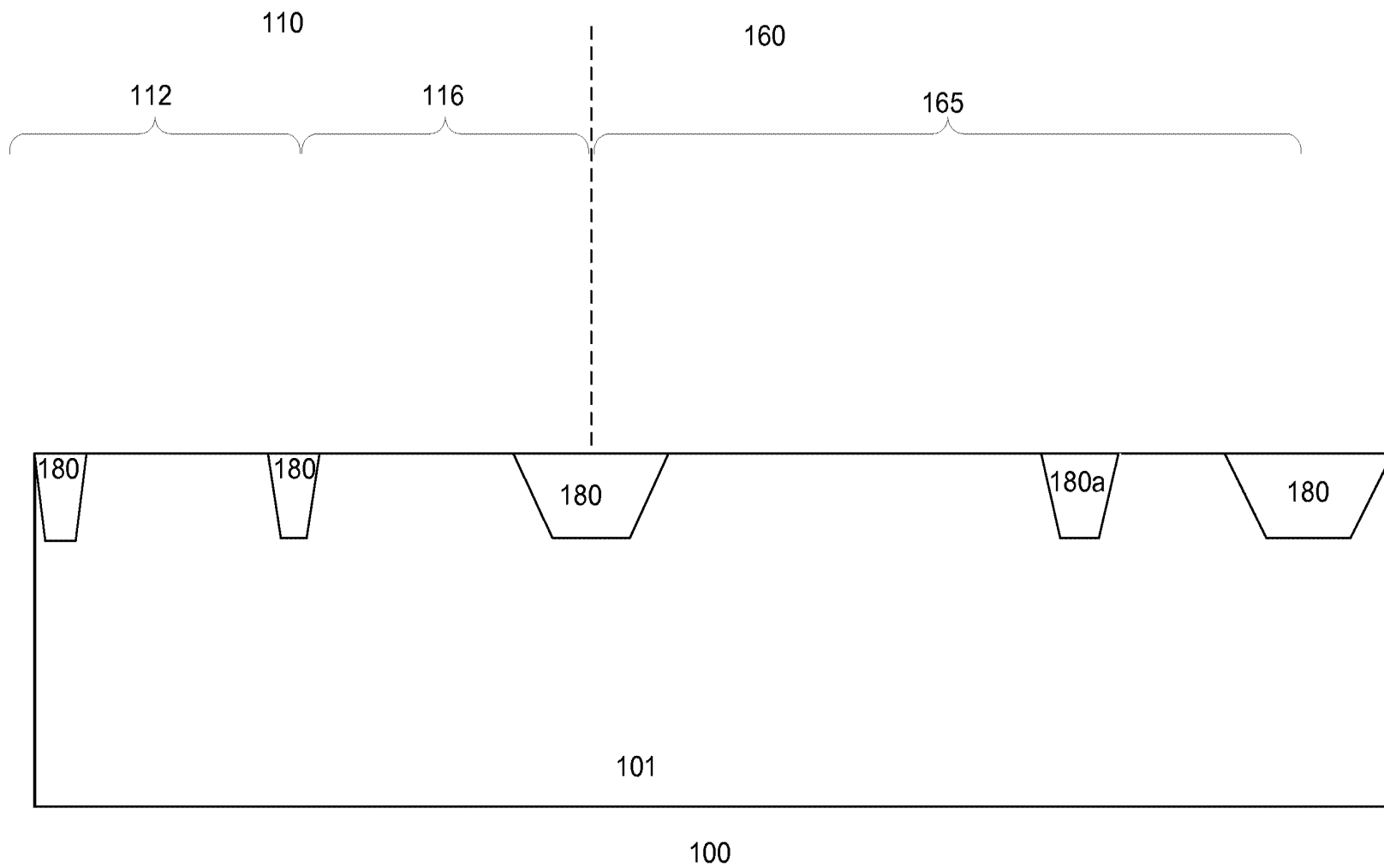


Fig. 2a

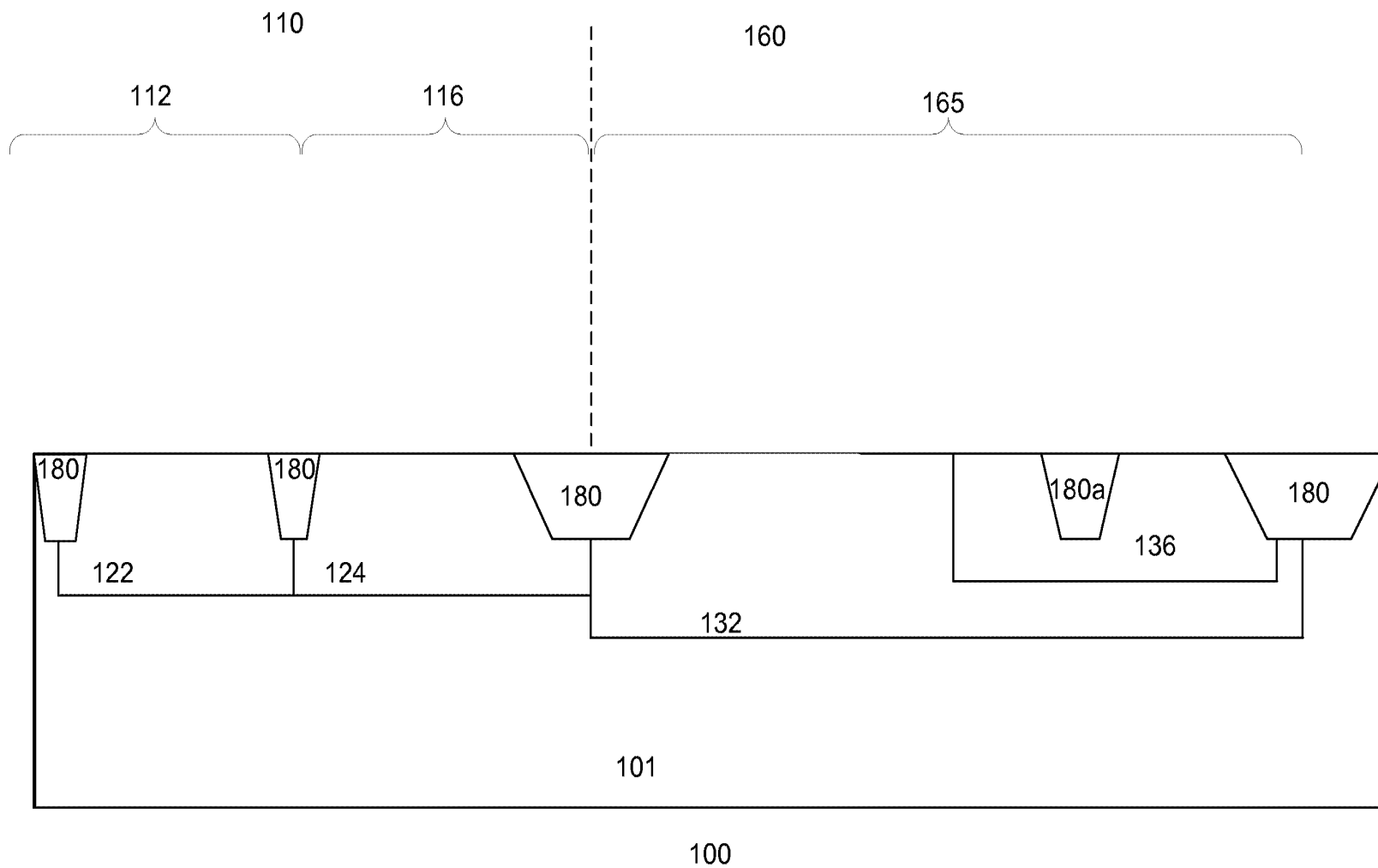


Fig. 2b

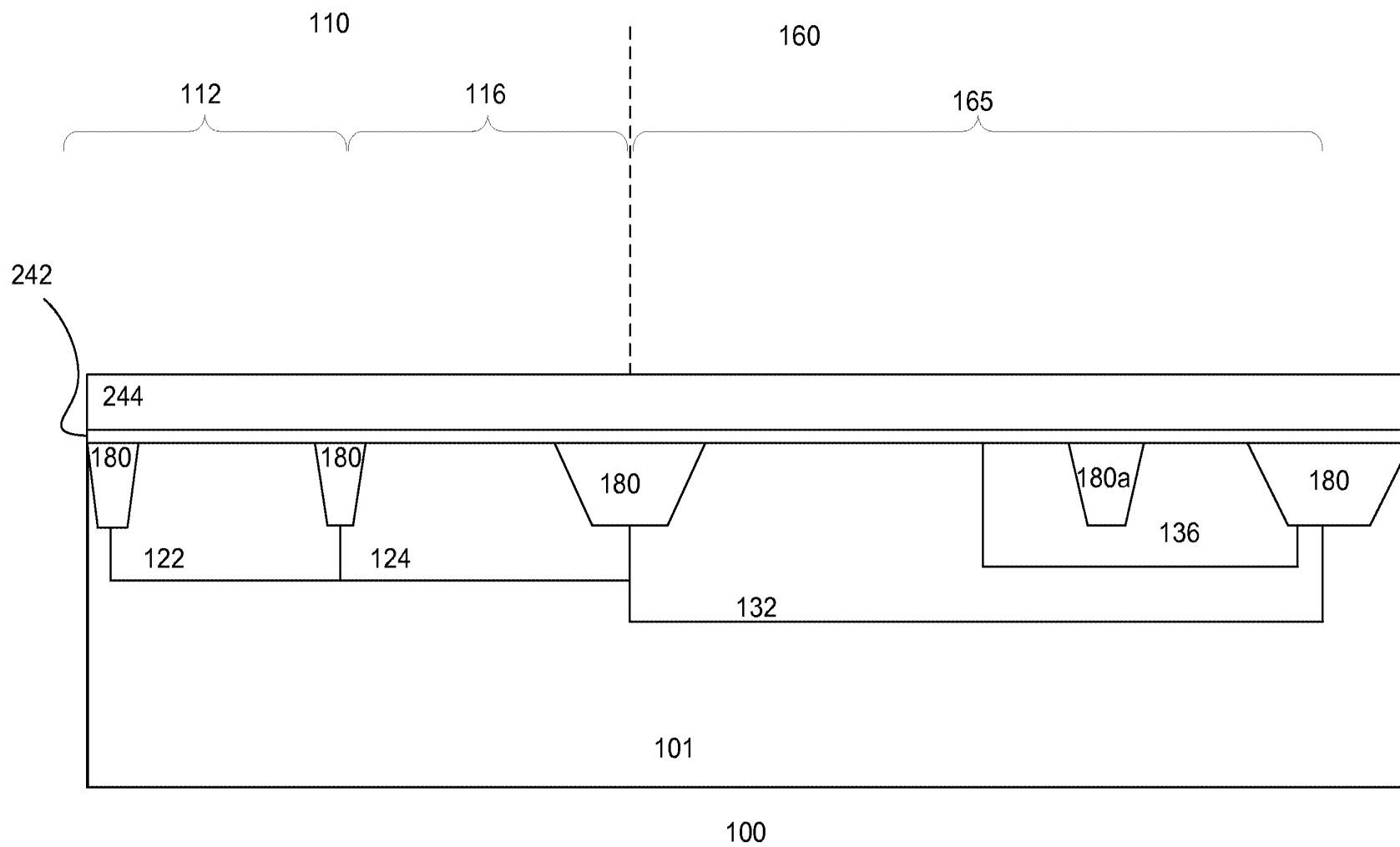


Fig. 2c

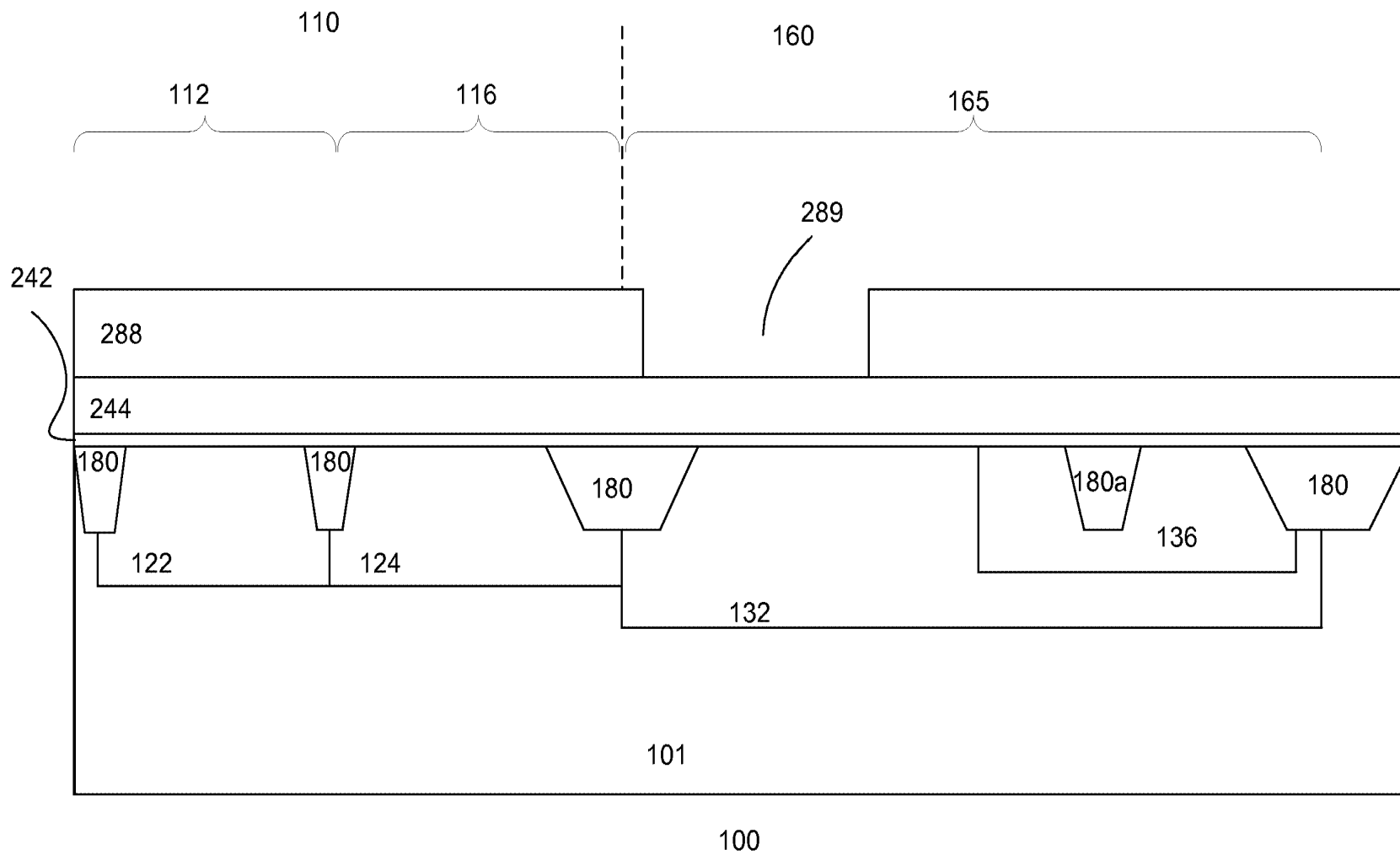


Fig. 2d

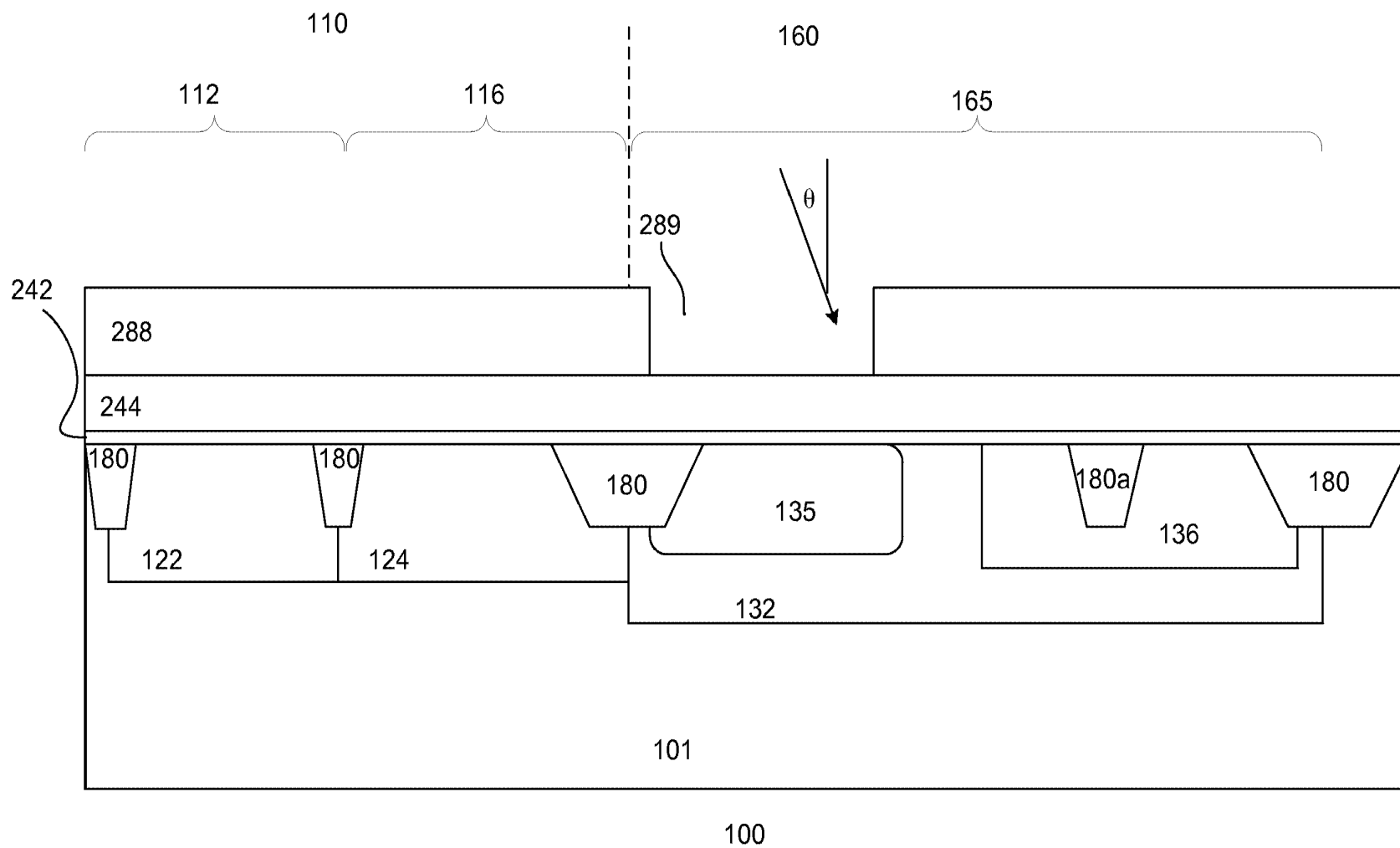


Fig. 2e

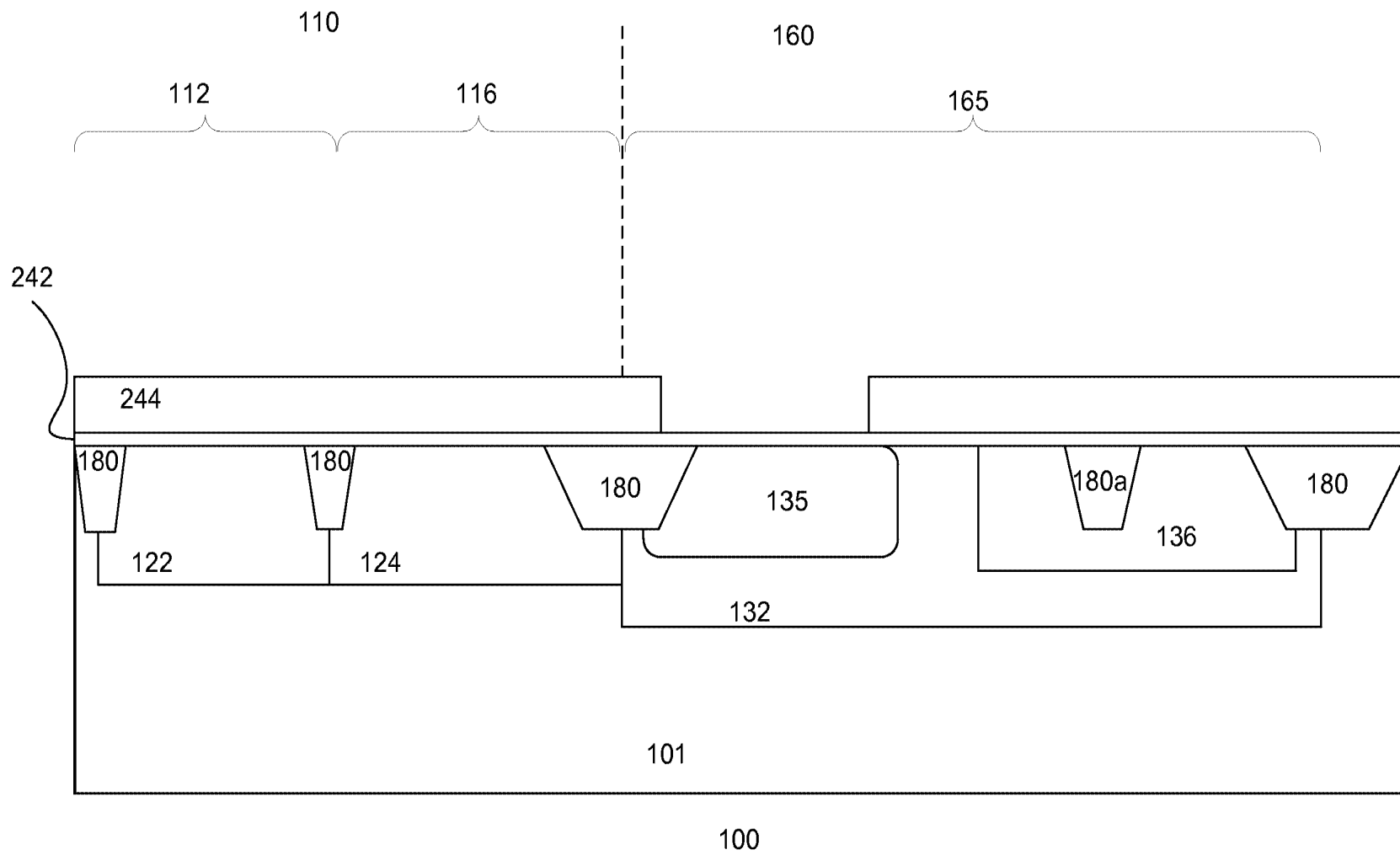


Fig. 2f

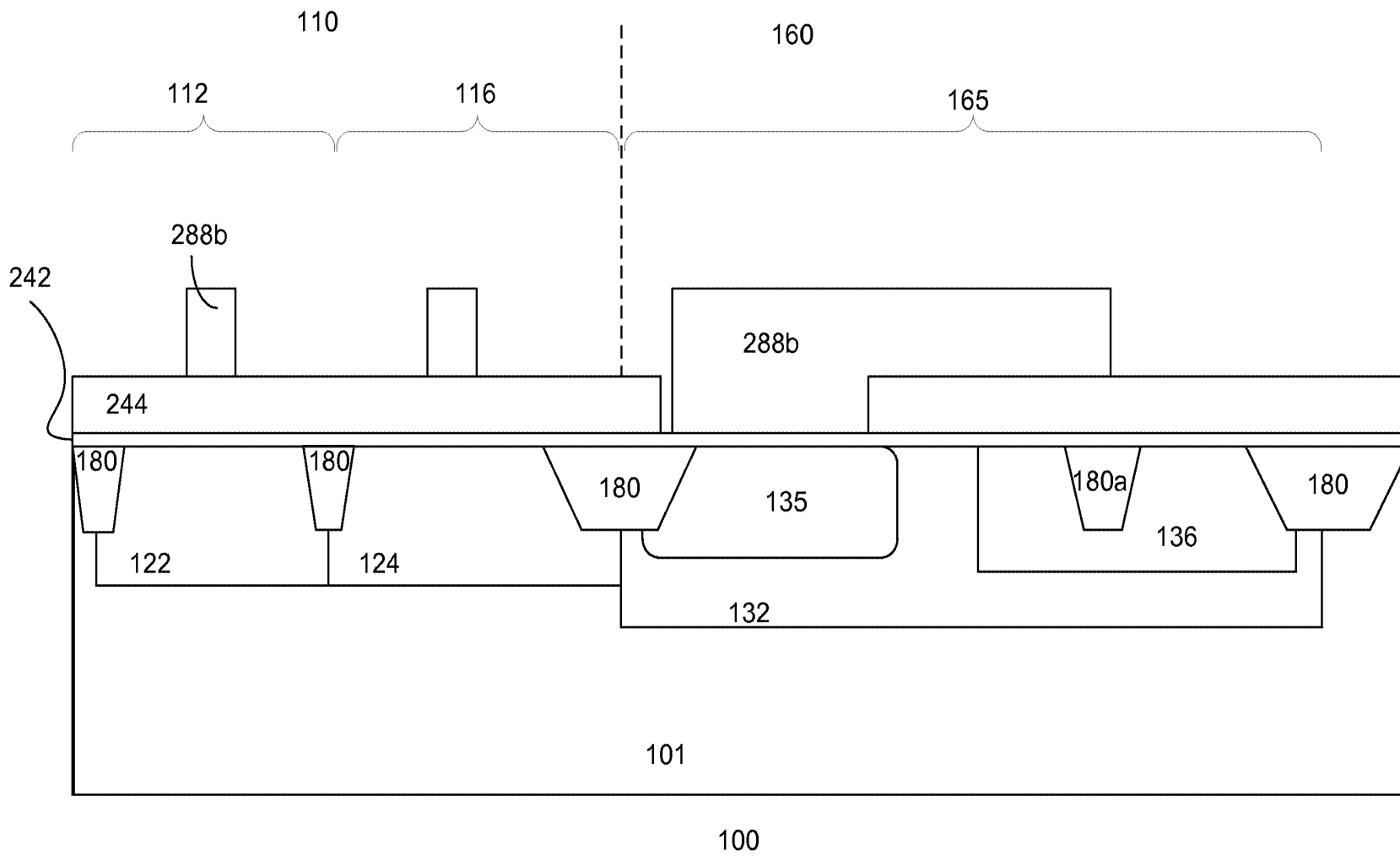


Fig. 2g

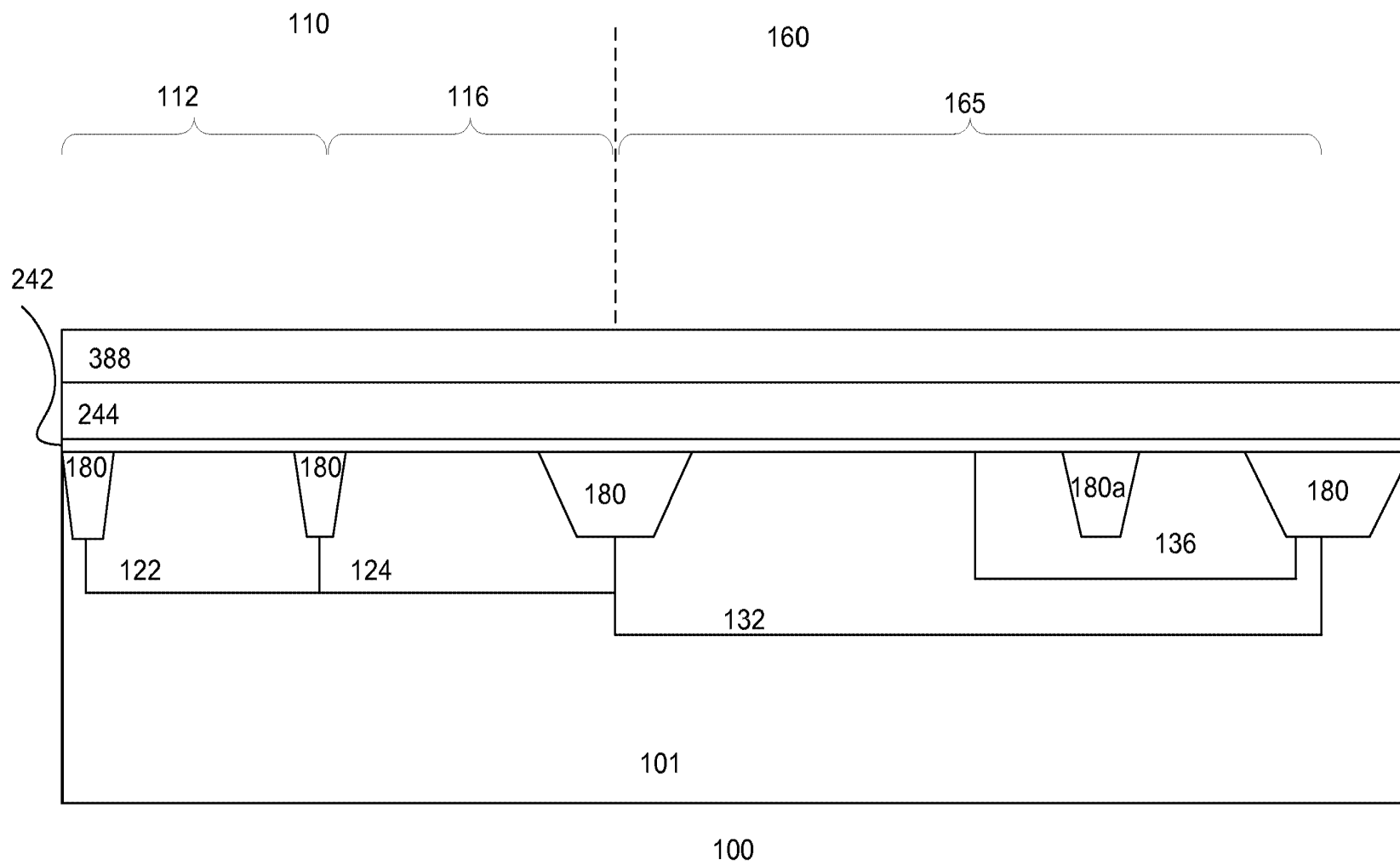


Fig. 3a

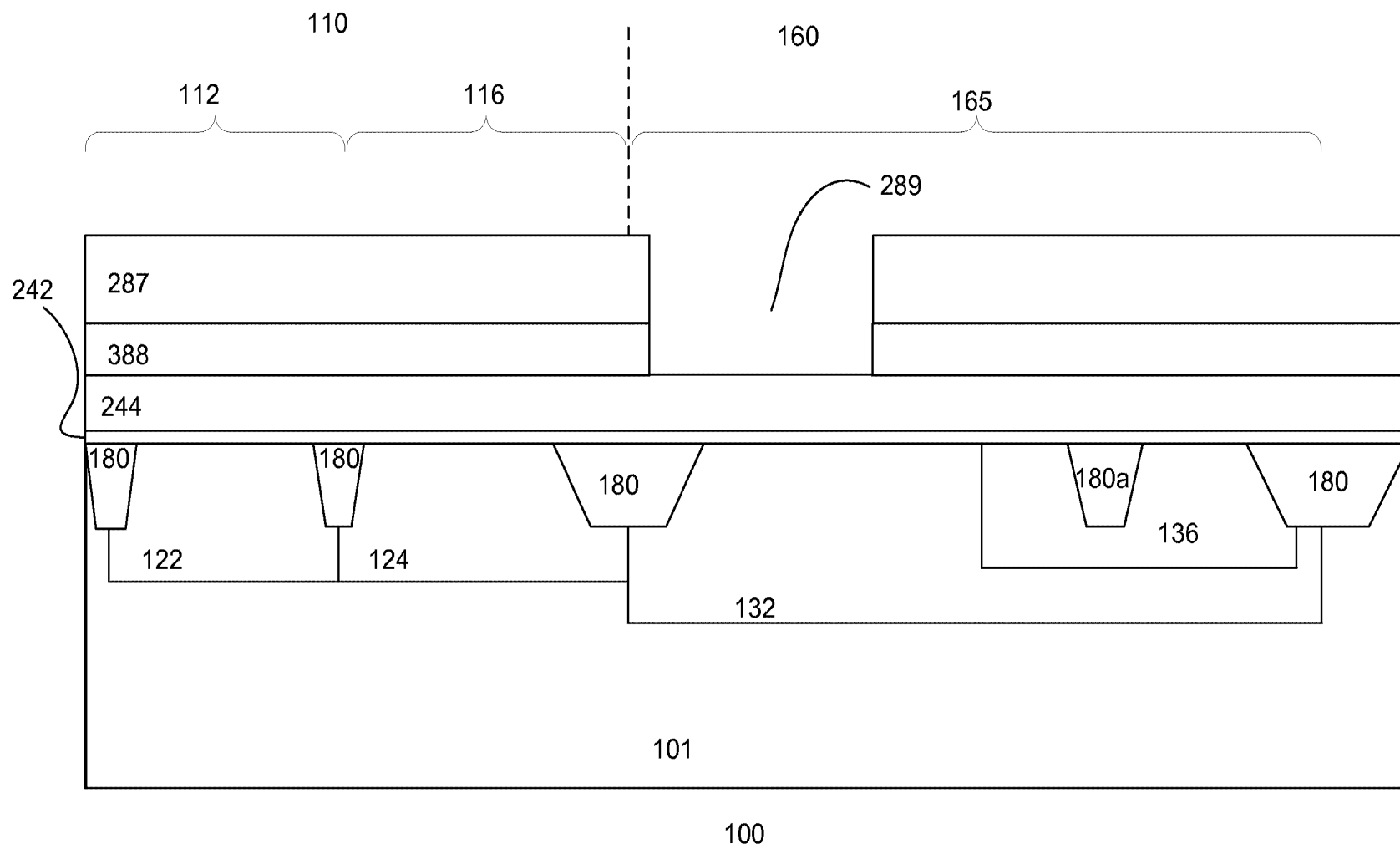


Fig. 3b

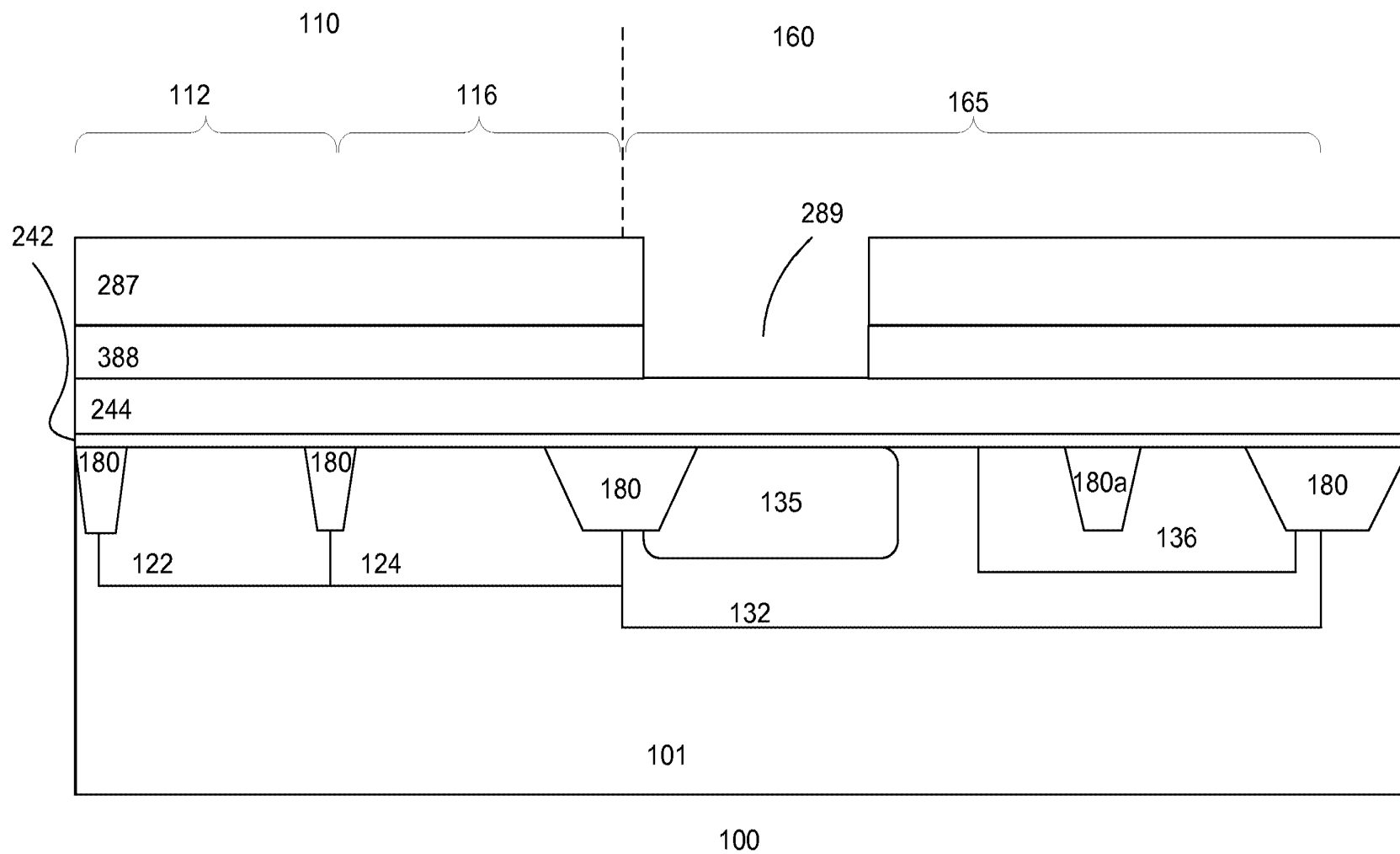


Fig. 3c

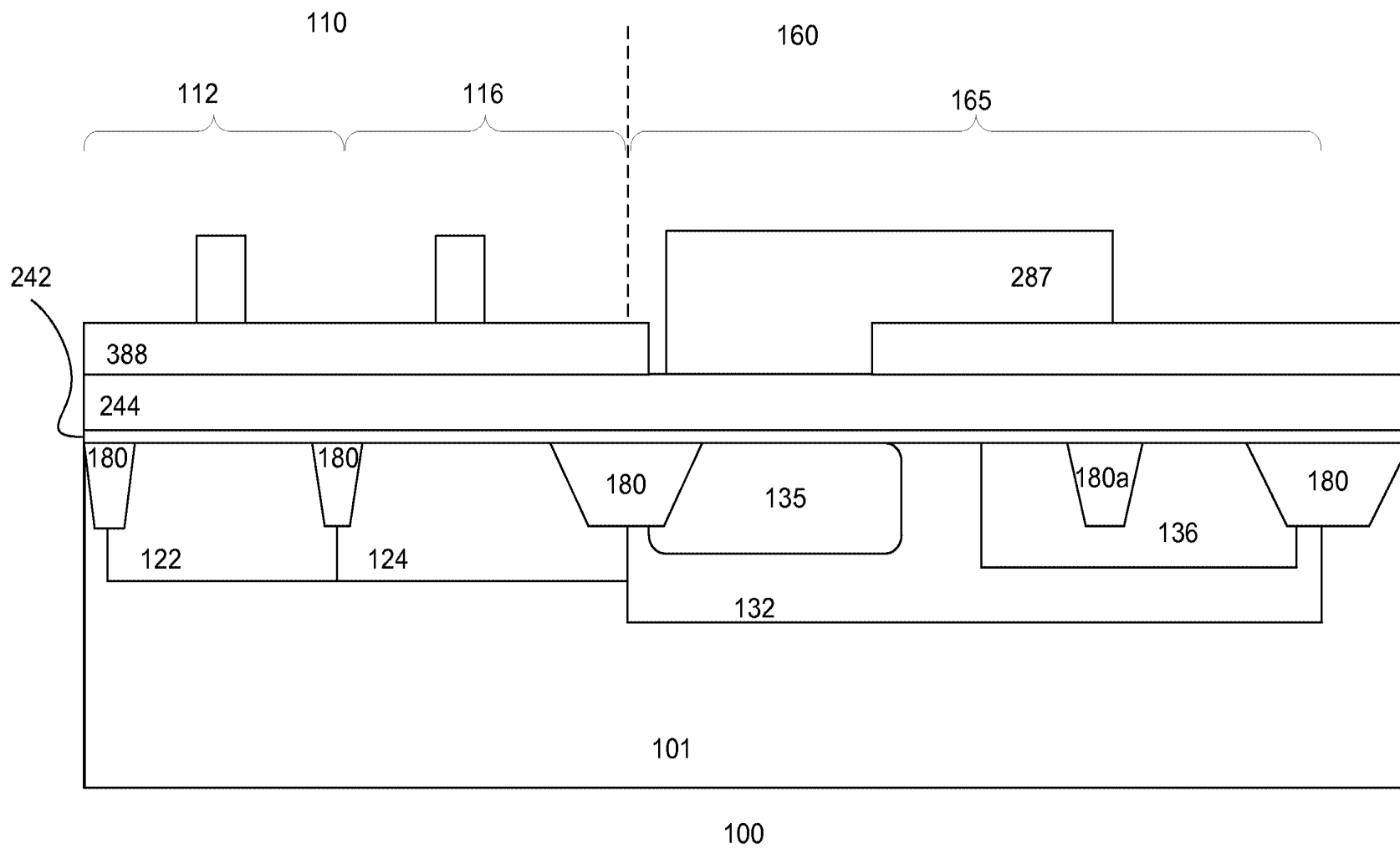


Fig. 3d

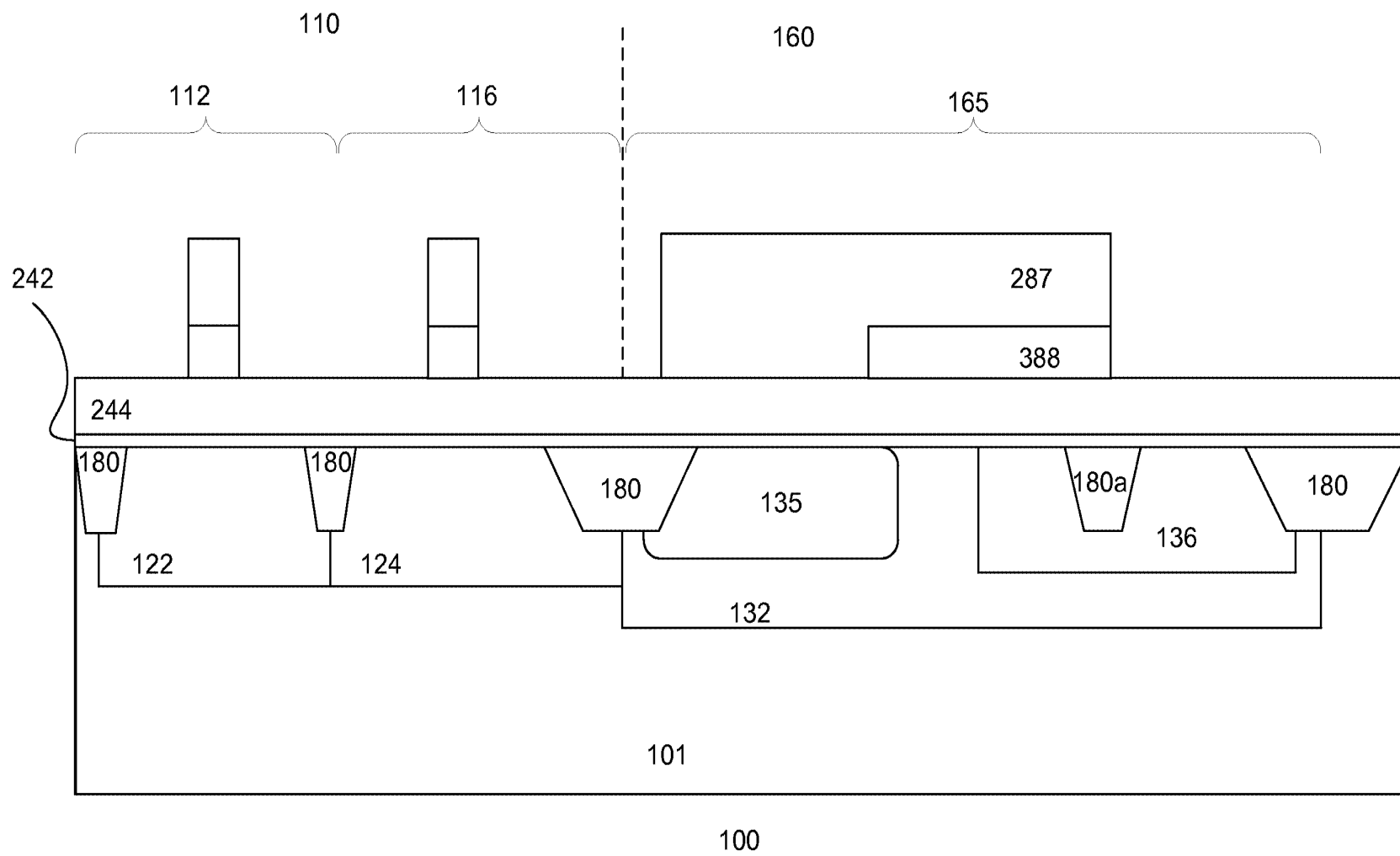


Fig. 3e

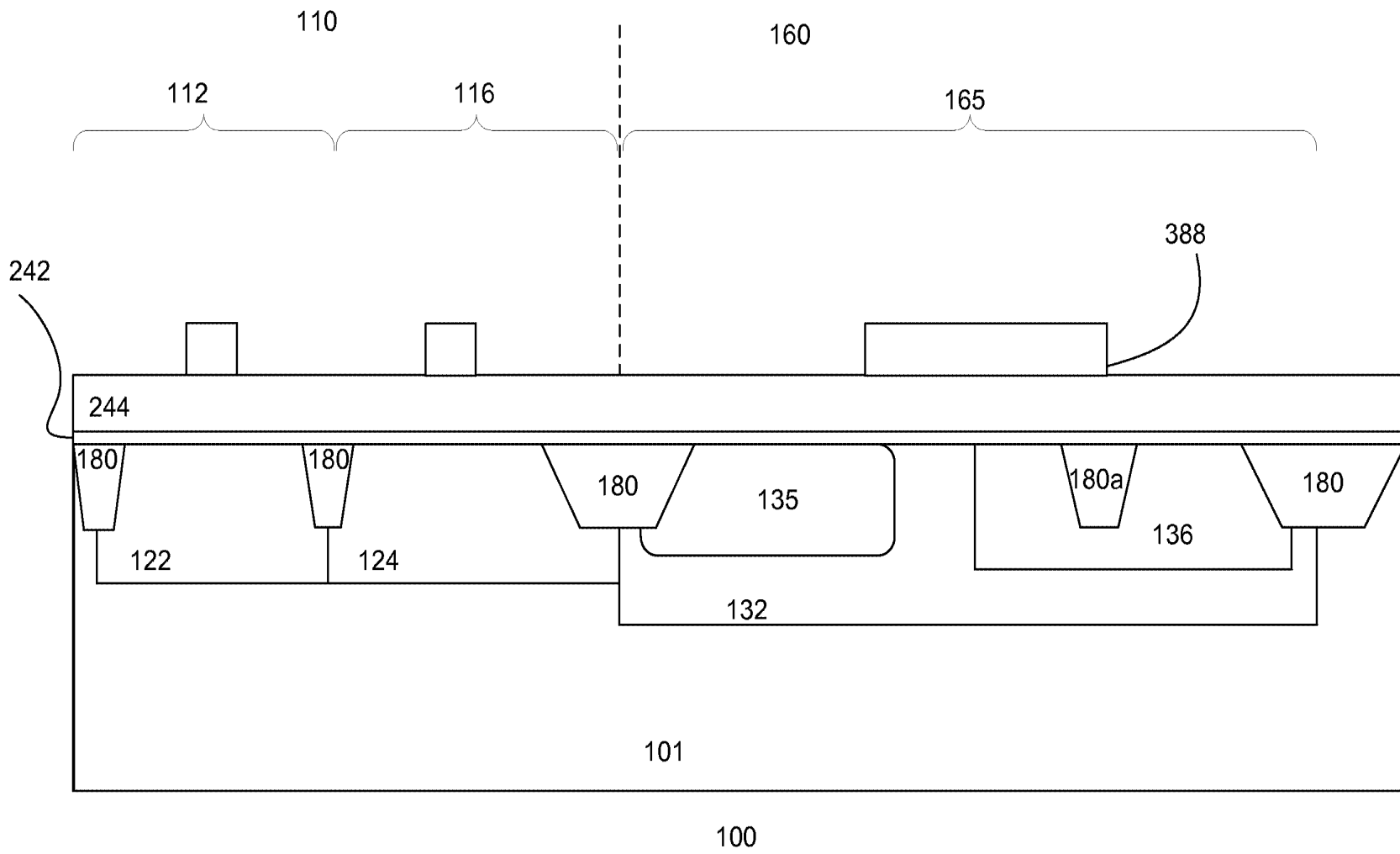


Fig. 3f

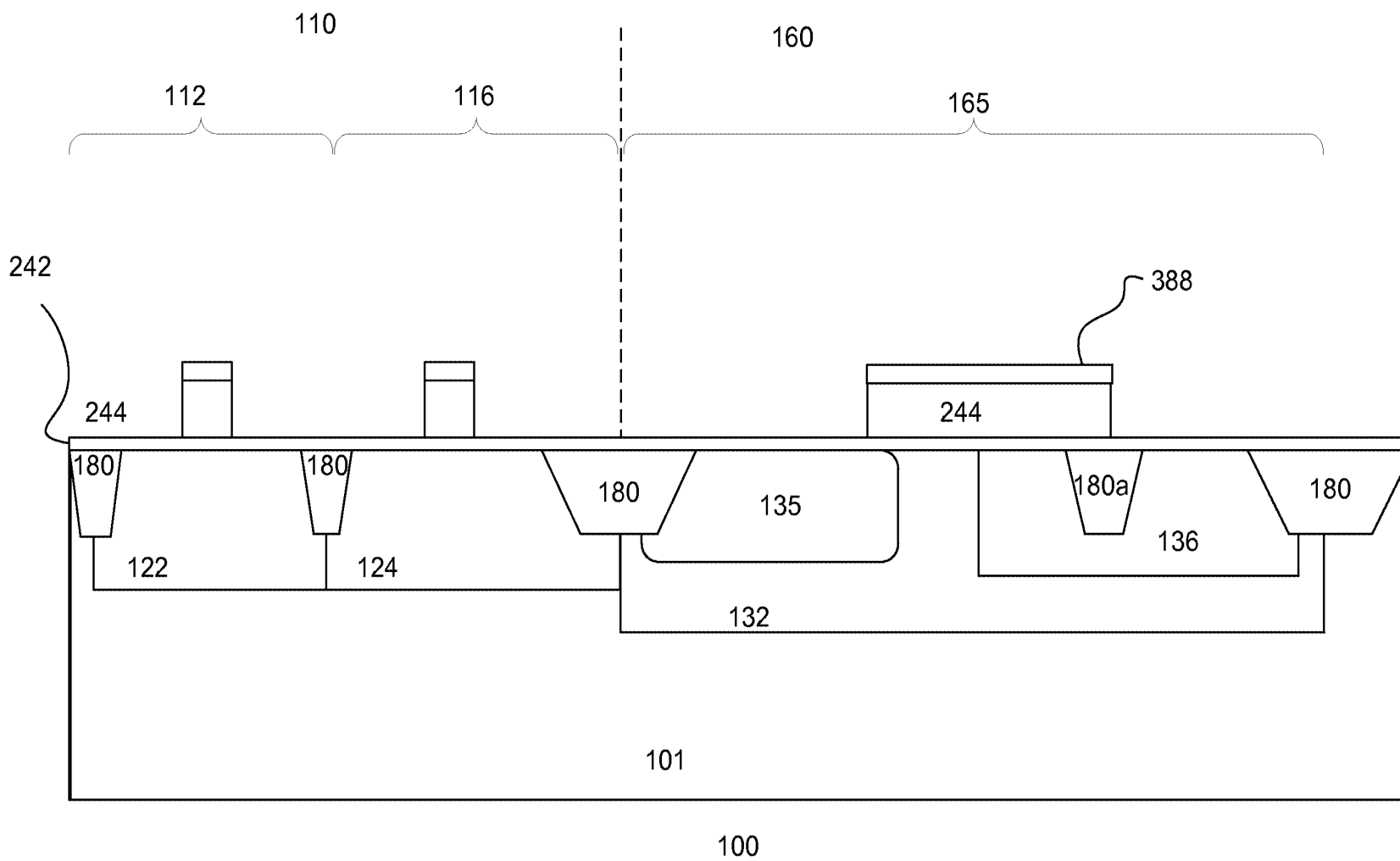


Fig. 3g

Electronic Patent Application Fee Transmittal

Application Number:				
Filing Date:				
Title of Invention:	HIGH VOLTAGE DEVICE			
First Named Inventor/Applicant Name:	Junwen LIU			
Filer:	Dexter Kam Chin			
Attorney Docket Number:	CSM P 2009 NAT 03 US 0			
Filed as Large Entity				
Utility under 35 USC 111(a) Filing Fees				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Utility application filing	1011	1	330	330
Utility Search Fee	1111	1	540	540
Utility Examination Fee	1311	1	220	220
Pages:				
Claims:				
Claims in excess of 20	1202	1	52	52
Miscellaneous-Filing:				
Petition:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				1142

Electronic Acknowledgement Receipt

EFS ID:	4834973
Application Number:	12390509
International Application Number:	
Confirmation Number:	5260
Title of Invention:	HIGH VOLTAGE DEVICE
First Named Inventor/Applicant Name:	Junwen LIU
Customer Number:	31366
Filer:	Dexter Kam Chin
Filer Authorized By:	
Attorney Docket Number:	CSM P 2009 NAT 03 US 0
Receipt Date:	23-FEB-2009
Filing Date:	
Time Stamp:	03:12:22
Application Type:	Utility under 35 USC 111(a)

Payment information:

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Payment Type	Deposit Account
Payment was successfully received in RAM	\$ 1142
RAM confirmation Number	5598
Deposit Account	502388
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File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1		CSMP2009NAT03US0-Specification.pdf	139690 5fa0dbb649b8ce04818690a02553d4698fa1c7fc	yes	23
	Multipart Description/PDF files in .zip description				
	Document Description		Start		End
	Specification		1		18
	Claims		19		22
	Abstract		23		23
Warnings:					
Information:					
2	Application Data Sheet	CSMP2009NAT03US0-ADS.pdf	1339540 a89478c88809679bce52f5d2356228ce83dfa750	no	5
Warnings:					
Information:					
3	Oath or Declaration filed	CSMP2009NAT03US0-Decl_POA.pdf	87560 be86a2bccaf767b6474a83da0f0c52a247f272ba	no	5
Warnings:					
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4	Power of Attorney	CSMP2009NAT03US0-Decl_POA_copy.pdf	87560 be86a2bccaf767b6474a83da0f0c52a247f272ba	no	5
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5	Information Disclosure Statement (IDS) Filed (SB/08)	CSMP2009NAT03US0-IDS.pdf	761051 34b3dae6ede77cbd1396b2cc14a96cd046340952	no	4
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6	NPL Documents	Narrow_n_p_isolation_in_retrograde_well_implants.pdf	241406 97cc6038cf672e4810a6110ff02e48cec4d3762c	no	8
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8	Fee Worksheet (PTO-06)	fee-info.pdf	36187 9ea21c9ccb07db06cbc54846182b48b468e64725	no	2
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HIGH VOLTAGE DEVICE

BACKGROUND

[0001] Lateral Double-Diffused (LD) transistors have been widely employed in high voltage applications. For high performance LD transistors, low drain-to-source on-resistance ($R_{ds_{on}}$) is desired to minimize its power dissipation when it is turned on, as well as high breakdown voltage to maximize its voltage capability. To achieve low $R_{ds_{on}}$, the channel of the high LD transistor should be as short as possible.

[0002] As technology evolves into era of deep sub-micron (e.g., beyond 0.35 μm) Very-Large-Scale Integration (VLSI), there is a desire for both high voltage (HV) LD transistors and low voltage (LV) transistors to be fabricated on the same substrate. Generally, the processes for forming LV devices, such as complementary metal-oxide-semiconductor (CMOS) processes, are used to form the HV devices. However, conventional CMOS processes are incompatible for forming HV transistors. For example, process overlay issues make it difficult for alignment of the HV channel well, which defines the channel length. Process variations in process overlay requires large process windows. This makes forming short channel lengths to achieve low $R_{ds_{on}}$ difficult if not impossible. Furthermore, the thin gate electrodes used prevent them serving as a hard mask, further exacerbating process control in forming the channel well.

[0003] From the foregoing discussion, it is desirable to provide reliable HV devices formed with short channel lengths to have low $R_{ds_{on}}$.

SUMMARY

[0004] A method of forming a device is presented. The method includes providing a substrate prepared with an active device region. The active device region includes gate stack layers of a gate stack comprising at least a gate electrode layer over a gate dielectric layer. An implant mask is formed on the substrate with an opening exposing a portion of a top gate stack

layers. Ions are implanted through the opening and gate stack layers into the substrate to form a channel well. The substrate is patterned to at least remove portion of a top gate stack layer unprotected by the implant mask.

[0005] In another embodiment, a device is disclosed. The device includes a substrate prepared with an active region. The active region includes a gate stack with a gate electrode layer over a gate dielectric layer. A first edge of a gate and a second edge of the gate are patterned separately. A doped channel well is disposed in the substrate adjacent to the first edge of the gate. The first edge of the gate overlaps the channel well with a channel edge of the channel well beneath the gate. The first edge of the gate and the channel edge define an effective length of the device. A doped drift well is adjacent to the second edge of the gate.

[0006] These and other objects, along with advantages and features of the present invention herein disclosed, will become apparent through reference to the following description and the accompanying drawings. Furthermore, it is to be understood that the features of the various embodiments described herein are not mutually exclusive and can exist in various combinations and permutations.

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] In the drawings, like reference characters generally refer to the same parts throughout the different views. Also, the drawings are not necessarily to scale, emphasis instead generally being placed upon illustrating the principles of the invention. In the following description, various embodiments of the present invention are described with reference to the following drawings, in which:

[0008] Fig. 1 shows an embodiment of a device;

[0009] Figs. 2a-g show an embodiment of a process for forming a device; and

[0010] Figs. 3a-g show an alternative embodiment of a process for forming a device.

DETAILED DESCRIPTION

[0011] Embodiments generally relate to semiconductor devices or integrated circuits. More particularly, some embodiments relate to high power devices. For example, high power devices include lateral double-diffused (LD) transistors, such as metal oxide transistors (MOS). The high power devices can be employed as switching voltage regulators for power management applications. The LD transistors can be easily integrated into devices or ICs. The devices or ICs can be incorporated into or used with, for example, consumer electronic products, particularly portable consumer products, such as cell phones, laptop computers and personal digital assistants (PDAs).

[0012] Fig. 1 shows a portion 100 of an embodiment of a device. As shown, the portion includes a substrate 101. The substrate, for example, comprises a silicon substrate. The substrate can be lightly doped with p-type dopants. N-type or other types of substrates, such as silicon germanium or silicon-on-insulator (SOI), are also useful.

[0013] The substrate may be prepared with first and second regions 110 and 160. In one embodiment, the first region comprises a low voltage (LV) device region while the second region comprises a high voltage (HV) device region. The regions may be other types of device regions or additional device regions may be provided. For example, a dual gate oxide (DGO) region may be provided for DGO devices. The DGO devices may be used for I/O circuitries. Alternatively, the device may include only a HV region.

[0014] The first region, in one embodiment, comprises first and second type active sub-regions 112 and 116. The first and second types are complementary types, forming a complementary type device. For example, the complementary type device comprises a complementary metal oxide semiconductor (CMOS) device. Forming non-complementary or other types of devices is also useful.

[0015] The first type active sub-region, for example, comprises a n-type active sub-region and the second type active sub-region comprises a p-type active region. The first type active sub-region comprises a second type doped well 122; the second type active sub-region comprises a first type doped well 124. For example, the first type active sub-region comprises a p-type doped well and the second type active sub-region comprises a n-type doped well. Other configurations of active sub-regions are also useful. P-type dopants can include boron (B), indium (In) or a combination thereof while n-type dopants can include phosphorus (P), arsenic (As), antimony (Sb) or a combination thereof.

[0016] A first type transistor 140 is provided in the first type active sub-region and a second type transistor 150 is provided in the second type active sub-region. For example, a n-type transistor is provided in the n-type active sub-region with a p-type doped well while a p-type transistor is provided in the p-type active sub-region with a n-type doped well. The transistors, in one embodiment, are MOS field effect transistors (MOSFETs). Other types of transistors may also be useful.

[0017] A transistor, for example, comprises a gate having a gate electrode 144 over a gate dielectric 142. The gate electrode, for example, comprises polysilicon. Other types of gate electrode materials, such as polysilicide or metal silicide, are also useful. Polysilicide or metal silicide gates can be formed by reacting the desired metal with polysilicon. Various types of metal, such as cobalt, titanium or nickel, can be used. Other types of gate electrode materials may also be useful. Furthermore, the gate electrode may be doped with dopants. Doping the gate electrode may depend on the technology. The gate electrode, for example, may be doped with the same dopant type of the transistor type. Doping the gate electrode with other dopant types is also useful. The gate dielectric can be silicon oxide. Other types of gate electrode or gate dielectric materials can also be useful. The material of the gate electrode and gate dielectric may depend on the technology. For example, cobalt silicide gate electrodes may be used for

0.18 μm and 0.13 μm technologies while nickel silicide may be used for 0.09 μm technologies. Other configurations of gate electrode and gate dielectric materials are also useful.

[0018] Sidewalls of the gate can be provided with sidewall spacers 148. The sidewall spacers, for example, comprise a dielectric material, such as silicon oxide, silicon nitride or a combination thereof. Other types of materials can be used for the sidewall spacers.

[0019] Adjacent to the gate are diffusion regions 146. The diffusion regions serve as, for example, source and drain of the transistor. Extension diffusion regions 147 can be provided. The extension diffusion regions, for example, are shallow diffusion regions which extend under the spacers. The diffusion regions (including the extension diffusion regions) of the first type transistor are doped with the first type dopants and the diffusion regions of the second type transistor are doped with the second type dopants. For example p-type diffusion regions are provided for p-type transistors and n-type diffusion regions are provided for n-type transistors. The depth and dopant concentration of the diffusion regions may depend, for example, on the application, such as voltage requirements.

[0020] The HV region includes a HV active region 165. In one embodiment, the HV active region comprises a first type active region. For example, the HV active region comprises a n-type active region. Providing a p-type HV active region is also useful. Furthermore, it is understood that the HV region is depicted with only one active region for purpose of simplification and that the HV region can include numerous HV active regions or sub-regions. For example, the HV active sub-regions can be complementary types of HV sub-regions for complementary types of HV devices.

[0021] Isolation regions 180, such as shallow trench isolation (STI) regions are provided to isolate the LV and HV regions as well as active sub-regions within the regions. Other types of isolation regions are also useful.

[0022] In one embodiment, the first type HV active region comprises at least first and second portions. The first portion comprises a second type HV channel (or body) well 135 and the second portion comprises a first type HV drift well 136. The HV drift well, for example, defines the type of HV device. For example, a p-type HV active region would have a n-type HV channel well and a p-type HV drift well. In one embodiment, a third or middle portion separates the channel and drift wells. In other embodiments, the first and second portions are contiguous portions. For example, the HV channel and HV drift wells contact each other.

[0023] A deep well 132 may be provided in the HV active region, encompassing the HV channel and drift wells. Generally, the deep well comprises dopants of the first type or same as the HV device type. The deep well may be used to isolate the drift well from the substrate. For example, a p-type device will have a n-type deep well. In some cases, the deep well can be optional or the deep well can comprise dopants of the second or opposite type of the HV device type. For example, in the case of a n-type HV device, the deep well can be a p-type or n-type deep well. The different types of deep well selected may depend on the type of substrate. Generally, p-type doped substrates are used for fabricating devices. When p-type substrates are used, the n-type deep well may be optional. This is because the n-type drift well is already different from the p-type substrate. For a n-type substrate, a p-type deep well may be used to isolate the channel well from the substrate. Other configurations of deep wells may also be useful.

[0024] As shown, a first type HV device is provided in the first type HV active region. The first type HV device, in one embodiment, comprises a LD device, such as a LDMOS device. The HV device comprises a gate with first and second edges 179a-b or sidewalls.

[0025] The gate includes a HV gate electrode 174 over a HV gate dielectric 172. The HV gate electrode, for example, comprises polysilicon. Other types of HV gate electrode materials, such as polysilicide or metal silicide, are also useful. Various types of metal, such as cobalt,

titanium or nickel, can be used. Other types of HV gate electrode materials may also be useful. Furthermore, the gate electrode may be doped with dopants. The HV gate electrode, for example, may be doped with the same dopant type of the transistor type. Doping the HV gate electrode with other dopant types is also useful. The HV gate dielectric can be silicon oxide. Other types of HV gate dielectric materials can also be useful.

[0026] In one embodiment, the material of the HV gate electrode and HV gate dielectric may depend on the CMOS process and/or technology. In one embodiment, the material of the HV gate electrode and HV gate dielectric may depend on the CMOS process for LV devices and/or technology. For example, cobalt silicide gate electrodes may be used for 0.18 μm and 0.13 μm technologies while nickel silicide may be used for 0.09 μm technologies. Other configurations of gate electrode and gate dielectric materials are also useful.

[0027] Sidewalls of the gate can be provided with sidewall spacers 178. The sidewall spacers, for example, comprise a dielectric material, such as silicon oxide, silicon nitride or a combination thereof. Other types of materials can be used for the sidewall spacers.

[0028] In one embodiment, the various gate elements of the HV gate, such as gate electrode, gate dielectric and spacers are of the same type of material as the LV devices in the LV region. This facilitates process compatibility for forming HV and LV devices.

[0029] The HV gate overlaps the channel and drift wells in the first and second portions of the HV active region. A drift isolation region 180a may be provided within the drift well 136 on the high voltage side of the HV device. The drift isolation region 180a can improve breakdown voltage of the HV device due to high voltage. For example, the drift isolation region is particularly useful to sustain high voltages for applications beyond about 10V. As shown, the drift isolation region 180a is disposed in the drift well 136 under the gate. Disposing the drift well at other positions, such as partially underlying the gate may also be useful.

[0030] A first type drain diffusion region 176b is provided between the drift isolation region 180a and isolation region 180 which isolates the HV active region from other sub-active regions in the HV region. A first type source diffusion region 176a is provided in the channel well 135 of the first type HV active region. A source extension region 177 can be provided which extends under the spacer on the channel side of the HV gate. The first type source and drain diffusion regions, for example, comprise a p-type diffusion regions for a p-type HV active region. Providing n-type diffusion regions for a n-type HV active region is also useful. The dopant depth and concentration of the diffusion regions may be selected depending on the application. In one embodiment, the dopant depth and concentration of diffusion regions may be the same as those of LV devices. Other dopant depths and concentrations are also useful.

[0031] A second type body contact region 171 may be provided in the first portion of the HV active region. As shown, the second type body contact region 171 is between the source diffusion region 176a and the isolation region 180. The second type body contact region comprises second type dopants. In one embodiment, dopant depth and concentration of the body contact region may be the same as the diffusion regions of the LV devices. Other dopant depths and concentrations are also useful. The body contact region 171 provides electrical coupling to the channel well 135.

[0032] An edge 133 of the channel well 135 extends below the gate on the source side of the gate. The edge 133, for example, can be referred to as the channel edge of the channel well. The channel edge of the channel well 135 and the first edge 179a of the gate on the source side define a channel of the HV transistor having an effective channel length L .

[0033] The effective channel length, in one embodiment, is controlled by ion implantation which is self-aligned to the gate edge. For example, dopant ions are implanted using an implant mask through the gate electrode or a hard mask and the gate electrode. The implant mask is also used to pattern the channel edge of the HV gate. The effective channel length may be controlled

by channel or body implant conditions selected to satisfy threshold voltage and breakdown voltage requirements. For example, shorter effective channel length may be achieved by decreasing tilt angle, implant energy or dosage and vice-versa.

[0034] With the channel well being self-aligned to the gate edge of the implant, process overlay issues are avoided. Very short effective channel lengths with very low $R_{ds_{on}}$ can be achieved, enabling high current to pass through a small area. This is particularly useful for technologies beyond 0.35 μm , such as, for example, 0.25 μm , 0.18 μm and 0.13 μm technologies.

[0035] Smaller area requirements for power management applications with reduced power dissipation increases performance while reducing costs.

[0036] In one embodiment, the effective channel length L is less than 0.4 μm . In other embodiments, the effective channel length is less than 0.3 μm . In yet another embodiment, the effective channel length is less than 0.25 μm . Other effective channel lengths may also be useful, for example, depending on the application.

[0037] Figs. 2a-g show cross-sectional views of an embodiment of a process for forming a device or IC 100. Referring to Fig. 2a, a substrate 101 is provided. The substrate can comprise a silicon substrate, such as a lightly p-type doped substrate. Other types of substrates, including n-type doped substrates, silicon germanium or silicon-on-insulator (SOI), are also useful.

[0038] As shown, first and second device regions 110 and 160 are defined on the substrate. The first region, for example, comprises a LV device region while the second region comprises a HV device region. Other types of device regions or additional device regions may be provided. Alternatively, the device may include only a HV region.

[0039] The LV device region 110 can be defined with first and second type active sub-regions 112 and 116. In one embodiment, the first and second type active sub-regions are

complementary type active sub-regions to form a complementary device such as a CMOS device. Forming non-complementary or other types of devices is also useful.

[0040] As for the HV device region 160, a HV active region 165 is defined therein. In one embodiment, the HV active region 165 can be a first or second type HV active region. Although only one HV active region is depicted, it is understood that the HV device region may include additional HV active regions or sub-regions. The HV active sub-regions can be complementary HV active sub-regions.

[0041] The substrate is also prepared with isolation regions 180 to separate the device regions as well as other active device sub-regions. In one embodiment, the isolation regions 180 comprise STIs. Various conventional processes can be employed to form the STI regions. For example, the substrate can be etched using conventional etch and mask techniques to form trenches which are then filled with dielectric materials such as silicon oxide. Chemical mechanical polishing (CMP) can be performed to remove the excess dielectric materials and provide a planar substrate top surface. Other processes or materials can also be used to form the STIs.

[0042] In Fig. 2b, the doped wells for the active regions of the device regions are formed. In one embodiment, a first type doped well 124 is formed in the second type LV active sub-region 116 and a second type doped well 122 is formed in the first type LV active sub-region 112. For example, a n-type doped well is formed in the p-type active sub-region and a p-type doped well is formed in the n-type active sub-region.

[0043] In the HV active region 165, it includes at least first and second portions. For example, the first portion corresponds to a drain portion and the second portion corresponds to a source portion. In some embodiments, a third or middle portion is provided to separate the first and second portions. Alternatively, the first and second portions are contiguous portions without a middle portion.

[0044] In one embodiment, a doped well 136 is formed in the drain portion of the HV active region. The doped well, for example, is a drift well of the HV device. The drift well comprises the same type as the HV active region. In the case of a first type HV active region, the drift well comprises a first type. For example, a n-type drift well is provided for a n-type HV active region.

[0045] A drift STI region 180a may be optionally provided in the drift well 136. The drift STI region is disposed in the drift well to improve reliability of high voltage devices. For example, the drift STI region improves breakdown reliability of the HV device. The drift STI region can be formed at the same time as the other STI regions.

[0046] A deep HV well 132 may be provided in the HV active region 165. As shown, the deep HV well 132 is formed in the whole HV active region 165 and encompasses the first and second portions of the HV active region 165. The deep HV well comprises an opposite type as the drift well and the HV active region. In the case of a first type HV active region, the deep well comprises a second type. For example, a p-type deep well is provided for a n-type HV active region.

[0047] The depth and concentration of dopants of the various wells may depend on, for example, application. For example, higher voltage applications may require lower doping concentration of dopants and greater depth of the wells. In one embodiment, the depth of the deep well 132 is about 3 μm with a dopant concentration of about $5\text{E}16\text{cm}^{-3}$. For the drift well 136, it may have a depth of about 1.5 μm with a dopant concentration of about $5\text{E}17\text{cm}^{-3}$. Other depths and dopant concentrations are also useful for the wells.

[0048] As described, the STI regions are formed prior to forming the various doped wells. Other process schemes, such as forming the STI regions after the various doped wells are formed may also be useful.

[0049] The doped wells can be formed by ion implantation. P-type dopants can include boron (B), BF₂ (boron compounded with fluorine), indium (In) or a combination thereof while n-type dopants can include phosphorus (P), arsenic (As), antimony (Sb) or a combination thereof. Generally, the first and second types of the active regions are formed selectively in separate processes. For example, the n-type regions can be implanted with n-type dopants while an implant mask prevents doping of the p-type regions.

[0050] In one embodiment, the wells of the HV region are formed separately from the LV regions. For example, the LV wells can be formed first followed by the HV wells. Forming the HV wells prior to the LV wells is also useful. The different types of LV wells can be formed in separate processes and the different types of HV wells can also be formed in separate processes. For example, the first type and second type wells in the LV region are formed in separate implant processes. Similarly, the different types of wells in the HV region are formed in separate implant processes.

[0051] A doped well can be formed by a single implant process or multiple implant processes. In multiple implant processes, dopants can be implanted at different energy levels to achieve the desired doping profiles. In the case of a single implant process, a high temperature drive-in process is performed to achieve the desired doping profile.

[0052] A single implant process may be employed to, for example, form deep wells. For a n-type deep well, phosphorus may be implanted with a dopant concentration of about $6 \times 10^{12} \text{ cm}^{-2}$ at an energy level of about 2000 KeV followed by a two-hour drive-in at a temperature of about 1100 °C. Single implant processes may also be used to form other types of wells. Forming wells using multiple implant processes may also be useful. For example, LV wells may be formed by multiple implant processes.

[0053] In Fig. 2c, various layers of the gate are formed on the substrate. In one embodiment, a gate dielectric layer 242 is formed over the substrate. The gate dielectric layer, for example,

comprises silicon oxide. Other types of dielectric materials, such as silicon oxynitride, may also be useful. Alternatively, high-k, low-k or composite of dielectric materials may be used. The thickness of the gate dielectric layer 242 may be about 30 Å for 1.8V gate voltage transistors. Other thicknesses may also be useful. The thickness, for example, may depend on the gate voltage applications. For example, higher gate voltages may require thicker gate dielectric layers. In one embodiment, the gate dielectric layer is formed by thermal oxidation. Other techniques, such as chemical vapor deposition (CVD), can also be used to form the gate dielectric layer.

[0054] A gate electrode layer 244 is deposited on the gate dielectric layer 242. The gate electrode layer comprises, in one embodiment, polysilicon (poly). The gate electrode layer 244 can be formed as an amorphous or non-amorphous layer. For amorphous deposited layer, subsequent processing can be performed to crystallize it. Other types of gate electrode materials are also useful. For example, the polysilicon can be subsequently processed to form polysilicide or metal gates. The thickness of the gate electrode layer may be about 2000 Å or less. In another embodiment, the thickness of the gate electrode layer is about 2500 Å or less. In yet another embodiment, the thickness of the gate electrode layer can be about 4000 Å or less. For example, in the case of polysilicide gates, the polysilicon may be about 2000 Å while the tungsten is about 2000 Å. Other thicknesses are also useful. Various techniques can be used to form the gate electrode layer. For example, polysilicon can be deposited by CVD while metal can be deposited by sputtering. Other techniques, depending on the material, may also be useful.

[0055] Referring to Fig. 2d, a mask layer 288 is formed on the substrate, covering the gate electrode layer. In one embodiment, the mask layer comprises photoresist. The mask is patterned as desired. To pattern the mask layer, photolithography can be employed. For example, the mask can be selectively exposed to an exposure source through a lithographic mask. Depending on whether a positive or negative resist is used, the exposed or unexposed

portions are removed by development. To enhance lithographic resolution, an ARC layer (not shown) may be provided beneath the mask layer.

[0056] The mask layer 288 is patterned to form an opening 289 to expose a portion of the gate electrode layer 244. In one embodiment, the opening corresponds to the opening for the channel well implant. For example, the mask layer serves as the channel well implant mask. The ARC layer may be patterned with the mask layer. Subsequently patterning the ARC layer may also be useful.

[0057] In Fig. 2c, the substrate is implanted with dopants to form a channel well 135. In one embodiment, dopants of the opposite type as the type of the HV active region are implanted. For example, p-type dopants are implanted into the p-type deep well of a n-type HV active region. In one embodiment, the dopants are implanted at an angle to form a channel from the gate edge to the edge of the channel well. The implant angle θ can be in a range of about 1 to 45°. Other implant angles may also be useful. The implant conditions can be varied depending on application requirements. For example, the angle, dose and energy can be selected to achieve the desired effective channel length.

[0058] In one embodiment, the channel well is formed by multiple implants. For example, the channel well can be formed by at least two tilt and rotate implants. In one embodiment, the implant comprises a quad implant. A quad implant comprises 4 tilted angled implants, each rotated by a rotation angle. For example, a p-type channel well can be formed with a quad implant at a tilt angle of 30° with rotation angle of 45° with a boron dose of about $2 \times 10^{13} \text{cm}^{-2}$ at an energy level of about 130 KeV. Other tilt angles, rotation angles, doses and energy levels are also useful. For example, a quad implant can comprise a tilt angle of 7° with rotation angle of 45° with a boron dose of about $2 \times 10^{13} \text{cm}^{-2}$ at an energy level of about 150 KeV.

[0059] After forming the channel doped well 135, exposed portions of the gate electrode layer 244 are removed, as shown in Fig. 2f. In one embodiment, an anisotropic etch, such as a

reactive ion etch (RIE) is performed to remove exposed portions of the gate electrode layer. The removal of the exposed portions of the gate electrode layer, in one embodiment, forms a first edge of a HV gate. The first edge, in one embodiment, corresponds to the source side of the HV gate. As shown, the gate dielectric layer 242 serves as an etch stop for etching the gate electrode layer. Leaving the gate dielectric layer can be advantageous since it can serve as an implant mask for subsequent ion implantation to protect the substrate from implant damage.

Alternatively, the exposed portion of the gate dielectric layer can be removed. Removing the gate dielectric layer 242 to expose the substrate may also be useful. After etching the gate electrode layer, the mask layer 288 is removed.

[0060] As described, the implant mask, in one embodiment, serves as a channel well implant mask as well as a mask for patterning the first edge (e.g., source side) of the HV gate. By using the same mask to form the channel well and source side of the gate, the channel is self-aligned to the gate edge. This enables the channel length of the HV device to be well controlled to produce very short effective channel lengths to reduce $R_{ds_{on}}$ performance. In one embodiment, the effective channel length L is less than $0.4\text{ }\mu\text{m}$. In other embodiments, the effective channel length is less than $0.3\text{ }\mu\text{m}$. In yet another embodiment, the effective channel length is less than $0.25\text{ }\mu\text{m}$.

[0061] Referring to Fig. 2g, another mask layer 288b is formed on the substrate and patterned. The mask layer is patterned to expose portions of the gate electrode layer 244 to be removed, protecting portions corresponding to gates of the LV devices. For example, the mask serves as a gate mask. Additionally, the mask layer 288b protects the source portion of the HV region and the gate of the HV device. The exposed portions of the gate electrode layer 244 are removed to form gates of both the LV and HV devices.

[0062] After the gates are formed, the process continues to form the device, as illustrated in Fig. 1. The process, for example, includes forming extension regions 147 by ion implantation,

followed by forming spacers 148 and 178 and diffusion regions 146 and 176. The different types of extension and diffusion regions can be formed in different processes. For example, n-type diffusion regions are formed in one implant process and p-type diffusion regions are formed in another process. Furthermore, the diffusion and extension regions of the different device regions can be formed separately. Silicide gates and contacts can be formed after the diffusion regions are formed. For example, metals, such as cobalt are deposited on the substrate 101 and reacted to form silicide contacts and gates. The contacts and gates can be formed in the same or different processes. Unreacted metals are removed. A pre-metal and an inter-level dielectric layer can be formed in which contacts and interconnects are formed. Additional interconnect levels can be formed by, for example, dual damascene techniques. Final passivation, dicing, assembly and testing may be performed to complete the IC.

[0063] An alternative embodiment of a process of forming a device 100 is shown in Figs. 3a-g. Referring to Fig. 3a, a partially processed device is shown. The partially processed device is similar to that shown in Fig. 2c. A hard mask 388 is formed on the gate electrode layer 244. The hard mask, for example, comprises silicon oxide. In one embodiment, the hard mask 388 is formed by CVD. The thickness of the hard mask, for example, is about 40nm. Other materials, techniques or thicknesses may also be useful. For example, the hard mask may be formed from silicon nitride or other types of materials.

[0064] Referring to Fig. 3b, a soft mask 287, such as photoresist, is formed on the hard mask 388. The soft mask can be patterned by photolithographic techniques to form an opening 289 which can serve as a channel implant mask. An ARC layer can be provided between the soft mask 287 and the hard mask 388. The pattern of the soft mask is transferred to the hard mask by, for example, RIE.

[0065] In one embodiment, the soft mask layer is removed after patterning of the hard mask, as shown in Fig. 3c. The substrate is implanted with dopants to form a channel doped well 135.

In one embodiment, dopants of the opposite type as the type of the HV active region are implanted. For example, p-type dopants are implanted into the p-type deep well of a n-type HV active region.

[0066] In one embodiment, the dopants are implanted at an angle to form a channel from the gate edge to the edge of the channel well. The implant angle θ can be in a range of about 1 to 45°. Other implant angles may also be useful. The implant conditions can be varied depending on application requirements. For example, the angle, dose and energy can be selected to achieve the desired effective channel length.

[0067] In one embodiment, the channel well is formed by multiple implants. For example, the channel well can be formed by at least two tilt and rotate implants. In one embodiment, the implant comprises a quad implant. A quad implant comprises 4 tilted angled implants, each rotated by a rotation angle. For example, a p-type channel well can be formed with a quad implant at a tilt angle of 30° with rotation angle of 45° with a boron dose of about $2 \times 10^{13} \text{cm}^{-2}$ at an energy level of about 130 KeV. Other tilt angles, rotation angles, doses and energy levels are also useful. For example, a quad implant can comprise a tilt angle of 7° with rotation angle of 45° with a boron dose of about $2 \times 10^{13} \text{cm}^{-2}$ at an energy level of about 150 KeV.

[0068] As shown in Fig. 3d, another soft mask layer 287 is deposited on the substrate, covering the hard mask layer and exposed gate electrode layer. The soft mask is patterned by lithography. The soft mask protects portions of the hard mask layer 388 corresponding to the transistor gates. A new ARC layer can be provided between the soft mask 287 and the hard mask 388. In Fig. 3e, exposed portions of the hard mask are removed, exposing portions of the gate electrode layer. Patterning of the hard mask layer can be achieved by, for example, RIE. The patterned hard mask serves as a gate mask. The soft mask, as shown in Fig. 3f, is removed after patterning the hard mask.

[0069] Referring to Fig. 3g, the gate electrode layer 244 is patterned to form gates in the LV and HV region by, for example etch. Patterning of the gate electrode layer can be achieved using, for example, RIE. The patterning of the gate electrode layer also erodes the hard mask, reducing its thickness. The hard mask 388 can be removed by, for example, a clean step after etching the gate electrode layer. The process continues as previously described to complete the device.

[0070] The invention may be embodied in other specific forms without departing from the spirit or essential characteristics thereof. The foregoing embodiments, therefore, are to be considered in all respects illustrative rather than limiting the invention described herein. Scope of the invention is thus indicated by the appended claims, rather than by the foregoing description, and all changes that come within the meaning and range of equivalency of the claims are intended to be embraced therein.

[0071] What is claimed is:

CLAIMS

1. A method of forming a device comprising:
 - providing a substrate prepared with an active device region, wherein the active device region includes gate stack layers of a gate stack comprising at least a gate electrode layer over a gate dielectric layer;
 - forming an implant mask on the substrate with an opening exposing a portion of a top gate stack layers;
 - implanting ions through the opening and gate stack layers into the substrate to form a channel well; and
 - patterning the substrate to at least remove portion of a top gate stack layer unprotected by the implant mask.
2. The method of claim 1 wherein the top gate stack layer comprises the gate electrode layer of the gate stack.
3. The method of claim 2 wherein the gate electrode layer comprises silicon, including amorphous or polysilicon.
4. The method of claim 2 wherein:
 - patterning the substrate forms a first edge of the gate; and
 - further comprises pattern the substrate to form a second edge of the gate.
5. The method of claim 4 wherein the first edge corresponds to a channel edge of the gate of the device and the second edge corresponds to the drain edge of the gate of the device.

6. The method of claim 1 wherein:
patterning the substrate forms a first edge of the gate; and
further comprises pattern the substrate to form a second edge of the gate.
7. The method of claim 6 wherein the first edge corresponds to a channel edge of the gate of the device and the second edge corresponds to the drain edge of the gate of the device.
8. The method of claim 1 wherein patterning the substrate leaves the gate dielectric layer remaining on the substrate.
9. The method of claim 1 wherein the gate stack layers comprises a hard mask layer over the gate electrode layer and patterning the gate stack removes the hard mask layer to expose the gate electrode layer.
10. The method of claim 9 wherein patterning the hard mask forms a first edge of hard mask corresponding to a first edge of the gate stack and further comprises patterning the hard mask to form a second edge of the hard mask corresponding to a second edge of the gate stack.
11. The method of claim 10 comprises patterning the gate stack layers with the patterned hard mask to form a gate with first and second edges.
12. The method of claim 11 wherein patterning the gate stack layers leaves the gate dielectric layer remaining on the substrate.
13. The method of claim 1 wherein implanting ions comprises a tilt angle implant.

14. The method of claim 1 wherein a tilt angle of the implant is about $1 - 45^\circ$.
15. The method of claim 1 wherein implanting ions comprises multiple tilt angle implants which are rotated about a plane of the substrate.
16. The method of claim 15 wherein a tilt angle of the implant is about $1 - 45^\circ$.
17. The method of claim 1 wherein implanting ions comprises a quad tilt angle implant which is rotated about a plane of the substrate.
18. A device comprising:
 - a substrate prepared with an active device region, wherein the active device region includes a gate stack with a gate electrode layer over a gate dielectric layer, wherein a first edge of a gate and a second edge of the gate are patterned separately;
 - a doped channel well disposed in the substrate adjacent to the first edge of the gate, wherein the first edge of the gate overlaps the channel well with a channel edge of the channel well beneath the gate, the first edge of the gate and channel edge defines an effective channel length of the device; and
 - a doped drift well adjacent to the second edge of the gate.
19. The device of claim 18 wherein the effective channel length is less than about $0.4\ \mu\text{m}$.

20. A method of forming a device comprising:

providing a substrate prepared with an active device region which includes a doped drift well in a first portion of the active device region and gate stack layers of a gate stack comprising at least a gate electrode layer over a gate dielectric layer on the surface of the substrate;

forming an implant mask on the substrate with an opening exposing a portion of a top gate stack layers;

implanting ions through the opening and gate stack layers into the substrate to form a channel well; and

patterning the substrate to at least remove portion of a top gate stack layer unprotected by the implant mask.

21. The method of claim 20 wherein the substrate comprises:

a drift isolation region in the doped drift well; and

a doped deep well in the substrate of the active device region encompassing the doped drift well.

HIGH VOLTAGE DEVICE

ABSTRACT

A method of forming a device is presented. A substrate prepared with an active device region is provided. The active device region includes gate stack layers of a gate stack that includes at least a gate electrode layer over a gate dielectric layer. An implant mask is formed on the substrate with an opening exposing a portion of a top gate stack layers. Ions are implanted through the opening and gate stack layers into the substrate to form a channel well. The substrate is patterned to at least remove portion of a top gate stack layer unprotected by the implant mask.

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Application Data Sheet 37 CFR 1.76		Attorney Docket Number	CSM P 2009 NAT 03 US 0
		Application Number	
Title of Invention	HIGH VOLTAGE DEVICE		
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Attorney Docket Number	CSM P 2009 NAT 03 US 0	Small Entity Status Claimed	☐
Application Type	Nonprovisional		
Subject Matter	Utility		
Suggested Class (if any)		Sub Class (if any)	
Suggested Technology Center (if any)			
Total Number of Drawing Sheets (if any)	15	Suggested Figure for Publication (if any)	

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Title of Invention	HIGH VOLTAGE DEVICE		

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Customer Number	31366		

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This section allows for the applicant to either claim benefit under 35 U.S.C. 119(e), 120, 121, or 365(c) or indicate National Stage entry from a PCT application. Providing this information in the application data sheet constitutes the specific reference required by 35 U.S.C. 119(e) or 120, and 37 CFR 1.78(a)(2) or CFR 1.78(a)(4), and need not otherwise be made part of the specification.			
Prior Application Status		Remove	
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)
Additional Domestic Benefit/National Stage Data may be generated within this form by selecting the Add button.			Add

Foreign Priority Information:

This section allows for the applicant to claim benefit of foreign priority and to identify any prior foreign application for which priority is not claimed. Providing this information in the application data sheet constitutes the claim for priority as required by 35 U.S.C. 119(b) and 37 CFR 1.55(a).			
		Remove	
Application Number	Country i	Parent Filing Date (YYYY-MM-DD)	Priority Claimed
			☐ Yes ☒ No
Additional Foreign Priority Data may be generated within this form by selecting the Add button.			Add

Assignee Information:

Providing this information in the application data sheet does not substitute for compliance with any requirement of part 3 of Title 37 of the CFR to have an assignment recorded in the Office.	
Assignee 1	Remove

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	CSM P 2009 NAT 03 US 0
		Application Number	
Title of Invention	HIGH VOLTAGE DEVICE		

If the Assignee is an Organization check here. ☒			
Organization Name	Chartered Semiconductor Manufacturing, Ltd.		
Mailing Address Information:			
Address 1	60 Woodlands Industrial Park D Street 2		
Address 2			
City	Singapore	State/Province	
Country i	SG	Postal Code	738406
Phone Number	+65 6362 2838	Fax Number	+65 6360 4970
Email Address			
Additional Assignee Data may be generated within this form by selecting the Add button. Add			

Signature:

A signature of the applicant or representative is required in accordance with 37 CFR 1.33 and 10.18. Please see 37 CFR 1.4(d) for the form of the signature.					
Signature	/dexter chin/			Date (YYYY-MM-DD)	2009-02-23
First Name	Dexter	Last Name	CHIN	Registration Number	38842

This collection of information is required by 37 CFR 1.76. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 23 minutes to complete, including gathering, preparing, and submitting the completed application data sheet form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

DECLARATION AND POWER OF ATTORNEY

As a below named inventor, I hereby declare that:

My residence, post office address, and citizenship are as stated below next to my name,

I believe I am the original, first and sole inventor (if only one name is listed below) or an original, first and joint inventor (if plural names are listed below) of the subject matter claimed and for which a patent is sought on the invention entitled:

HIGH VOLTAGE DEVICE

the specification of which

☒ is attached hereto

☐ was filed on ____ as Application Serial No. ____ and was amended on ____ (if applicable).

I hereby state that I have reviewed and understand the contents of the above-identified specification, including the claims, as amended by any amendment referred to above.

I do not know and do not believe the same was ever known or used in the United States of America before this invention thereof or more than one year prior to this application, and that the same was not in public use or on sale in the United States of America more than one year prior to this application. I acknowledge the duty to disclose information which is known to me to be material to patentability in accordance with title 37, Code of Federal Regulations, section 1.56.

I hereby claim foreign priority benefits under Title 35, United States Code, Section 119(a)-(d) or Section 365(b) of any foreign application(s) for patent or inventor's certificate, or Section 365(a) of any PCT international application which designated at least one country other than the United States, listed below and have also identified below any foreign application for patent or inventor's certificate having a filing date before that of the application on which priority is claimed:

Prior Foreign Application(s):

<u>Number</u>	<u>Country</u>	<u>Day/Month/Year filed</u>	<u>Priority Claimed</u>	
			<u>Yes</u>	<u>No</u>
				☒

I hereby claim the benefit under 35 USC 119(e) of any United States provisional application(s) listed below:

Prior Provisional Application(s):

<u>Application Number</u>	<u>Filing Date</u>

I hereby claim the benefit under Title 35, United States Code, Section 120 of any United States application(s), or Section 365(c) of any PCT international application designating the United States, listed below and, insofar as the subject matter of each of the claims of this application is not disclosed in the prior United States or PCT international application in the manner provided by the first paragraph of Title 35, United States Code, Section 112, I acknowledge the duty to disclose material information as defined in Title 37, Code of Federal

Docket No.: CSM P 2009 NAT 03 US 0

Regulations, Section 1.56 which occurred between the filing date of the prior application and the national or PCT international filing date of this application:

Prior U.S. Application(s):

<u>Serial No.</u>	<u>Filing Date</u>	<u>Status: Patented, Pending, Abandoned</u>
-------------------	--------------------	---

I hereby declare that all statements made herein of my own knowledge are true and that all statements made on information and belief are believed to be true; and further that these statements were made with the knowledge that willful false statements and the like so made are punishable by fine or imprisonment, or both, under Section 1001 of Title 18 of the United States Code and that such willful false statements may jeopardize the validity of the application or any patent issued thereon.

I hereby revoke any previous Powers of Attorney and appoint the following attorney(s) and/or agent(s), each said individual being a member or associate of **Horizon IP Pte. Ltd.** and associated with:

Customer Number: **31366**

for so long as they remain with such law offices or company with full power of substitution and revocation, to prosecute this application and to transact in connection therewith all business in the Patent and Trademark Office and before competent International Authorities; said appointment to be to the exclusion of myself and my other attorney(s).

Correspondence should be addressed to the address associated with:

Customer Number: **31366**

.....

Inventor's signature: Li Junwen Date: 23/02/2009

Full name of sole or first inventor: Junwen LIU

Citizenship: CN

Residence Address: Blk 429 Woodlands St 41, #12-250, Singapore
730429

P.O. Address: Chartered Semiconductor Manufacturing, Ltd.
60 Woodlands Industrial Park D Street 2
Singapore 738406

Docket No.: CSM P 2009 NAT 03 US 0

Inventor's signature:  Date: 23/02/09

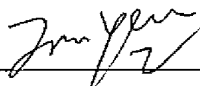
Full name of second inventor: Purakh Raj VERMA

Citizenship: IN

Residence Address: 29 Rosewood Drive, #04-23, Casablanca,
Singapore 737291

P.O. Address: Chartered Semiconductor Manufacturing, Ltd.
60 Woodlands Industrial Park D Street 2
Singapore 738406

Docket No.: CSM P 2009 NAT 03 US 0

Inventor's signature:  Date: 23/02/2009
Full name of second inventor: Yan JIN
Citizenship: CN
Residence Address: Blk 712 Woodlands Dr. 70, #07-101, Singapore
730712
P.O. Address: Chartered Semiconductor Manufacturing, Ltd.
60 Woodlands Industrial Park D Street 2
Singapore 738406

Docket No.: CSM P 2009 NAT 03 US 0

Inventor's signature: Zhu Baofu Date: 23/02/2009

Full name of second inventor: Baofu ZHU

Citizenship: CN

Residence Address: Blk 690E Woodlands Drive 75, #15-200,
Singapore 735690

P.O. Address: Chartered Semiconductor Manufacturing, Ltd.
60 Woodlands Industrial Park D Street 2
Singapore 738406

DECLARATION AND POWER OF ATTORNEY

As a below named inventor, I hereby declare that:

My residence, post office address, and citizenship are as stated below next to my name,

I believe I am the original, first and sole inventor (if only one name is listed below) or an original, first and joint inventor (if plural names are listed below) of the subject matter claimed and for which a patent is sought on the invention entitled:

HIGH VOLTAGE DEVICE

the specification of which

☒ is attached hereto

☐ was filed on ___ as Application Serial No. ___ and was amended on ___ (if applicable).

I hereby state that I have reviewed and understand the contents of the above-identified specification, including the claims, as amended by any amendment referred to above.

I do not know and do not believe the same was ever known or used in the United States of America before this invention thereof or more than one year prior to this application, and that the same was not in public use or on sale in the United States of America more than one year prior to this application. I acknowledge the duty to disclose information which is known to me to be material to patentability in accordance with title 37, Code of Federal Regulations, section 1.56.

I hereby claim foreign priority benefits under Title 35, United States Code, Section 119(a)-(d) or Section 365(b) of any foreign application(s) for patent or inventor's certificate, or Section 365(a) of any PCT international application which designated at least one country other than the United States, listed below and have also identified below any foreign application for patent or inventor's certificate having a filing date before that of the application on which priority is claimed:

Prior Foreign Application(s):

<u>Number</u>	<u>Country</u>	<u>Day/Month/Year filed</u>	<u>Priority Claimed</u>	
			<u>Yes</u>	<u>No</u>
				☒

I hereby claim the benefit under 35 USC 119(e) of any United States provisional application(s) listed below:

Prior Provisional Application(s):

<u>Application Number</u>	<u>Filing Date</u>

I hereby claim the benefit under Title 35, United States Code, Section 120 of any United States application(s), or Section 365(c) of any PCT international application designating the United States, listed below and, insofar as the subject matter of each of the claims of this application is not disclosed in the prior United States or PCT international application in the manner provided by the first paragraph of Title 35, United States Code, Section 112, I acknowledge the duty to disclose material information as defined in Title 37, Code of Federal

Docket No.: CSM P 2009 NAT 03 US 0

Regulations, Section 1.56 which occurred between the filing date of the prior application and the national or PCT international filing date of this application:

Prior U.S. Application(s):

<u>Serial No.</u>	<u>Filing Date</u>	<u>Status: Patented, Pending, Abandoned</u>
-------------------	--------------------	---

I hereby declare that all statements made herein of my own knowledge are true and that all statements made on information and belief are believed to be true; and further that these statements were made with the knowledge that willful false statements and the like so made are punishable by fine or imprisonment, or both, under Section 1001 of Title 18 of the United States Code and that such willful false statements may jeopardize the validity of the application or any patent issued thereon.

I hereby revoke any previous Powers of Attorney and appoint the following attorney(s) and/or agent(s), each said individual being a member or associate of **Horizon IP Pte. Ltd.** and associated with:

Customer Number: **31366**

for so long as they remain with such law offices or company with full power of substitution and revocation, to prosecute this application and to transact in connection therewith all business in the Patent and Trademark Office and before competent International Authorities; said appointment to be to the exclusion of myself and my other attorney(s).

Correspondence should be addressed to the address associated with:

Customer Number: **31366**

.....

Inventor's signature:	<u>Li Junwen</u>	Date:	<u>23/02/2009</u>
Full name of sole or first inventor:	Junwen <u>LIU</u>		
Citizenship:	CN		
Residence Address:	Blk 429 Woodlands St 41, #12-250, Singapore 730429		
P.O. Address:	<u>Chartered Semiconductor Manufacturing, Ltd.</u> <u>60 Woodlands Industrial Park D Street 2</u> <u>Singapore 738406</u>		

Docket No.: CSM P 2009 NAT 03 US 0

Inventor's signature: _____



Date: 23/02/09

Full name of second inventor: .

Purakh Raj **VERMA**

Citizenship:

IN

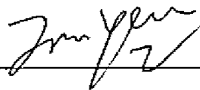
Residence Address:

29 Rosewood Drive, #04-23, Casablanca,
Singapore 737291

P.O. Address:

Chartered Semiconductor Manufacturing, Ltd.
60 Woodlands Industrial Park D Street 2
Singapore 738406

Docket No.: CSM P 2009 NAT 03 US 0

Inventor's signature:  Date: 23/02/2009
Full name of second inventor: Yan JIN
Citizenship: CN
Residence Address: Blk 712 Woodlands Dr. 70, #07-101, Singapore
730712
P.O. Address: Chartered Semiconductor Manufacturing, Ltd.
60 Woodlands Industrial Park D Street 2
Singapore 738406

Docket No.: CSM P 2009 NAT 03 US 0

Inventor's signature: Zhu Baofu Date: 23/02/2009

Full name of second inventor: Baofu ZHU

Citizenship: CN

Residence Address: Blk 690E Woodlands Drive 75, #15-200,
Singapore 735690

P.O. Address: Chartered Semiconductor Manufacturing, Ltd.
60 Woodlands Industrial Park D Street 2
Singapore 738406

Filing Date: 02/23/09

Approved for use through 7/31/2006. OMB 0651-0032

U.S. Patent and Trademark Office; U.S. DEPARTMENT OF COMMERCE

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PATENT APPLICATION FEE DETERMINATION RECORD

Substitute for Form PTO-875

Application or Docket Number

12/390,509**APPLICATION AS FILED – PART I**

(Column 1) (Column 2)

FOR	NUMBER FILED	NUMBER EXTRA
BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A
SEARCH FEE (37 CFR 1.16(k), (l), or (m))	N/A	N/A
EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A
TOTAL CLAIMS (37 CFR 1.16(i))	21 minus 20 =	1
INDEPENDENT CLAIMS (37 CFR 1.16(h))	3 minus 3 =	*
APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$260 (\$130 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR	
MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))		

SMALL ENTITY

OR

OTHER THAN SMALL ENTITY

RATE (\$)	FEE (\$)
N/A	
N/A	
N/A	
x\$26	
x\$110	
195	
TOTAL	

OR

RATE (\$)	FEE (\$)
N/A	330
N/A	540
N/A	220
x\$52	52
x\$220	
390	
TOTAL	1142

* If the difference in column 1 is less than zero, enter "0" in column 2.

APPLICATION AS AMENDED – PART II

(Column 1) (Column 2) (Column 3)

AMENDMENT A		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA
Total (37 CFR 1.16(i))	*	Minus	**	=	
Independent (37 CFR 1.16(h))	*	Minus	***	=	
Application Size Fee (37 CFR 1.16(s))					
FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))					

SMALL ENTITY

OR

OTHER THAN SMALL ENTITY

RATE (\$)	ADDITIONAL FEE (\$)
X =	
X =	
N/A	
TOTAL ADD'T FEE	

OR

RATE (\$)	ADDITIONAL FEE (\$)
X =	
X =	
N/A	
TOTAL ADD'T FEE	

(Column 1) (Column 2) (Column 3)

AMENDMENT B		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA
Total (37 CFR 1.16(i))	*	Minus	**	=	
Independent (37 CFR 1.16(h))	*	Minus	***	=	
Application Size Fee (37 CFR 1.16(s))					
FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))					

RATE (\$)	ADDITIONAL FEE (\$)
X =	
X =	
N/A	
TOTAL ADD'T FEE	

OR

RATE (\$)	ADDITIONAL FEE (\$)
X =	
X =	
N/A	
TOTAL ADD'T FEE	

* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.

** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".

*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".

The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.

This collection of information is required by 37 CFR 1.16. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
12/390,509	02/23/2009	2812	1142	CSM P 2009 NAT 03 US 0	21	3

CONFIRMATION NO. 5260

31366
HORIZON IP PTE LTD
7500A Beach Road
#04-306/308 The Plaza
SINGAPORE 199591,
SINGAPORE

FILING RECEIPT



Date Mailed: 03/12/2009

Receipt is acknowledged of this non-provisional patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF APPLICANT, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection. Please verify the accuracy of the data presented on this receipt. **If an error is noted on this Filing Receipt, please submit a written request for a Filing Receipt Correction. Please provide a copy of this Filing Receipt with the changes noted thereon. If you received a "Notice to File Missing Parts" for this application, please submit any corrections to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections**

Applicant(s)

Junwen LIU, Singapore, SINGAPORE;
Purakh Raj VERMA, Singapore, SINGAPORE;
Yan JIN, Singapore, SINGAPORE;
Baofu ZHU, Singapore, SINGAPORE;

Assignment For Published Patent Application

CHARTERED SEMICONDUCTOR MANUFACTURING, LTD., Singapore, SINGAPORE

Power of Attorney: The patent practitioners associated with Customer Number 31366

Domestic Priority data as claimed by applicant

Foreign Applications

If Required, Foreign Filing License Granted: 03/04/2009

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 12/390,509**

Projected Publication Date: 08/26/2010

Non-Publication Request: No

Early Publication Request: No

Title

HIGH VOLTAGE DEVICE

Preliminary Class

438

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

Almost every country has its own patent law, and a person desiring a patent in a particular country must make an application for patent in that country in accordance with its particular laws. Since the laws of many countries differ in various respects from the patent law of the United States, applicants are advised to seek guidance from specific foreign countries to ensure that patent rights are not lost prematurely.

Applicants also are advised that in the case of inventions made in the United States, the Director of the USPTO must issue a license before applicants can apply for a patent in a foreign country. The filing of a U.S. patent application serves as a request for a foreign filing license. The application's filing receipt contains further information and guidance as to the status of applicant's license for foreign filing.

Applicants may wish to consult the USPTO booklet, "General Information Concerning Patents" (specifically, the section entitled "Treaties and Foreign Patents") for more information on timeframes and deadlines for filing foreign patent applications. The guide is available either by contacting the USPTO Contact Center at 800-786-9199, or it can be viewed on the USPTO website at <http://www.uspto.gov/web/offices/pac/doc/general/index.html>.

For information on preventing theft of your intellectual property (patents, trademarks and copyrights), you may wish to consult the U.S. Government website, <http://www.stopfakes.gov>. Part of a Department of Commerce initiative, this website includes self-help "toolkits" giving innovators guidance on how to protect intellectual property in specific countries such as China, Korea and Mexico. For questions regarding patent enforcement issues, applicants may call the U.S. Government hotline at 1-866-999-HALT (1-866-999-4158).

LICENSE FOR FOREIGN FILING UNDER**Title 35, United States Code, Section 184****Title 37, Code of Federal Regulations, 5.11 & 5.15****GRANTED**

The applicant has been granted a license under 35 U.S.C. 184, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" followed by a date appears on this form. Such licenses are issued in all applications where the conditions for issuance of a license have been met, regardless of whether or not a license may be required as

set forth in 37 CFR 5.15. The scope and limitations of this license are set forth in 37 CFR 5.15(a) unless an earlier license has been issued under 37 CFR 5.15(b). The license is subject to revocation upon written notification. The date indicated is the effective date of the license, unless an earlier license of similar scope has been granted under 37 CFR 5.13 or 5.14.

This license is to be retained by the licensee and may be used at any time on or after the effective date thereof unless it is revoked. This license is automatically transferred to any related applications(s) filed under 37 CFR 1.53(d). This license is not retroactive.

The grant of a license does not in any way lessen the responsibility of a licensee for the security of the subject matter as imposed by any Government contract or the provisions of existing laws relating to espionage and the national security or the export of technical data. Licensees should apprise themselves of current regulations especially with respect to certain countries, of other agencies, particularly the Office of Defense Trade Controls, Department of State (with respect to Arms, Munitions and Implements of War (22 CFR 121-128)); the Bureau of Industry and Security, Department of Commerce (15 CFR parts 730-774); the Office of Foreign Assets Control, Department of Treasury (31 CFR Parts 500+) and the Department of Energy.

NOT GRANTED

No license under 35 U.S.C. 184 has been granted at this time, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" DOES NOT appear on this form. Applicant may still petition for a license under 37 CFR 5.12, if a license is desired before the expiration of 6 months from the filing date of the application. If 6 months has lapsed from the filing date of this application and the licensee has not received any indication of a secrecy order under 35 U.S.C. 181, the licensee may foreign file the application pursuant to 37 CFR 5.15(b).



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
12/390,509	02/23/2009	Junwen LIU	CSM P 2009 NAT 03 US 0

CONFIRMATION NO. 5260

PUBLICATION NOTICE



OC000000043223794

31366
HORIZON IP PTE LTD
7500A Beach Road
#04-306/308 The Plaza
SINGAPORE 199591,
SINGAPORE

Title:HIGH VOLTAGE DEVICE

Publication No.US-2010-0213544-A1

Publication Date:08/26/2010

NOTICE OF PUBLICATION OF APPLICATION

The above-identified application will be electronically published as a patent application publication pursuant to 37 CFR 1.211, et seq. The patent application publication number and publication date are set forth above.

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In addition, information on the status of the application, including the mailing date of Office actions and the dates of receipt of correspondence filed in the Office, may also be accessed via the Internet through the Patent Electronic Business Center at www.uspto.gov using the public side of the Patent Application Information and Retrieval (PAIR) system. The direct link to access this status information is currently <http://pair.uspto.gov/>. Prior to publication, such status information is confidential and may only be obtained by applicant using the private side of PAIR.

Further assistance in electronically accessing the publication, or about PAIR, is available by calling the Patent Electronic Business Center at 1-866-217-9197.

Office of Data Management, Application Assistance Unit (571) 272-4000, or (571) 272-4200, or 1-888-786-0101



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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
12/390,509	02/23/2009	Junwen LIU	CSM P 2009 NAT 03 US 0	5260
31366	7590	12/13/2010	EXAMINER	
HORIZON IP PTE LTD 7500A Beach Road #04-306/308 The Plaza SINGAPORE 199591, SINGAPORE			DANG, PHUC T	
			ART UNIT	PAPER NUMBER
			2892	
			NOTIFICATION DATE	DELIVERY MODE
			12/13/2010	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

dexter.chin@horizonip.com.sg
nannan.chen@horizonip.com.sg
oa.notification@horizonip.com.sg

Office Action Summary	Application No. 12/390,509	Applicant(s) LIU ET AL.	
	Examiner Phuc T. Dang	Art Unit 2892	

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 1 MONTH(S) OR THIRTY (30) DAYS, WHICHEVER IS LONGER, FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

- 1) ☒ Responsive to communication(s) filed on 23 February 2009.
- 2a) ☐ This action is **FINAL**. 2b) ☒ This action is non-final.
- 3) ☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims

- 4) ☒ Claim(s) 1-21 is/are pending in the application.
- 4a) Of the above claim(s) _____ is/are withdrawn from consideration.
- 5) ☐ Claim(s) _____ is/are allowed.
- 6) ☐ Claim(s) _____ is/are rejected.
- 7) ☐ Claim(s) _____ is/are objected to.
- 8) ☒ Claim(s) 1-21 are subject to restriction and/or election requirement.

Application Papers

- 9) ☐ The specification is objected to by the Examiner.
- 10) ☐ The drawing(s) filed on _____ is/are: a) ☐ accepted or b) ☐ objected to by the Examiner.
Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).
Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).
- 11) ☐ The oath or declaration is objected to by the Examiner. Note the attached Office Action or form PTO-152.

Priority under 35 U.S.C. § 119

- 12) ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).
- a) ☐ All b) ☐ Some * c) ☐ None of:
1. ☐ Certified copies of the priority documents have been received.
 2. ☐ Certified copies of the priority documents have been received in Application No. _____.
 3. ☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

* See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

- | | |
|---|---|
| 1) ☐ Notice of References Cited (PTO-892) | 4) ☐ Interview Summary (PTO-413) |
| 2) ☐ Notice of Draftperson's Patent Drawing Review (PTO-948) | Paper No(s)/Mail Date. _____ |
| 3) ☐ Information Disclosure Statement(s) (PTO/SB/08) | 5) ☐ Notice of Informal Patent Application |
| Paper No(s)/Mail Date _____ | 6) ☐ Other: _____ |

DETAILED ACTION

Election/Restriction

Restriction to one of the following inventions is required under 35 U.S.C. 121:

- I. Group I, Claims 1-17 and 20-21, drawn to a method of forming a device, classified in class 438, subclass 286.
- II. Group II, Claims 18-19, drawn to a device, classified in class 257, subclass 339.

The inventions are distinct, each from the other because of the following reasons:

1. Inventions I and II are related as method of making and product made.

The inventions are distinct if either or both of the following can be shown: (1) that the process as claimed can be used to make other and materially different product or (2) that the product as claimed can be made by another and materially different process (MPEP § 806.05(f)). In the instant case, a device can be fabricated with different method other than the method steps recited in Group I or vice-versa such as a step of diffusion or doping through the opening and gate stack layers can be used into the substrate to form a channel well instead implanting as disclosed in the method of Group I.

However, the issues of method and product claims are divergent.

Furthermore, there may be some overlap in the searches of the two groups, but there is no reason to believe that the searches would be identical. Therefore, based on the additional work involved in searching and examination of the two inventions together, restriction of distinct inventions is clearly proper.

2. Applicant is advised that the response to this requirement to be complete must include an election of the invention to be examined even though the requirement be traverse (37 CFR 1.143).
3. Applicant is reminded that upon the cancellation of claims to a non-elected invention, the inventorship must be amended in compliance with 37 CFR 1.48(b) if one or more of the currently named inventors is no longer an inventor of at least one claim remaining in the application. Any amendment of inventorship must be accompanied by a diligently-filed petition under 37 CFR 1.48(b) and by the fee required under 37 CFR 1.17(h).
4. Any inquiry concerning this communication or earlier communication from the examiner should be direct to Phuc T. Dang whose telephone number (571) 272-1776. The examiner can normally be reached on Monday through Friday from 8:00am to 5:00pm.

Application/Control Number: 12/390,509
Art Unit: 2892

Page 3

/Phuc T Dang/
Primary Examiner, Art Unit 2892

<i>Index of Claims</i> 	Application/Control No. 12390509	Applicant(s)/Patent Under Reexamination LIU ET AL.
	Examiner Phuc T Dang	Art Unit 2892

✓	Rejected	-	Cancelled	N	Non-Elected	A	Appeal
=	Allowed	÷	Restricted	I	Interference	O	Objected

☐ Claims renumbered in the same order as presented by applicant ☐ CPA ☐ T.D. ☐ R.1.47										
CLAIM		DATE								
Final	Original	12/07/2010								
	1	÷								
	2	÷								
	3	÷								
	4	÷								
	5	÷								
	6	÷								
	7	÷								
	8	÷								
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	19	÷								
	20	÷								
	21	÷								

Appl. No. 12/390,509
Amdt dated January 07, 2011
Reply to Office Action dated December 13, 2010

PATENT

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Appl. No. : 12/390,509 Confirmation No. : 5260
Applicant : Junwen LIU et al.
Filed : 02/23/2009
TC / A.U. : 2892
Examiner : DANG, Phuc T
Docket No. : CSM P 2009 NAT 03 US 0
Customer No. : 031366
Title : HIGH VOLTAGE DEVICE

CERTIFICATE OF TRANSMISSION

I hereby certify that this correspondence is being electronically transmitted to the United States Patent and Trademark Office, Alexandria, VA 22313-1450 on January 07, 2011.

Signed: /nannan chen/

Nannan CHEN

Mail Stop AMENDMENT
Commissioner For Patents
P.O. Box 1450
Alexandria, VA 22313-1450

AMENDMENT

Sir:

In response to the Office Action mailed on December 13, 2010, please amend the above-identified application as follows:

- a) **Amendments to the Claims** are reflected in the listing of claims beginning on page 2 of this paper; and
- b) **Remarks/Arguments** begin on page 7 of this paper.

Amendments to Claims

This listing of claims will replace all prior versions, and listings, of claims in the application:

Listing of Claims:

1. (currently amended) A method of forming a device comprising:

providing a substrate prepared with an active device region, wherein the active device region includes gate stack layers of a gate stack comprising at least a gate electrode layer over a gate dielectric layer;

forming an implant mask on the substrate with an opening exposing a portion of a top layer of the gate stack layers;

implanting ions through the opening and gate stack layers into the substrate to form a channel well; and

patterning the substrate to at least remove portion of ~~[[a]]~~ the top gate stack layer unprotected by the implant mask.
2. (original) The method of claim 1 wherein the top gate stack layer comprises the gate electrode layer of the gate stack.
3. (original) The method of claim 2 wherein the gate electrode layer comprises silicon, including amorphous or polysilicon.
4. (currently amended) The method of claim 2 wherein:

patterning the substrate forms a first edge of the gate; and

further comprises ~~pattern~~ patterning the substrate to form a second edge of the gate.

5. (original) The method of claim 4 wherein the first edge corresponds to a channel edge of the gate of the device and the second edge corresponds to the drain edge of the gate of the device.
6. (currently amended) The method of claim 1 wherein:
patterning the substrate forms a first edge of the gate; and
further comprises ~~pattern~~ patterning the substrate to form a second edge of the gate.
7. (original) The method of claim 6 wherein the first edge corresponds to a channel edge of the gate of the device and the second edge corresponds to the drain edge of the gate of the device.
8. (original) The method of claim 1 wherein patterning the substrate leaves the gate dielectric layer remaining on the substrate.
9. (currently amended) The method of claim 1 wherein the gate stack layers comprise[[s]] a hard mask layer over the gate electrode layer and patterning the ~~gate stack~~ hard mask removes a portion of the hard mask layer to expose the gate electrode layer.
10. (currently amended) The method of claim 9 wherein patterning the hard mask forms a first edge of the hard mask corresponding to a first edge of the gate stack and further comprises patterning the hard mask to form a second edge of the hard mask corresponding to a second edge of the gate stack.

11. (original) The method of claim 10 comprises patterning the gate stack layers with the patterned hard mask to form a gate with first and second edges.

12. (original) The method of claim 11 wherein patterning the gate stack layers leaves the gate dielectric layer remaining on the substrate.

13. (original) The method of claim 1 wherein implanting ions comprises a tilt angle implant.

14. (original) The method of claim 1 wherein a tilt angle of the implant is about 1 – 45°.

15. (original) The method of claim 1 wherein implanting ions comprises multiple tilt angle implants which are rotated about a plane of the substrate.

16. (original) The method of claim 15 wherein a tilt angle of the implant is about 1 – 45°.

17. (original) The method of claim 1 wherein implanting ions comprises a quad tilt angle implant which is rotated about a plane of the substrate.

18-19. (canceled)

20. (currently amended) A method of forming a device comprising:

providing a substrate prepared with an active device region which includes a doped drift well in a first portion of the active device region and gate stack layers of a gate stack comprising at least a gate electrode layer over a gate dielectric layer on the surface of the substrate;

forming an implant mask on the substrate with an opening exposing a portion of a top layer of the gate stack layers;

implanting ions through the opening and gate stack layers into the substrate to form a channel well; and

patterning the substrate to at least remove portion of ~~[[a]]~~ the top gate stack layer unprotected by the implant mask.

21. (original) The method of claim 20 wherein the substrate comprises:

a drift isolation region in the doped drift well; and

a doped deep well in the substrate of the active device region encompassing the doped drift well.

22. (new) A method of forming a device comprising:

providing a substrate prepared with an active device region which includes a doped drift well in a first portion of the active device region, a doped channel well in a second portion of the active region and gate stack layers of a gate stack comprising at least a gate electrode layer over a gate dielectric layer on the surface of the substrate;

forming an implant mask on the substrate with an opening exposing a portion of a top layer of the gate stack layers;

implanting ions through the opening and gate stack layers into the substrate to form a channel well; and

patterning the substrate to at least remove portion of the top gate stack layer unprotected by the implant mask.

23. (new) The method of claim 22 wherein the doped drift well in the first portion of the active region is contiguous with the doped channel well in the second portion of the active region.

24. (new) The method of claim 22 wherein the doped drift well in the first portion of the active region is separated from the doped channel well in the second portion of the active region by a third portion.

REMARKS AND ARGUMENTS

Claim Amendments and Newly Added Claims

Without conceding the propriety of the rejections herein and in the interest of expediting prosecution, Applicants amend some claims herein. Applicants amend claims to clarify claimed features. Such amendments are made to expedite prosecution and to more quickly identify allowable subject matter. Such amendments are merely intended to clarify the claimed features, and should not be construed as further limiting the claimed invention in response to the cited references. Applicants also add new claims 22-24. Applicants submit that all amendments and newly added claims are fully supported by the Application and therefore do not constitute new matter.

Election/Restriction

The Examiner has indicated that the claims of the present invention are drawn to two distinct inventions and required election of one of the groups pursuant to 35 U.S.C. § 121. In particular, group I claims (claims 1-17 and 20-21) are drawn to a method of forming a device while group II claims (claims 18 and 19) are drawn to a device. In response, Applicants elect group I invention for prosecution on the merits and cancel group II invention from consideration without prejudice.

In addition, Applicants also add new claims (claims 22-24). Applicants believe that the new claims are directed to group I invention. All features of the newly added claims are fully supported by the Application and therefore do not constitute new matter.

Appl. No. 12/390,509
Amdt dated January 07, 2011
Reply to Office Action dated December 13, 2010

Conclusion

It is respectfully submitted that this application is in condition for allowance and the issuance of a formal Notice of Allowance at an early date is respectfully requested. Should the Examiner believe that a telephone conference would expedite prosecution of this application, please telephone the undersigned attorney at his number set out below.

Date: January 07, 2011

Respectfully submitted,

/dexter chin/

Dexter CHIN
Attorney for Applicants
Reg. No.: 38,842

Horizon IP Pte Ltd
7500A Beach Road
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Tel.: (65) 9836 9908
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Email: dexter.chin@horizonip.com.sg

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

TRANSMITTAL FORM <i>(to be used for all correspondence after initial filing)</i>	Application Number	12/390,509	
	Filing Date	02/23/2009	
	First Named Inventor	Junwen LIU	
	Art Unit	2892	
	Examiner Name	DANG, PHUC T	
Total Number of Pages in This Submission	12	Attorney Docket Number	CSM P 2009 NAT 03 US 0

ENCLOSURES (Check all that apply)		
☒ Fee Transmittal Form ☒ Fee Attached ☒ Amendment/Reply ☐ After Final ☐ Affidavits/declaration(s) ☐ Extension of Time Request ☐ Express Abandonment Request ☐ Information Disclosure Statement ☐ Certified Copy of Priority Document(s) ☐ Reply to Missing Parts/ Incomplete Application ☐ Reply to Missing Parts under 37 CFR 1.52 or 1.53	☐ Drawing(s) ☐ Licensing-related Papers ☐ Petition ☐ Petition to Convert to a Provisional Application ☐ Power of Attorney, Revocation Change of Correspondence Address ☐ Terminal Disclaimer ☐ Request for Refund ☐ CD, Number of CD(s) _____ ☐ Landscape Table on CD	☐ After Allowance Communication to TC ☐ Appeal Communication to Board of Appeals and Interferences ☐ Appeal Communication to TC (Appeal Notice, Brief, Reply Brief) ☐ Proprietary Information ☐ Status Letter ☐ Other Enclosure(s) (please identify below):
Remarks		

SIGNATURE OF APPLICANT, ATTORNEY, OR AGENT			
Firm Name	Horizon IP Pte. Ltd.		
Signature	/dexter chin/		
Printed name	Dexter CHIN		
Date	01/07/2011	Reg. No.	38,842

CERTIFICATE OF TRANSMISSION/MAILING			
I hereby certify that this correspondence is being facsimile transmitted to the USPTO or deposited with the United States Postal Service with sufficient postage as first class mail in an envelope addressed to: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450 on the date shown below:			
Signature	/nannan chen/		
Typed or printed name	Nannan CHEN	Date	01/07/2011

This collection of information is required by 37 CFR 1.5. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to 2 hours to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

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The **Privacy Act of 1974 (P.L. 93-579)** requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (*i.e.*, GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Electronic Patent Application Fee Transmittal

Application Number:	12390509			
Filing Date:	23-Feb-2009			
Title of Invention:	HIGH VOLTAGE DEVICE			
First Named Inventor/Applicant Name:	Junwen LIU			
Filer:	Dexter Kam Chin			
Attorney Docket Number:	CSM P 2009 NAT 03 US 0			
Filed as Large Entity				
Utility under 35 USC 111(a) Filing Fees				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Claims in excess of 20	1202	1	52	52
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Miscellaneous:				
Total in USD (\$)				52

Electronic Acknowledgement Receipt

EFS ID:	9183911
Application Number:	12390509
International Application Number:	
Confirmation Number:	5260
Title of Invention:	HIGH VOLTAGE DEVICE
First Named Inventor/Applicant Name:	Junwen LIU
Customer Number:	31366
Filer:	Dexter Kam Chin
Filer Authorized By:	
Attorney Docket Number:	CSM P 2009 NAT 03 US 0
Receipt Date:	07-JAN-2011
Filing Date:	23-FEB-2009
Time Stamp:	01:58:35
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	Deposit Account
Payment was successfully received in RAM	\$ 52
RAM confirmation Number	6287
Deposit Account	502388
Authorized User	

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

Charge any Additional Fees required under 37 C.F.R. Section 1.21 (Miscellaneous fees and charges)

File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Response to Election / Restriction Filed	CSMP2009NAT03US0- ResponsetoER.pdf	105692	no	8
			1e86df0a0b678e48f54cd73996c3e509c04156df		

Warnings:**Information:**

2	Miscellaneous Incoming Letter	CSMP2009NAT03US0-TF.pdf	277623	no	2
			7c848eae0cc1a3b1530e02265f58b2ab7b14b284		

Warnings:**Information:**

3	Fee Worksheet (PTO-875)	fee-info.pdf	29908	no	2
			7a5f7f15e171d906e0989bdd76bc0d1f70fa8444		

Warnings:**Information:**

Total Files Size (in bytes):			413223
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This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

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PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875					Application or Docket Number 12/390,509		Filing Date 02/23/2009		☐ To be Mailed	
APPLICATION AS FILED – PART I										
(Column 1)			(Column 2)		SMALL ENTITY ☐ OR			OTHER THAN SMALL ENTITY		
FOR	NUMBER FILED	NUMBER EXTRA	RATE (\$)	FEE (\$)	OR	RATE (\$)	FEE (\$)			
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A			N/A				
☐ SEARCH FEE (37 CFR 1.16(k), (l), or (m))	N/A	N/A	N/A			N/A				
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A			N/A				
TOTAL CLAIMS (37 CFR 1.16(i))	minus 20 =	*	X \$	=		X \$	=			
INDEPENDENT CLAIMS (37 CFR 1.16(h))	minus 3 =	*	X \$	=		X \$	=			
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$250 (\$125 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).									
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))										
* If the difference in column 1 is less than zero, enter "0" in column 2.			TOTAL			TOTAL				
APPLICATION AS AMENDED – PART II										
(Column 1)			(Column 2)		SMALL ENTITY OR			OTHER THAN SMALL ENTITY		
AMENDMENT	01/07/2011	CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)	OR	RATE (\$)	ADDITIONAL FEE (\$)	
	Total (37 CFR 1.16(i))	* 22	Minus	** 21	=	1		X \$52=	52	
	Independent (37 CFR 1.16(h))	* 3	Minus	*** 3	=	0		X \$220=	0	
☐ Application Size Fee (37 CFR 1.16(s))										
☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))										
					TOTAL ADD'L FEE			TOTAL ADD'L FEE	52	
(Column 1)			(Column 2)		SMALL ENTITY OR			OTHER THAN SMALL ENTITY		
AMENDMENT		CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)	OR	RATE (\$)	ADDITIONAL FEE (\$)	
	Total (37 CFR 1.16(i))	*	Minus	**	=			X \$	=	
	Independent (37 CFR 1.16(h))	*	Minus	***	=			X \$	=	
☐ Application Size Fee (37 CFR 1.16(s))										
☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))										
					TOTAL ADD'L FEE			TOTAL ADD'L FEE		
* If the entry in column 1 is less than the entry in column 2, write "0" in column 3. ** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20". *** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3". The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.										

Legal Instrument Examiner:
/NICOLE C. LAWRENCE/

This collection of information is required by 37 CFR 1.16. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.



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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
12/390,509	02/23/2009	Junwen LIU	CSM P 2009 NAT 03 US 0	5260
31366	7590	02/08/2011	EXAMINER	
HORIZON IP PTE LTD 7500A Beach Road #04-306/308 The Plaza SINGAPORE 199591, SINGAPORE			DANG, PHUC T	
			ART UNIT	PAPER NUMBER
			2892	
			NOTIFICATION DATE	DELIVERY MODE
			02/08/2011	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

dexter.chin@horizonip.com.sg
nannan.chen@horizonip.com.sg
oa.notification@horizonip.com.sg

Office Action Summary	Application No.	Applicant(s)	
	12/390,509	LIU ET AL.	
	Examiner	Art Unit	
	Phuc T. Dang	2892	

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTH(S) OR THIRTY (30) DAYS, WHICHEVER IS LONGER, FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

- 1) ☒ Responsive to communication(s) filed on election filed on 01/07/2011.
- 2a) ☐ This action is **FINAL**. 2b) ☒ This action is non-final.
- 3) ☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims

- 4) ☒ Claim(s) 1-17 and 20-24 is/are pending in the application.
- 4a) Of the above claim(s) _____ is/are withdrawn from consideration.
- 5) ☐ Claim(s) _____ is/are allowed.
- 6) ☒ Claim(s) 1-17 and 20-24 is/are rejected.
- 7) ☒ Claim(s) 1 is/are objected to.
- 8) ☐ Claim(s) _____ are subject to restriction and/or election requirement.

Application Papers

- 9) ☒ The specification is objected to by the Examiner.
- 10) ☒ The drawing(s) filed on 23 February 2009 is/are: a) ☒ accepted or b) ☐ objected to by the Examiner.
Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).
Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).
- 11) ☐ The oath or declaration is objected to by the Examiner. Note the attached Office Action or form PTO-152.

Priority under 35 U.S.C. § 119

- 12) ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).
- a) ☐ All b) ☐ Some * c) ☐ None of:
1. ☐ Certified copies of the priority documents have been received.
2. ☐ Certified copies of the priority documents have been received in Application No. _____.
3. ☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

* See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

- | | |
|--|---|
| 1) ☒ Notice of References Cited (PTO-892) | 4) ☐ Interview Summary (PTO-413) |
| 2) ☐ Notice of Draftsperson's Patent Drawing Review (PTO-948) | Paper No(s)/Mail Date. _____ |
| 3) ☒ Information Disclosure Statement(s) (PTO/SB/08) | 5) ☐ Notice of Informal Patent Application |
| Paper No(s)/Mail Date <u>02/23/2009</u> | 6) ☐ Other: _____ |

DETAILED ACTION

Election/Restrictions

1. Applicant's election without traverse of Group I (claims 1-17, 20-21 and new claims 21-24) filed on 01/07/2011 and canceled Group II (claims 18-19) has been acknowledged and considered. Applicants have the right to file a divisional application covering the subject matter of the non-elected claims (claims 18-19).

Claims 1-17 and 20-24 are currently pending in the application.

Oath/Declaration

2. The oath/declaration filed on 02/23/2009 is acceptable.

Information Disclosure Statement

3. The office acknowledges receipt of the following items from the applicant:

Information Disclosure Statement (IDS) filed on 02/23/2009.

Specification

4. The specification is objected to for the following reason: The title of the invention is not descriptive. A new title is required that is clearly indicative of the invention to which the claims are directed (see MPEP 606.01).

A title such as -- METHOD OF FORMING A DEVICE -- or is suggested by the applicant.

Note that the claims are directed to a method of making a semiconductor device and the title of the application should match the title in the claims.

The specification has been updated.

Claim Objections

5. Claim 1 is objected to because of the following reason:

In claim 1, line 9, a term of "... the top gate stack layer" should amend to -- the top gate stack layers --.

Claim Rejections-35 USC § 112

The following is a quotation of the second paragraph of 35 U.S.C. 112:

The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the application regards as his invention.

6. Claims 1-17 and 20-24 are rejected under 35 U.S.C. 112, second paragraph, as being indefinite for failing to particularly point out and distinctly claim the subject matter which the applicant regards as the invention.

In claims 1-17 and 20-24, step of "patterning the substrate..." is confusing and indefinite because the gate stack is patterned by the implant mask, not the substrate? Applicant should clarify this limitation in the claims. For a purpose of examination, the examiner supposes that patterning the gate stack layers in the claims.

Claim Rejections - 35 USC § 102

(b) the invention was patented or described in a printed publication in this or a foreign country or in public use or on sale in this country, more than one year prior to the date of application for patent in the United States.

This application currently names joint inventors. In considering patentability of the claims under 35 U.S.C. 103(a), the examiner presumes that the subject matter of the various claims was commonly owned at the time any inventions covered therein were made absent any evidence to the contrary. Applicant is advised of the obligation under 37 CFR 1.56 to point out the inventor and invention dates of each claim that was not commonly owned at the time a later invention was

Art Unit: 2892

made in order for the examiner to consider the applicability of 35 U.S.C. 103(c) and potential 35 U.S.C. 102(f) or (g) prior art under 35 U.S.C. 103(a).

7. Claims 1-3 and 9-12 are rejected under 35 U.S.C. 102 (b) as being anticipated by Wann et al., hereafter “Wann” (U.S. Patent No. 5,589,414).

Regarding claims 1-3 and 9-12, Wann discloses a method of forming a device comprising:

providing a substrate 32 prepared with an active device region, wherein the active device region includes gate stack layers of a gate stack comprising at least a gate electrode layer 38 over a gate dielectric layer 36;

forming an implant mask 47 on the substrate 32 with an opening 47' exposing a portion of a top layer of the gate stack layers;

implanting ions 48 through the opening 47' and gate stack layers into the substrate 32 to form a channel well 48'; and

patterning the gate stack layers to at least remove portion of the top gate stack layers unprotected by the implant mask 48 (Figs. 2A-2F).

Claim Rejections - 35 USC § 103

The following is a quotation of 35 U.S.C. 103(a) which forms the basis for all obviousness rejections set forth in this Office action:

(a) A patent may not be obtained though the invention is not identically disclosed or described as set forth in section 102 of this title, if the differences between the subject matter sought to be patented and the prior art are such that the subject matter as a whole would have been obvious at the time the invention was made to a person having ordinary skill in the art to which said subject matter pertains. Patentability shall not be negated by the manner in which the invention was made.

8. Claims 13-17 are rejected under 35 U.S.C. 103(a) as being unpatentable over Wann in view of Csutak et al., hereafter “Csutak” (U.S. Patent No. 6,759,675 B2).

Wann discloses the process of the claimed invention as discussed above, but does not disclose how are the implanting ions implanted to form the channel well into the substrate.

However, Csutak, in the same field of endeavor, discloses how are the implanting ions implanted to form the channel well 18 into the substrate (Fig. 20)

It would have been obvious to one having ordinary skilled in the art at the time the invention was made to modify the teaching of Wann to form the implanting ions implanted to form the channel well into the substrate as taught by Csutak for a purpose of improving the effectiveness of the field isolation doping.

9. Claims 20 and 22-24 are rejected under 35 U.S.C. 103(a) as being unpatentable over Wann in view of Cai et al., hereafter "Cai" (U.S. Patent No. 6,180,132 B2).

Regarding claims 20 and 22, claims 20 and 22 are rejected similar to claim 1 above except for providing a doped drift well in a first portion of the active region and a doped channel well in a second portion of the active region.

However, Cai, in the same field of endeavor, discloses a doped drift well in a first portion of the active region and a doped channel well in a second portion of the active region (Figs. 1-5).

It would have been obvious to one having ordinary skilled in the art at the time the invention was made to modify the teaching of Wann to form a doped drift well in a first portion of the active region and a doped channel well in a second portion of the active region as taught by Cai for a purpose of improving the breakdown voltage of the semiconductor device.

10. Claim 21 is rejected under 35 U.S.C. 103(a) as being unpatentable over Wann and Cai in view of Jang (U.S. Publication No. 2009/0059111 A1).

Regarding claim 21, Wann and Cai disclose the process of the claimed invention as discussed above, but do not disclose a drift isolation region in the doped drift well; and a doped deep well in the substrate of the active device region encompassing the doped drift well.

However, Jang, in the same field of endeavor, discloses a drift isolation region (140a/140b) in the doped drift well (130); and a doped deep well (120) in the substrate of the active device region encompassing the doped drift well (Fig. 3).

It would have been obvious to one having ordinary skilled in the art at the time the invention was made to modify the teaching of Wann to form a drift isolation region in the doped drift well; and a doped deep well in the substrate of the active device region encompassing the doped drift well as taught by Cai for a purpose of improving the high current performance of the semiconductor device.

Cited Prior Arts

11. The prior art made of record and not relied upon is considered pertinent to applicant's disclosure Shimizu et al. (U.S. Patent No. 6,894,350 B2) and Negoro et al. (U.S. Patent No 6,979,864 B2).

Conclusion

12. Any inquiry concerning this communication or earlier communications from the examiner should be directed to Phuc T. Dang whose telephone number is 571-272-1776. The examiner can normally be reached on 8:00 am-5:00 pm.

13. If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Thao X. Le can be reached on (571) 272-1708. The fax phone numbers for the organization

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where this application or proceeding is assigned are 703-872-9306 for regular communications and Final communications.

14. Any inquiry of a general nature or relating to the status of this application or proceeding should be directed to the receptionist whose telephone number is 703-308-0956.

/Phuc T Dang/

Primary Examiner, Art Unit 2892

Notice of References Cited	Application/Control No. 12/390,509		Applicant(s)/Patent Under Reexamination LIU ET AL.	
	Examiner Phuc T. Dang		Art Unit 2892	Page 1 of 1

U.S. PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Name	Classification
*	A	US-5,589,414	12-1996	Wann et al.	438/289
*	B	US-6,759,675 B2	07-2004	Csutak et al.	257/21
*	C	US-7,180,132 B2	02-2007	Cai et al.	257/342
*	D	US-2009/0059111 A1	03-2009	JANG, DUCK KI	349/46
*	E	US-6,894,350 B2	05-2005	Shimizu et al.	257/343
*	F	US-6,979,864 B2	12-2005	Negoro et al.	257/335
	G	US-			
	H	US-			
	I	US-			
	J	US-			
	K	US-			
	L	US-			
	M	US-			

FOREIGN PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Country	Name	Classification
	N					
	O					
	P					
	Q					
	R					
	S					
	T					

NON-PATENT DOCUMENTS

*		Include as applicable: Author, Title Date, Publisher, Edition or Volume, Pertinent Pages)
	U	
	V	
	W	
	X	

*A copy of this reference is not being furnished with this Office action. (See MPEP § 707.05(a).)
Dates in MM-YYYY format are publication dates. Classifications may be US or foreign.



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BIB DATA SHEET

CONFIRMATION NO. 5260

SERIAL NUMBER 12/390,509	FILING or 371(c) DATE 02/23/2009 RULE	CLASS 257	GROUP ART UNIT 2892	ATTORNEY DOCKET NO. CSM P 2009 NAT 03 US 0		
APPLICANTS Junwen LIU, Singapore, SINGAPORE; Purakh Raj VERMA, Singapore, SINGAPORE; Yan JIN, Singapore, SINGAPORE; Baofu ZHU, Singapore, SINGAPORE; ** CONTINUING DATA ***** ** FOREIGN APPLICATIONS ***** ** IF REQUIRED, FOREIGN FILING LICENSE GRANTED ** 03/04/2009						
Foreign Priority claimed ☐ Yes ☒ No 35 USC 119(a-d) conditions met ☐ Yes ☒ No Verified and Acknowledged <u>/PHUC T DANG/</u> Examiner's Signature		☐ Met after Allowance Initials	STATE OR COUNTRY SINGAPORE	SHEETS DRAWINGS 15	TOTAL CLAIMS 21	INDEPENDENT CLAIMS 3
ADDRESS HORIZON IP PTE LTD 7500A Beach Road #04-306/308 The Plaza SINGAPORE 199591, SINGAPORE						
TITLE HIGH VOLTAGE DEVICE						
FILING FEE RECEIVED 1194	FEES: Authority has been given in Paper No. _____ to charge/credit DEPOSIT ACCOUNT No. _____ for following:			☐ All Fees		
				☐ 1.16 Fees (Filing)		
				☐ 1.17 Fees (Processing Ext. of time)		
				☐ 1.18 Fees (Issue)		
				☐ Other _____		
				☐ Credit		

<i>Index of Claims</i> 	Application/Control No. 12390509	Applicant(s)/Patent Under Reexamination LIU ET AL.
	Examiner Phuc T Dang	Art Unit 2892

✓	Rejected	-	Cancelled	N	Non-Elected	A	Appeal
=	Allowed	÷	Restricted	I	Interference	O	Objected

☐ Claims renumbered in the same order as presented by applicant				☐ CPA		☐ T.D.		☐ R.1.47	
CLAIM		DATE							
Final	Original	12/07/2010	02/02/2011						
	1	÷	✓						
	2	÷	✓						
	3	÷	✓						
	4	÷	✓						
	5	÷	✓						
	6	÷	✓						
	7	÷	✓						
	8	÷	✓						
	9	÷	✓						
	10	÷	✓						
	11	÷	✓						
	12	÷	✓						
	13	÷	✓						
	14	÷	✓						
	15	÷	✓						
	16	÷	✓						
	17	÷	✓						
	18	÷	-						
	19	÷	-						
	20	÷	✓						
	21	÷	✓						
	22		✓						
	23		✓						
	24		✓						

Search Notes 	Application/Control No. 12390509	Applicant(s)/Patent Under Reexamination LIU ET AL.
	Examiner Phuc T Dang	Art Unit 2892

SEARCHED			
Class	Subclass	Date	Examiner
438	286,289,294,302	02/02/2011	PD
257	339,E21.427,E29.255	02/02/2011	PD

SEARCH NOTES		
Search Notes	Date	Examiner
INVENTOR SEARCH	02/02/2011	PD
EAST SEARCH	02/02/2011	PD

INTERFERENCE SEARCH			
Class	Subclass	Date	Examiner

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EAST Search History

EAST Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L1	5	((substrate) and (active near (region area)) and (gate near electrode) and (gate near dielectric) and (mask) and (implant\$4) and (opening) and (channel near well) and (remov\$3)).clm.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2011/01/31 13:04
L2	59	(implant\$3) with (ion) with (channel adj1 well)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2011/01/31 13:11
L3	55	2 and @ad<="20090223"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2011/01/31 13:11
L4	160	(implant\$3) with (channel adj1 well)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2011/01/31 13:47
L5	151	4 and @ad<="20090223"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2011/01/31 13:47
L6	74	5 and (gate adj1 electrode)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2011/01/31 13:47
L7	24	6 and (gate adj1 dielectric)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2011/01/31 13:47

L8	2	"12390509"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2011/01/31 13:51
L9	2	8 and channel adj1 well	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2011/01/31 13:51
L10	5783	(implant\$3) with (channel with well)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2011/01/31 13:55
L11	944	438/286.ccls.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2011/01/31 13:56
L12	80	10 and 11	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2011/01/31 13:56
L13	70	12 and @ad<="20090223"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2011/01/31 13:56
L14	16849	(channel adj1 well)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2011/01/31 14:35
L15	36	14 and (implanting) with (channel adj1 well)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2011/01/31 14:36

1/ 31/ 2011 2:54:53 PM

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EAST Search History

EAST Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L1	24645	(remov\$3) with (gate) with (mask\$3)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2011/02/02 15:31
L2	88	(remov\$3) with (gate) with (unprotect\$3) with (mask\$3)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2011/02/02 15:31
L3	81	2 and @ad<="20090223"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2011/02/02 15:31
L4	267364	"438"/\$.ccls.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2011/02/02 16:12
L5	13636	1 and 4	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2011/02/02 16:12
L6	12676	5 and @ad<="20090223"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2011/02/02 16:12
L7	944	438/286.ccls.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2011/02/02 16:13

L8	270	1 and 7	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2011/02/02 16:13
L9	249	8 and @ad<="20090223"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2011/02/02 16:13
L10	7	"5589414"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2011/02/02 16:30
L11	1155	438/289.ccls.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2011/02/02 16:31
L12	206	1 and 11	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2011/02/02 16:32
L13	839	257/339.ccls.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2011/02/02 16:33
L14	54	1 and 13	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2011/02/02 16:33

2/ 2/ 2011 4:37:07 PM

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Receipt date: 02/23/2009

Doc code: IDS

Doc description: Information Disclosure Statement (IDS) Filed

12390509 - GAI: 2892

Approved for use through 02/28/2009. OMB 0651-0031

U.S. Patent and Trademark Office; U.S. DEPARTMENT OF COMMERCE

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		12/390,509	
	Filing Date		02/23/2009	
	First Named Inventor	Junwen LIU		
	Art Unit	2892		
	Examiner Name	Not Assigned yet P. Dang		
	Attorney Docket Number	CSM P 2009 NAT 03 US 0		

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Receipt date: 02/23/2009 INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		12390509 - GAU: 2892
	Filing Date		02/23/2009
	First Named Inventor	Junwen LIU	
	Art Unit	2892	
	Examiner Name	Not Assigned Yet P. Dang	
	Attorney Docket Number	CSM P 2009 NAT 03 US 0	

/PD/	1	LEONARD RUBIN ET AL., Narrow n+/p+ isolation in retrograde well implants, Solid State Technology Magazine, 01 July 2003, pages 119-125, Volume 46 Issue 7, Solid State Technology, USA.	☐
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Appl. No. 12/390,509
Amdt dated May 9, 2011
Reply to Office Action dated February 8, 2011

PATENT

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Appl. No. : 12/390,509 Confirmation No. : 5260
Applicant : Junwen LIU et al.
Filed : 02/23/2009
TC / A.U. : 2892
Examiner : DANG, PHUC T
Docket No. : CSM P 2009 NAT 03 US 0
Customer No. : 031366
Title : HIGH VOLTAGE DEVICE

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I hereby certify that this correspondence is being electronically
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Signed: /nannan chen/

Nannan CHEN

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AMENDMENT

Sir:

In response to the Office Action ("Action") mailed on February 8, 2011, please amend
the above-identified application as follows:

- a) **Amendments to the Specification** begin on page 2 of this paper;
- b) **Amendments to the Claims** are reflected in the listing of claims which begins on
page 3 of this paper; and
- c) **Remarks/Arguments** begin on page 9 of this paper.

Appl. No. 12/390,509
Amdt dated May 9, 2011
Reply to Office Action dated February 8, 2011

Amendments to the Specification

Please replace the title with the following rewritten title:

--METHOD OF FORMING A HIGH VOLTAGE DEVICE--

Amendments to the Claims

The listing of claims will replace all prior versions, and listings, of claims in the application:

Listing of Claims:

1. (currently amended) A method of forming a device comprising:

providing a substrate prepared with an active device region, wherein the active device region includes gate stack layers of a gate stack comprising at least a gate electrode layer over a gate dielectric layer;

forming an implant mask on the substrate with ~~[[an]]~~ a mask opening exposing a portion of a top layer of the gate stack layers, the mask opening having first and second opposing edges;

implanting ions through the mask opening and gate stack layers into the substrate to form a channel well in the substrate, wherein a first channel edge of the channel well extends beyond the first mask opening edge, wherein a first portion of the channel well is between the first channel edge and the first mask opening edge, the first portion defines a channel with a length L of a device; and

~~patterning the substrate to at least remove portion of the top gate stack layer unprotected by the implant mask~~ wherein the channel of the device is disposed under a mask portion and a second portion of the channel well is disposed below the mask opening.

2. (currently amended) The method of claim 1 wherein ~~the~~ a top layer of the gate stack layers comprises the gate electrode layer of the gate stack.

3. (original) The method of claim 2 wherein the gate electrode layer comprises silicon, including amorphous or polysilicon.

4. (currently amended) The method of claim 2 ~~wherein~~ comprises:

patterning the ~~substrate~~ gate stack layers to form a first gate edge of the gate stack; and

further comprises patterning the ~~substrate~~ gate stack layers to form a second gate edge of the gate stack after forming the first gate edge.

5. (currently amended) The method of claim 4 wherein the first gate edge corresponds to a source channel edge of the gate of the device and the second gate edge corresponds to the drain edge of the gate of the device.

6. (currently amended) The method of claim 1 ~~wherein~~:

~~patterning the substrate forms a first edge of the gate; and~~

further comprises patterning the ~~substrate~~ gate stack to at least remove a portion of the gate electrode layer after forming a first gate edge of the gate to form a second gate edge of the gate.

7. (currently amended) The method of claim 6 wherein the first gate edge corresponds to a source channel edge of the gate of the device and the second gate edge corresponds to the drain edge of the gate of the device.

8. (currently amended) The method of claim 1 wherein patterning the ~~substrate~~ gate stack layers leaves the gate dielectric layer remaining on the substrate.

9. (currently amended) The method of claim 1 wherein:
the gate stack layers comprise a hard mask layer over the gate electrode layer; ~~and~~
patterning the hard mask using the implant mask to remove ~~removes~~ a portion of the
hard mask layer to expose the gate electrode layer; and
implanting ions to form the channel well after patterning the hard mask.
10. (currently amended) The method of claim 9 wherein the hard mask is further
patterned to serve as an etch mask of a gate ~~patterning the hard mask forms a first edge of the~~
~~hard mask corresponding to a first edge of the gate stack and further comprises patterning the~~
~~hard mask to form a second edge of the hard mask corresponding to a second edge of the gate~~
~~stack.~~
11. (currently amended) The method of claim 10 further comprises patterning the gate
stack layers with the patterned hard mask to form a gate with first and second gate edges.
12. (original) The method of claim 11 wherein patterning the gate stack layers leaves
the gate dielectric layer remaining on the substrate.
13. (original) The method of claim 1 wherein implanting ions comprises a tilt angle
implant.
14. (original) The method of claim 1 wherein a tilt angle of the implant is about 1 –
45°.

15. (original) The method of claim 1 wherein implanting ions comprises multiple tilt angle implants which are rotated about a plane of the substrate.

16. (original) The method of claim 15 wherein a tilt angle of the implant is about 1 – 45°.

17. (original) The method of claim 1 wherein implanting ions comprises a quad tilt angle implant which is rotated about a plane of the substrate.

18-19. (canceled)

20. (currently amended) A method of forming a device comprising:

providing a substrate prepared with an active device region which includes a doped drift well in a first portion of the active device region and gate stack layers of a gate stack comprising at least a gate electrode layer over a gate dielectric layer on the surface of the substrate;

forming an implant mask on the substrate with ~~[[an]]~~ a mask opening exposing a portion of a top ~~layer~~ of the gate stack ~~layers~~, the mask opening having first and second opposing edges;

implanting ions through the mask opening and gate stack layers into the substrate to form a channel well in the substrate, wherein a first channel edge of the channel well extends beyond the first mask opening edge, wherein a first portion of the channel well is between the first channel edge and the first mask opening edge, the first portion defines a channel with a length L of a device; and

~~patterning the substrate to at least remove portion of the top gate electrode stack layer~~
~~unprotected by the implant mask~~ wherein the channel of the device is disposed under a mask
portion and a second portion of the channel well is disposed below the mask opening.

21. (original) The method of claim 20 wherein the substrate comprises:
a drift isolation region in the doped drift well; and
a doped deep well in the substrate of the active device region encompassing the doped drift well.

22. (currently amended) A method of forming a device comprising:
providing a substrate prepared with an active device region which includes a doped drift well in a first portion of the active device region, ~~a doped channel well in a second~~
~~portion of the active region~~ and gate stack layers of a gate stack comprising at least a gate electrode layer over a gate dielectric layer on the surface of the substrate;

forming an implant mask on the substrate with ~~[[an]]~~ a mask opening exposing a portion of a top layer of the gate stack layers, the mask opening having first and second opposing edges;

implanting ions through the mask opening and gate stack layers into the substrate to form a channel well in the substrate in a second portion of the active region, wherein a first channel edge of the channel well extends beyond the first mask opening edge, wherein a first portion of the channel well is between the first channel edge and the first mask opening edge, the first portion defines a channel with a length L of a device; and

~~patterning the substrate to at least remove portion of the top gate stack layers~~
~~unprotected by the implant mask~~ wherein the channel of the device is disposed under a mask
portion and a second portion of the channel well is disposed below the mask opening.

23. (previously presented) The method of claim 22 wherein the doped drift well in the first portion of the active region is contiguous with the doped channel well in the second portion of the active region.

24. (previously presented) The method of claim 22 wherein the doped drift well in the first portion of the active region is separated from the doped channel well in the second portion of the active region by a third portion.

REMARKS/ARGUMENTS

Claim Amendments

Without conceding the propriety of the rejections herein and in the interest of expediting prosecution, Applicants amend some claims herein. Applicants amend claims to clarify claimed features. Such amendments are made to expedite prosecution and more quickly identify allowable subject matter. Such amendments are merely intended to clarify the claimed features, and should not be construed as further limiting the claimed invention in response to the cited references. All amendments are fully supported by the Application and therefore do not constitute new matter.

Specification

The Specification has been objected to by the Examiner because the title of the invention is not descriptive. Applicants have amended the title as suggested by the Examiner. Applicants therefore submit that this basis of objection has been traversed.

Claim Objections

Claim 1 has been objected to by the Examiner for informalities. Applicants have amended the claim to correct the informality defects. Applicants therefore submit that this basis of objection has been traversed and respectfully request its withdrawal.

Claim Rejections – 35 USC § 112

Claims 1-17 and 20-24 are rejected under 35 USC §112, second paragraph, as being indefinite for failing to particularly point out and distinctly claim the subject matter which the Applicants regard as the invention. In particular, the Examiner indicated that in claims 1-17

and 20-24, step of “patterning the substrate...” is indefinite because the gate stack is patterned by the implant mask and not the substrate. Applicants respectfully disagree with this basis of rejection.

However, in the interest of expediting prosecution and without conceding the propriety of the rejections, Applicants amend claims 1, 20 and 22. Therefore, the amendments render claims 1, 20 and 22 clear and definite.

Claim Rejections – 35 USC § 102

Claims 1-3 and 9-12 are rejected under 35 USC § 102(b) as being anticipated by US Patent No. 5,589,414 (Wann). Applicants respectfully disagree with this basis of rejection.

Independent claim 1 requires that an implant mask with a mask opening having opposing first and second edges is provided over a gate stack on a substrate. Ions are implanted through the mask opening and gate stack into the substrate to form a channel well in the substrate. The channel well extends beyond the first mask opening edge. A first portion of the channel well is between the first channel edge and the first mask opening edge. The first portion defines a channel with a channel length L of a device. The channel of the device is disposed under a portion of the gate stack and a second portion of the channel well is disposed below the mask opening.

As for Wann, it discloses a method of forming a mask Rom with two layer gate electrode. A resist mask which over the lower polysilicon layer of a two layer gate electrode. The resist mask is patterned to form a code implant opening between the FOX regions. *See, e.g., Wann*, col. 4, lines 63-67 and Fig. 2C. The code implant opening corresponds to the desired location of the gate. Ions are implanted into the substrate to form a code implant region which corresponds to the channel. *See, e.g., Wann*, col. 5, lines 1-5 and Fig. 2D. The

gate, as clearly shown by Wann, is formed completely over the code implant region. *See, e.g., Wann, Fig. 2F.*

However, as required by present claim 1, a second portion of the channel well is below the mask opening and a first portion is under the mask. Clearly, as described by Wann, the code implant region is completely under the mask opening. *See, e.g., Wann, Fig. 2F* (elements 47' and 48'). Furthermore, present claim 1 requires that the first portion of the channel well which is under the mask defines the channel with a channel length L of the device. For example, the first portion of the channel well from the first channel edge to the first mask edge is the channel of the device. This is in contrast to Wann, whose channel is completely under the mask opening, not the mask.

Based on at least the foregoing reasons, Applicants submit that claim 1 and its directly and indirectly dependent claims are patentable over Wann. Furthermore, these claims may be patentable for additional independent reasons.

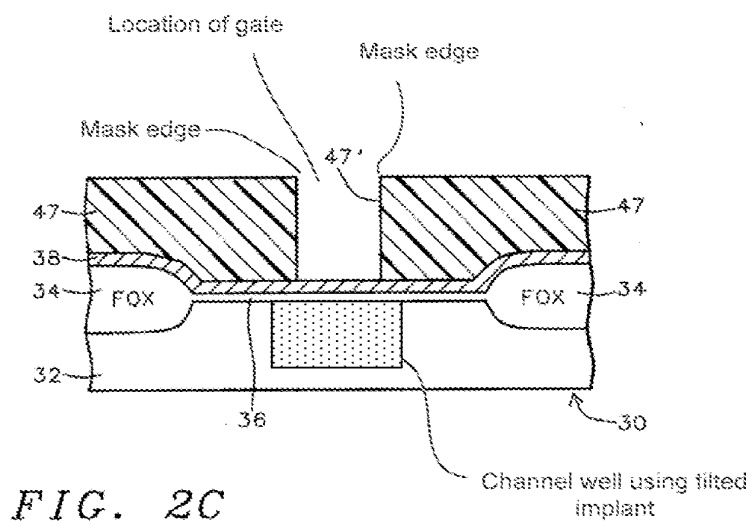
With respect to independent claims 20 and 22, they require similar limitations as claim 1. Applicants therefore submit Claims 20 and 22 as well as their directly and indirectly dependent claims are also patentable over Wann for similar reasons as claim 1.

Claim Rejections – 35 USC § 103

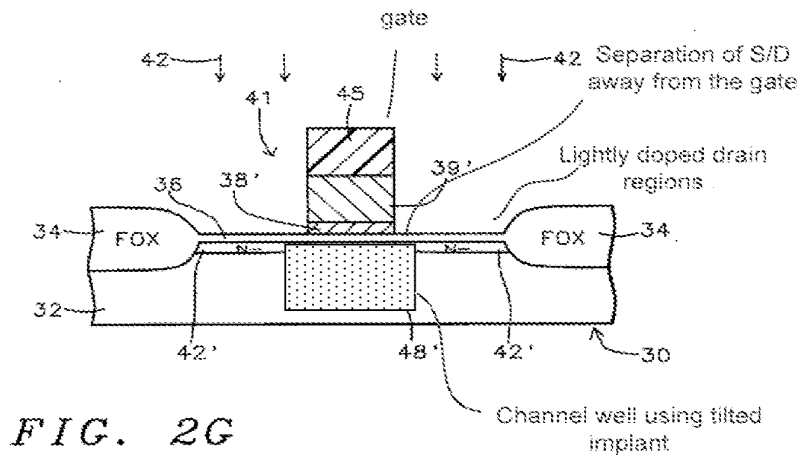
Claims 13-17 are rejected under 35 USC § 103(a) as being unpatentable over Wann in view of US Patent No. 6,759,675 (Csutak). Applicants respectfully disagree with this basis of rejection.

Claims 13-17 are directly or indirectly dependent on independent claim 1. As already discussed, independent claim 1 is patentable over Wann. The Examiner relies on Csutak to cure the defects of Wann.

Applicants submit that there is no suggestion or motivation to combine the references as suggested by the Examiner. The Examiner, in the Action, relies on Csutak to teach the use of tilted implants. However, applicants submit that if tilted implants were employed by Wann, its device would be rendered inoperable. For example, if tilted implants were employed by Wann to form the code implant region, this would result in the code implant region extending beyond the mask edges. Applicants have provided for convenience of the Examiner Fig. 2C of Wann below with annotation in red to illustrate this point.



The opening of the code implant corresponds to the location of the gate. As the process continues to form the gate and lightly doped drain regions, they would be prevented for extending up to the sidewalls of the gate as a result of the channel well formed by tilted implants. Applicants have provided for convenience of the Examiner Fig. 2G of Wann below with annotation in red to illustrate this point.



Clearly, not having the lightly doped drain regions extending up to the gate sidewalls is contrary to the intent of Wann, as well as what is well understood in the art. Separation of the lightly doped drain regions from the sidewalls of the gate may render the device inoperable.

Applicants therefore submit that based on the foregoing discussions, one or ordinary skilled in the art would not be led to combine the references as suggested by the Examiner.

Assuming arguendo, even if the references were combined, such a combination would still fail to render claims 13-17 unpatentable. As required by claim 1, the first portion of the channel well which is under the mask, not the mask opening, defines the channel of the device. Clearly, as indicated by annotated Fig. 2C, the portion of the channel well under the mask is the portions which extend laterally away from the sidewalls of the gate. These portions cannot be the channel. As is well know in the art, the channel is under the gate, not away from the gate.

Based on the foregoing discussion, Applicants submit that claims 13-17 are patentable over Wann and Csutak, alone or in combination. Furthermore, these claims may be patentable for additional independent reasons.

Claims 20 and 22-24 are rejected under 35 USC § 103(a) as being unpatentable over Wann in view of US Patent No. 6,180,132 (Cai). Applicants respectfully disagree with this basis of rejection.

Independent claims 20 and 22 include similar limitations as independent claim 1. For example, these independent claims require that the channel well has a second portion under the mask opening and a first portion which defines the channel with a channel length L of the device under the mask. This limitation of claims 20 and 22 is missing from Wann. The Examiner relies on Cai to cure the defects of Wann.

Applicants submit that there is no suggestion or motivation to combine the references as suggested by the Examiner. Even if the references were combined, such a combination still fail to render the present independent claims unpatentable.

The Examiner, in the Action, relies on Cai to teach a doped drift well in a first portion of the active region and a doped channel well in a second portion of the active region. However, Applicants submit that neither reference teaches or suggests that the channel well has a portion under the mask which defines a channel with a channel length L. Clearly, as described by Cai, the channel length is defined by the gate itself. *See, e.g., Cai*, col. 4, lines 65-67.

Based on the foregoing discussion, Applicants submit that independent claims 1, 20 and 22 as well as their directly and indirectly dependent claims, including claims 23-24 are patentable over Wann and Cai, alone or in combination. Furthermore, these claims may be patentable for additional independent reasons.

Claim 21 is rejected under 35 USC § 103(a) as being unpatentable over Wann and Cai in view of US Publication No. 2009/0059111 (Jang). Applicants respectfully disagree with this basis of rejection.

Claim 21 depends on independent claim 20. As previously discussed, the combination of Wann and Cai, alone or in combination, fails to teach or suggest forming a channel well which has a second portion under the mask opening and a first portion which defines the channel with a channel length L of the device under the mask.

Applicants submit that the addition of Jang fails to cure the defects of Wann and Cai. Jang, like Wann and Cai, nowhere teaches a channel well which has a second portion under the mask opening and a first portion which defines the channel with a channel length L of the device under the mask. This is evidenced by the fact that the Examiner merely relies on Jang to disclose a doped deep well which encompass the doped drift well. *See, e.g., Action*, page 6.

Based on the foregoing discussion, Applicants submit that independent claim 22 and its dependent claim 21 are patentable over Wann, Cai and Jang, alone or in any combination. Furthermore, these claims may be patentable for additional independent reasons.

Appl. No. 12/390,509
Amdt dated May 9, 2011
Reply to Office Action dated February 8, 2011

Conclusion

It is respectfully submitted that this application is in condition for allowance and the issuance of a formal Notice of Allowance at an early date is respectfully requested.

Should the examiner believe that a telephone conference would expedite prosecution of this application, please telephone the undersigned attorney at his number set out below.

Date: May 9, 2011

Respectfully submitted,

/dexter chin/

Dexter CHIN
Attorney for Applicants
Reg. No.: 38,842

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TRANSMITTAL FORM <i>(to be used for all correspondence after initial filing)</i>	Application Number	12/390,509	
	Filing Date	02/23/2009	
	First Named Inventor	Junwen LIU	
	Art Unit	2892	
	Examiner Name	DANG, PHUC T	
Total Number of Pages in This Submission	18	Attorney Docket Number	CSM P 2009 NAT 03 US 0

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EFS ID:	10053704
Application Number:	12390509
International Application Number:	
Confirmation Number:	5260
Title of Invention:	HIGH VOLTAGE DEVICE
First Named Inventor/Applicant Name:	Junwen LIU
Customer Number:	31366
Filer:	Dexter Kam Chin
Filer Authorized By:	
Attorney Docket Number:	CSM P 2009 NAT 03 US 0
Receipt Date:	09-MAY-2011
Filing Date:	23-FEB-2009
Time Stamp:	23:05:05
Application Type:	Utility under 35 USC 111(a)

Payment information:

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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Amendment/Req. Reconsideration-After Non-Final Reject	CSMP2009NAT03US0- ResponsetoOA.pdf	243342 861dc544fbac7cea592c0512590c6cb74223 9954	no	16

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2	Miscellaneous Incoming Letter	CSMP2009NAT063US0-TF.pdf	277459 c3076bb11ca07d1caa6a396b7927982076c6b213	no	2
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APPLICATION AS FILED – PART I										
(Column 1)			(Column 2)			SMALL ENTITY ☐ OR		OTHER THAN SMALL ENTITY		
FOR	NUMBER FILED	NUMBER EXTRA	RATE (\$)	FEE (\$)	OR	RATE (\$)	FEE (\$)			
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A			N/A				
☐ SEARCH FEE (37 CFR 1.16(k), (l), or (m))	N/A	N/A	N/A			N/A				
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A			N/A				
TOTAL CLAIMS (37 CFR 1.16(j))	minus 20 =	*	X \$	=		X \$	=			
INDEPENDENT CLAIMS (37 CFR 1.16(h))	minus 3 =	*	X \$	=		X \$	=			
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$250 (\$125 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).									
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))										
			TOTAL			TOTAL				
* If the difference in column 1 is less than zero, enter "0" in column 2.										
APPLICATION AS AMENDED – PART II										
(Column 1)			(Column 2)			SMALL ENTITY OR		OTHER THAN SMALL ENTITY		
AMENDMENT	05/09/2011	CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)	OR	RATE (\$)	ADDITIONAL FEE (\$)	
	Total (37 CFR 1.16(i))	* 22	Minus	** 22	=	0		X \$52=	0	
	Independent (37 CFR 1.16(h))	* 3	Minus	***3	=	0		X \$220=	0	
☐ Application Size Fee (37 CFR 1.16(s))										
☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))										
					TOTAL ADD'L FEE			TOTAL ADD'L FEE	0	
(Column 1)			(Column 2)			SMALL ENTITY OR		OTHER THAN SMALL ENTITY		
AMENDMENT		CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)	OR	RATE (\$)	ADDITIONAL FEE (\$)	
	Total (37 CFR 1.16(i))	*	Minus	**	=			X \$	=	
	Independent (37 CFR 1.16(h))	*	Minus	***	=			X \$	=	
☐ Application Size Fee (37 CFR 1.16(s))										
☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))										
					TOTAL ADD'L FEE			TOTAL ADD'L FEE		
* If the entry in column 1 is less than the entry in column 2, write "0" in column 3. ** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20". *** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3". The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.										

Legal Instrument Examiner:
/GYZELL JOHNSON SMITH/

This collection of information is required by 37 CFR 1.16. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

NOTICE OF ALLOWANCE AND FEE(S) DUE

31366 7590 07/12/2011
HORIZON IP PTE LTD
7500A Beach Road
#04-306/308 The Plaza
SINGAPORE 199591,
SINGAPORE

EXAMINER

DANG, PHUC T

ART UNIT

PAPER NUMBER

2892

DATE MAILED: 07/12/2011

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
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12/390,509

02/23/2009

Junwen LIU

CSM P 2009 NAT 03 US 0

5260

TITLE OF INVENTION: HIGH VOLTAGE DEVICE

APPLN. TYPE	SMALL ENTITY	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	NO	\$1510	\$300	\$0	\$1810	10/12/2011

THE APPLICATION IDENTIFIED ABOVE HAS BEEN EXAMINED AND IS ALLOWED FOR ISSUANCE AS A PATENT. PROSECUTION ON THE MERITS IS CLOSED. THIS NOTICE OF ALLOWANCE IS NOT A GRANT OF PATENT RIGHTS. THIS APPLICATION IS SUBJECT TO WITHDRAWAL FROM ISSUE AT THE INITIATIVE OF THE OFFICE OR UPON PETITION BY THE APPLICANT. SEE 37 CFR 1.313 AND MPEP 1308.

THE ISSUE FEE AND PUBLICATION FEE (IF REQUIRED) MUST BE PAID WITHIN THREE MONTHS FROM THE MAILING DATE OF THIS NOTICE OR THIS APPLICATION SHALL BE REGARDED AS ABANDONED. THIS STATUTORY PERIOD CANNOT BE EXTENDED. SEE 35 U.S.C. 151. THE ISSUE FEE DUE INDICATED ABOVE DOES NOT REFLECT A CREDIT FOR ANY PREVIOUSLY PAID ISSUE FEE IN THIS APPLICATION. IF AN ISSUE FEE HAS PREVIOUSLY BEEN PAID IN THIS APPLICATION (AS SHOWN ABOVE), THE RETURN OF PART B OF THIS FORM WILL BE CONSIDERED A REQUEST TO REAPPLY THE PREVIOUSLY PAID ISSUE FEE TOWARD THE ISSUE FEE NOW DUE.

HOW TO REPLY TO THIS NOTICE:

I. Review the SMALL ENTITY status shown above.

If the SMALL ENTITY is shown as YES, verify your current SMALL ENTITY status:

A. If the status is the same, pay the TOTAL FEE(S) DUE shown above.

B. If the status above is to be removed, check box 5b on Part B - Fee(s) Transmittal and pay the PUBLICATION FEE (if required) and twice the amount of the ISSUE FEE shown above, or

If the SMALL ENTITY is shown as NO:

A. Pay TOTAL FEE(S) DUE shown above, or

B. If applicant claimed SMALL ENTITY status before, or is now claiming SMALL ENTITY status, check box 5a on Part B - Fee(s) Transmittal and pay the PUBLICATION FEE (if required) and 1/2 the ISSUE FEE shown above.

II. PART B - FEE(S) TRANSMITTAL, or its equivalent, must be completed and returned to the United States Patent and Trademark Office (USPTO) with your ISSUE FEE and PUBLICATION FEE (if required). If you are charging the fee(s) to your deposit account, section "4b" of Part B - Fee(s) Transmittal should be completed and an extra copy of the form should be submitted. If an equivalent of Part B is filed, a request to reapply a previously paid issue fee must be clearly made, and delays in processing may occur due to the difficulty in recognizing the paper as an equivalent of Part B.

III. All communications regarding this application must give the application number. Please direct all communications prior to issuance to Mail Stop ISSUE FEE unless advised to the contrary.

IMPORTANT REMINDER: Utility patents issuing on applications filed on or after Dec. 12, 1980 may require payment of maintenance fees. It is patentee's responsibility to ensure timely payment of maintenance fees when due.

PART B - FEE(S) TRANSMITTAL

**Complete and send this form, together with applicable fee(s), to: Mail Mail Stop ISSUE FEE
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450
or Fax (571)-273-2885**

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence including the Patent, advance orders and notification of maintenance fees will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications.

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

31366 7590 07/12/2011
HORIZON IP PTE LTD
7500A Beach Road
#04-306/308 The Plaza
SINGAPORE 199591,
SINGAPORE

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being facsimile transmitted to the USPTO (571) 273-2885, on the date indicated below.

(Depositor's name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
12/390,509	02/23/2009	Junwen LIU	CSM P 2009 NAT 03 US 0	5260

TITLE OF INVENTION: HIGH VOLTAGE DEVICE

APPLN. TYPE	SMALL ENTITY	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	NO	\$1510	\$300	\$0	\$1810	10/12/2011

EXAMINER	ART UNIT	CLASS-SUBCLASS
DANG, PHUC T	2892	438-286000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

- ☐ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached.
☐ "Fee Address" indication (or "Fee Address" Indication form PTO/SB/47; Rev 03-02 or more recent) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list

- (1) the names of up to 3 registered patent attorneys or agents OR, alternatively, 1 _____
(2) the name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed. 2 _____
3 _____

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document has been filed for recordation as set forth in 37 CFR 3.11. Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE (B) RESIDENCE: (CITY and STATE OR COUNTRY)

Please check the appropriate assignee category or categories (will not be printed on the patent) : ☐ Individual ☐ Corporation or other private group entity ☐ Government

4a. The following fee(s) are submitted:

- ☐ Issue Fee
☐ Publication Fee (No small entity discount permitted)
☐ Advance Order - # of Copies _____

4b. Payment of Fee(s); (Please first reapply any previously paid issue fee shown above)

- ☐ A check is enclosed.
☐ Payment by credit card. Form PTO-2038 is attached.
☐ The Director is hereby authorized to charge the required fee(s), any deficiency, or credit any overpayment, to Deposit Account Number _____ (enclose an extra copy of this form).

5. Change in Entity Status (from status indicated above)

- ☐ a. Applicant claims SMALL ENTITY status. See 37 CFR 1.27. ☐ b. Applicant is no longer claiming SMALL ENTITY status. See 37 CFR 1.27(g)(2).

NOTE: The Issue Fee and Publication Fee (if required) will not be accepted from anyone other than the applicant; a registered attorney or agent; or the assignee or other party in interest as shown by the records of the United States Patent and Trademark Office.

Authorized Signature _____ Date _____

Typed or printed name _____ Registration No. _____

This collection of information is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450.

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.



UNITED STATES PATENT AND TRADEMARK OFFICE

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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
12/390,509	02/23/2009	Junwen LIU	CSM P 2009 NAT 03 US 0	5260

31366 7590 07/12/2011
HORIZON IP PTE LTD
7500A Beach Road
#04-306/308 The Plaza
SINGAPORE 199591,
SINGAPORE

EXAMINER

DANG, PHUC T

ART UNIT	PAPER NUMBER
----------	--------------

2892

DATE MAILED: 07/12/2011

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (application filed on or after May 29, 2000)

The Patent Term Adjustment to date is 233 day(s). If the issue fee is paid on the date that is three months after the mailing date of this notice and the patent issues on the Tuesday before the date that is 28 weeks (six and a half months) after the mailing date of this notice, the Patent Term Adjustment will be 233 day(s).

If a Continued Prosecution Application (CPA) was filed in the above-identified application, the filing date that determines Patent Term Adjustment is the filing date of the most recent CPA.

Applicant will be able to obtain more detailed information by accessing the Patent Application Information Retrieval (PAIR) WEB site (<http://pair.uspto.gov>).

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Customer Service Center of the Office of Patent Publication at 1-(888)-786-0101 or (571)-272-4200.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Notice of Allowability	Application No.	Applicant(s)	
	12/390,509	LIU ET AL.	
	Examiner	Art Unit	
	PHUC DANG	2892	

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. ☒ This communication is responsive to amendment filed on 05/09/2011.
2. ☒ The allowed claim(s) is/are 1-17 and 20-24 (renumbered as in new claims 1-22).
3. ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).
 - a) ☐ All b) ☐ Some* c) ☐ None of the:
 1. ☐ Certified copies of the priority documents have been received.
 2. ☐ Certified copies of the priority documents have been received in Application No. _____.
 3. ☐ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).

* Certified copies not received: _____.

Applicant has THREE MONTHS FROM THE "MAILING DATE" of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.
THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.

4. ☐ A SUBSTITUTE OATH OR DECLARATION must be submitted. Note the attached EXAMINER'S AMENDMENT or NOTICE OF INFORMAL PATENT APPLICATION (PTO-152) which gives reason(s) why the oath or declaration is deficient.
5. ☐ CORRECTED DRAWINGS (as "replacement sheets") must be submitted.
 - (a) ☐ including changes required by the Notice of Draftsperson's Patent Drawing Review (PTO-948) attached
 - 1) ☐ hereto or 2) ☐ to Paper No./Mail Date _____.
 - (b) ☐ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date _____.

Identifying indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).
6. ☐ DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

Attachment(s)

- | | |
|---|---|
| <ol style="list-style-type: none"> 1. ☐ Notice of References Cited (PTO-892) 2. ☐ Notice of Draftsperson's Patent Drawing Review (PTO-948) 3. ☐ Information Disclosure Statements (PTO/SB/08),
Paper No./Mail Date _____ 4. ☐ Examiner's Comment Regarding Requirement for Deposit
of Biological Material | <ol style="list-style-type: none"> 5. ☐ Notice of Informal Patent Application 6. ☐ Interview Summary (PTO-413),
Paper No./Mail Date _____. 7. ☐ Examiner's Amendment/Comment 8. ☐ Examiner's Statement of Reasons for Allowance 9. ☐ Other _____. |
|---|---|

/Phuc T Dang/
Primary Examiner, Art Unit 2892

<i>Index of Claims</i> 	Application/Control No. 12390509	Applicant(s)/Patent Under Reexamination LIU ET AL.
	Examiner PHUC DANG	Art Unit 2892

✓	Rejected	-	Cancelled	N	Non-Elected	A	Appeal
=	Allowed	÷	Restricted	I	Interference	O	Objected

☐ Claims renumbered in the same order as presented by applicant					☐ CPA					☐ T.D.					☐ R.1.47				
CLAIM		DATE																	
Final	Original	12/07/2010	02/02/2011	06/29/2011															
1	1	÷	✓	=															
2	2	÷	✓	=															
3	3	÷	✓	=															
4	4	÷	✓	=															
5	5	÷	✓	=															
6	6	÷	✓	=															
7	7	÷	✓	=															
8	8	÷	✓	=															
9	9	÷	✓	=															
10	10	÷	✓	=															
11	11	÷	✓	=															
12	12	÷	✓	=															
13	13	÷	✓	=															
14	14	÷	✓	=															
15	15	÷	✓	=															
16	16	÷	✓	=															
17	17	÷	✓	=															
	18	÷	-	-															
	19	÷	-	-															
18	20	÷	✓	=															
19	21	÷	✓	=															
20	22		✓	=															
21	23		✓	=															
22	24		✓	=															

Search Notes 	Application/Control No. 12390509	Applicant(s)/Patent Under Reexamination LIU ET AL.
	Examiner Phuc T Dang	Art Unit 2892

SEARCHED			
Class	Subclass	Date	Examiner
438	286,289,294,302	02/02/2011	PD
257	339,E21.427,E29.255	02/02/2011	PD
same	as above	06/29/2011	PD

SEARCH NOTES		
Search Notes	Date	Examiner
INVENTOR SEARCH	02/02/2011	PD
EAST SEARCH	02/02/2011	PD
EAST SEARCH (Interference Search History Printout)	06/29/2011	PD

INTERFERENCE SEARCH			
Class	Subclass	Date	Examiner
same	as above	06/29/2011	PD

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UNITED STATES DEPARTMENT OF COMMERCE
 United States Patent and Trademark Office
 Address: COMMISSIONER FOR PATENTS
 P.O. Box 1450
 Alexandria, Virginia 22313-1450
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BIB DATA SHEET

CONFIRMATION NO. 5260

SERIAL NUMBER 12/390,509	FILING or 371(c) DATE 02/23/2009 RULE	CLASS 257	GROUP ART UNIT 2892	ATTORNEY DOCKET NO. CSM P 2009 NAT 03 US 0		
APPLICANTS Junwen LIU, Singapore, SINGAPORE; Purakh Raj VERMA, Singapore, SINGAPORE; Yan JIN, Singapore, SINGAPORE; Baofu ZHU, Singapore, SINGAPORE; ** CONTINUING DATA ***** ** FOREIGN APPLICATIONS ***** ** IF REQUIRED, FOREIGN FILING LICENSE GRANTED ** 03/04/2009						
Foreign Priority claimed ☐ Yes ☒ No 35 USC 119(a-d) conditions met ☐ Yes ☒ No Verified and Acknowledged <u>/PHUC T DANG/</u> Examiner's Signature		☐ Met after Allowance Initials	STATE OR COUNTRY SINGAPORE	SHEETS DRAWINGS 15	TOTAL CLAIMS 21	INDEPENDENT CLAIMS 3
ADDRESS HORIZON IP PTE LTD 7500A Beach Road #04-306/308 The Plaza SINGAPORE 199591, SINGAPORE						
TITLE HIGH VOLTAGE DEVICE						
FILING FEE RECEIVED 1194	FEES: Authority has been given in Paper No. _____ to charge/credit DEPOSIT ACCOUNT No. _____ for following:			☐ All Fees		
				☐ 1.16 Fees (Filing)		
				☐ 1.17 Fees (Processing Ext. of time)		
				☐ 1.18 Fees (Issue)		
				☐ Other _____		
			☐ Credit			

EAST Search History**EAST Search History (Prior Art)**

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L1	983	438/286. ccls.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2011/06/29 13:12
L2	3	1 and (mask) with (channel adj1 well)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2011/06/29 13:13
L3	1179	438/289. ccls.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2011/06/29 13:14
L4	2	3 and (mask) with (channel adj1 well)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2011/06/29 13:14
L5	669	438/294. ccls.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2011/06/29 13:14
L6	0	5 and (mask) with (channel adj1 well)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2011/06/29 13:14

L7	857	257/339. ccls.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2011/06/29 13:15
L8	3	7 and (mask) with (channel adj1 well)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2011/06/29 13:15
L9	1318	257/E21.427. ccls.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2011/06/29 13:15
L10	4	9 and (mask) with (channel adj1 well)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2011/06/29 13:15

EAST Search History (I nterference)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L11	66	((substrate) and (mask) and (implant\$3) and (channel adj1 well) and (gate near electrode) and (opening aperture hole) and (gate near dielectric))	USPAT; UPAD	OR	ON	2011/06/29 13:17

L12	0	((substrate) and (mask) and (implant\$3) and (channel adj1 well) and (gate near electrode) and (opening aperture hole) and (gate near dielectric) and (active near region) and (first near channel near edge))	USPAT; UPAD	OR	ON	2011/06/29 13:18
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6/ 29/ 2011 1:18:33 PM

**C:\ Documents and Settings\ pdang1\ My Documents\ EAST\ Workspaces
\ 12390509.wsp**

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P.O.BOX 1450
ALEXANDRIA VA 22313-1451

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POSTEDIGITAL
NNNNN

Horizon IP Pte Ltd.
7500A Beach Road, #04-306/308
The Plaza
Singapore, 199591
SINGAPORE



**Courtesy Reminder for
Application Serial No: 12/390,509**

Attorney Docket No: CSM P 2009 NAT 03 US 0
Customer Number: 16162
Date of Electronic Notification: 07/12/2011

This is a courtesy reminder that new correspondence is available for this application. The official date of notification of the outgoing correspondence will be indicated on the form PTOL-90 accompanying the correspondence.

An email notification regarding the correspondence was sent to the following email address(es) associated with your customer number:

oa.notification@horizonip.com.sg
dexter.chin@horizonip.com.sg
nannan.chen@horizonip.com.sg

Please verify that these email addresses are correct.

To view your correspondence online or update your email addresses, please visit us anytime at <https://sportal.uspto.gov/secure/myportal/privatepair>. If you have any questions, please email the Electronic Business Center (EBC) at EBC@uspto.gov or call 1-866-217-9197.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	PATENT NUMBER	GROUP ART UNIT	FILE WRAPPER LOCATION
12/390,509		2892	



Correspondence Address/Fee Address Change

The following fields have been set to Customer Number 16162 on 07/19/2011

- Correspondence Address

The address of record for Customer Number 16162 is:

16162
Horizon IP Pte Ltd.
7500A Beach Road, #04-306/308
The Plaza
Singapore, 199591
SINGAPORE



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United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
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Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	PATENT NUMBER	GROUP ART UNIT	FILE WRAPPER LOCATION
12/390,509		2892	



Correspondence Address/Fee Address Change

The following fields have been set to Customer Number 16162 on 07/20/2011

- Correspondence Address

The address of record for Customer Number 16162 is:

16162
Horizon IP Pte Ltd.
7500A Beach Road, #04-306/308
The Plaza
Singapore, 199591
SINGAPORE

PART B - FEE(S) TRANSMITTAL

**Complete and send this form, together with applicable fee(s), to: Mail Mail Stop ISSUE FEE
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450
or Fax (571)-273-2885**

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence including the Patent, advance orders and notification of maintenance fees will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications.

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

31366 7590 07/12/2011
HORIZON IP PTE LTD
7500A Beach Road
#04-306/308 The Plaza
SINGAPORE 199591,
SINGAPORE

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being facsimile transmitted to the USPTO (571) 273-2885, on the date indicated below.

Nannan CHEN	(Depositor's name)
/nannan chen/	(Signature)
09/25/2011	(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
-----------------	-------------	----------------------	---------------------	------------------

12/390,509	02/23/2009	Junwen LIU	CSM P 2009 NAT 03 US 0	5260
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TITLE OF INVENTION: HIGH VOLTAGE DEVICE

APPLN. TYPE	SMALL ENTITY	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	NO	\$1510	\$300	\$0	\$1810	10/12/2011

EXAMINER	ART UNIT	CLASS-SUBCLASS
DANG, PHUC T	2892	438-286000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

- ☐ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached.
☐ "Fee Address" indication (or "Fee Address" Indication form PTO/SB/47; Rev 03-02 or more recent) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list

- (1) the names of up to 3 registered patent attorneys or agents OR, alternatively,
(2) the name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed.

1 Horizon IP Pte Ltd
2
3

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document has been filed for recordation as set forth in 37 CFR 3.11. Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE GLOBALFOUNDRIES SINGAPORE PTE. LTD.
(B) RESIDENCE: (CITY and STATE OR COUNTRY) SINGAPORE

Please check the appropriate assignee category or categories (will not be printed on the patent): ☐ Individual ☒ Corporation or other private group entity ☐ Government

4a. The following fee(s) are submitted:

- ☒ Issue Fee
☒ Publication Fee (No small entity discount permitted)
☐ Advance Order - # of Copies

4b. Payment of Fee(s): (Please first reapply any previously paid issue fee shown above)

- ☐ A check is enclosed.
☒ Payment by credit card. Form PTO-2038 is attached.
☒ The Director is hereby authorized to charge the required fee(s), any deficiency, or credit any overpayment, to Deposit Account Number 502388 (enclose an extra copy of this form).

5. Change in Entity Status (from status indicated above)

- ☐ a. Applicant claims SMALL ENTITY status. See 37 CFR 1.27. ☐ b. Applicant is no longer claiming SMALL ENTITY status. See 37 CFR 1.27(g)(2).

NOTE: The Issue Fee and Publication Fee (if required) will not be accepted from anyone other than the applicant; a registered attorney or agent; or the assignee or other party in interest as shown by the records of the United States Patent and Trademark Office.

Authorized Signature /dexter chin/ Date 09/25/2011
Typed or printed name Dexter CHIN Registration No. 38,842

This collection of information is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450.

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PART B - FEE(S) TRANSMITTAL

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CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

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31366 7590 07/12/2011
HORIZON IP PTE LTD
7500A Beach Road
#04-306/308 The Plaza
SINGAPORE 199591,
SINGAPORE

Certificate of Mailing or Transmission

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Nannan CHEN	(Depositor's name)
/nannan chen/	(Signature)
09/25/2011	(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
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12/390,509	02/23/2009	Junwen LIU	CSM P 2009 NAT 03 US 0	5260
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TITLE OF INVENTION: HIGH VOLTAGE DEVICE

APPLN. TYPE	SMALL ENTITY	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	NO	\$1510	\$300	\$0	\$1810	10/12/2011

EXAMINER	ART UNIT	CLASS-SUBCLASS
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☐ "Fee Address" indication (or "Fee Address" Indication form PTO/SB/47; Rev 03-02 or more recent) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list

- (1) the names of up to 3 registered patent attorneys or agents OR, alternatively,
(2) the name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed.

1 Horizon IP Pte Ltd
2
3

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document has been filed for recordation as set forth in 37 CFR 3.11. Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE GLOBALFOUNDRIES SINGAPORE PTE. LTD.
(B) RESIDENCE: (CITY and STATE OR COUNTRY) SINGAPORE

Please check the appropriate assignee category or categories (will not be printed on the patent): ☐ Individual ☒ Corporation or other private group entity ☐ Government

4a. The following fee(s) are submitted:

- ☒ Issue Fee
☒ Publication Fee (No small entity discount permitted)
☐ Advance Order - # of Copies

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- ☐ A check is enclosed.
☒ Payment by credit card. Form PTO-2038 is attached.
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- ☐ a. Applicant claims SMALL ENTITY status. See 37 CFR 1.27. ☐ b. Applicant is no longer claiming SMALL ENTITY status. See 37 CFR 1.27(g)(2).

NOTE: The Issue Fee and Publication Fee (if required) will not be accepted from anyone other than the applicant; a registered attorney or agent; or the assignee or other party in interest as shown by the records of the United States Patent and Trademark Office.

Authorized Signature /dexter chin/ Date 09/25/2011
Typed or printed name Dexter CHIN Registration No. 38,842

This collection of information is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450.

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TRANSMITTAL FORM <i>(to be used for all correspondence after initial filing)</i>	Application Number	12/390,509	
	Filing Date	02/23/2009	
	First Named Inventor	Junwen LIU	
	Art Unit	2892	
	Examiner Name	DANG, PHUC T	
Total Number of Pages in This Submission	6	Attorney Docket Number	CSM P 2009 NAT 03 US 0

ENCLOSURES (Check all that apply)		
☒ Fee Transmittal Form ☒ Fee Attached ☐ Amendment/Reply ☐ After Final ☐ Affidavits/declaration(s) ☐ Extension of Time Request ☐ Express Abandonment Request ☐ Information Disclosure Statement ☐ Certified Copy of Priority Document(s) ☐ Reply to Missing Parts/ Incomplete Application ☐ Reply to Missing Parts under 37 CFR 1.52 or 1.53	☐ Drawing(s) ☐ Licensing-related Papers ☐ Petition ☐ Petition to Convert to a Provisional Application ☐ Power of Attorney, Revocation Change of Correspondence Address ☐ Terminal Disclaimer ☐ Request for Refund ☐ CD, Number of CD(s) _____ ☐ Landscape Table on CD	☐ After Allowance Communication to TC ☐ Appeal Communication to Board of Appeals and Interferences ☐ Appeal Communication to TC (Appeal Notice, Brief, Reply Brief) ☐ Proprietary Information ☐ Status Letter ☒ Other Enclosure(s) (please identify below):
Remarks Part B - Issue Fee(s) Transmittal (1 original & 1 copy)		

SIGNATURE OF APPLICANT, ATTORNEY, OR AGENT			
Firm Name	Horizon IP Pte. Ltd.		
Signature	/dexter chin/		
Printed name	Dexter CHIN		
Date	09/25/2011	Reg. No.	38,842

CERTIFICATE OF TRANSMISSION/MAILING			
I hereby certify that this correspondence is being facsimile transmitted to the USPTO or deposited with the United States Postal Service with sufficient postage as first class mail in an envelope addressed to: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450 on the date shown below:			
Signature	/nannan chen/		
Typed or printed name	Nannan CHEN	Date	09/25/2011

This collection of information is required by 37 CFR 1.5. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to 2 hours to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

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The **Privacy Act of 1974 (P.L. 93-579)** requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
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3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (*i.e.*, GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Electronic Patent Application Fee Transmittal

Application Number:	12390509			
Filing Date:	23-Feb-2009			
Title of Invention:	METHOD OF FORMING A HIGH VOLTAGE DEVICE			
First Named Inventor/Applicant Name:	Junwen LIU			
Filer:	Dexter Kam Chin			
Attorney Docket Number:	CSM P 2009 NAT 03 US 0			
Filed as Large Entity				
Utility under 35 USC 111(a) Filing Fees				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Utility Appl issue fee	1501	1	1510	1510
Publ. Fee- early, voluntary, or normal	1504	1	300	300

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				1810

Electronic Acknowledgement Receipt

EFS ID:	11044641
Application Number:	12390509
International Application Number:	
Confirmation Number:	5260
Title of Invention:	METHOD OF FORMING A HIGH VOLTAGE DEVICE
First Named Inventor/Applicant Name:	Junwen LIU
Customer Number:	16162
Filer:	Dexter Kam Chin
Filer Authorized By:	
Attorney Docket Number:	CSM P 2009 NAT 03 US 0
Receipt Date:	25-SEP-2011
Filing Date:	23-FEB-2009
Time Stamp:	23:34:11
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	Credit Card
Payment was successfully received in RAM	\$ 1810
RAM confirmation Number	18666
Deposit Account	502388
Authorized User	CHIN,DEXTER

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

Charge any Additional Fees required under 37 C.F.R. Section 1.20 (Post Issuance fees)

Charge any Additional Fees required under 37 C.F.R. Section 1.21 (Miscellaneous fees and charges)

File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Issue Fee Payment (PTO-85B)	CSMP2009NAT03US0-PartBTransmittal_copy.pdf	90033	no	1
			53bbc38451f5425fdf7e82d7fec6085c0c1e7dd		
Warnings:					
Information:					
2	Issue Fee Payment (PTO-85B)	CSMP2009NAT03US0-PartBTransmittal_org.pdf	90033	no	1
			53bbc38451f5425fdf7e82d7fec6085c0c1e7dd		
Warnings:					
Information:					
3	Miscellaneous Incoming Letter	CSMP2009NAT03US0-TF.pdf	278080	no	2
			33d7fbdb968b059d1fc029d5ab9ee3f050a2318b		
Warnings:					
Information:					
4	Fee Worksheet (SB06)	fee-info.pdf	31866	no	2
			729a55cb59de79d80b2a8d09cb0d7a3bc140f63f		
Warnings:					
Information:					
Total Files Size (in bytes):			490012		

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New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



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 Alexandria, Virginia 22313-1450
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Bib Data Sheet

CONFIRMATION NO. 5260

SERIAL NUMBER 12/390,509	FILING OR 371(c) DATE 02/23/2009 RULE	CLASS 438	GROUP ART UNIT 2892	ATTORNEY DOCKET NO. CSM P 2009 NAT 03 US 0
------------------------------------	---	---------------------	-------------------------------	---

APPLICANTS

Junwen LIU, Singapore, SINGAPORE;
 Purakh Raj VERMA, Singapore, SINGAPORE;
 Yan JIN, Singapore, SINGAPORE;
 Baofu ZHU, Singapore, SINGAPORE;

** CONTINUING DATA *****

** FOREIGN APPLICATIONS *****

IF REQUIRED, FOREIGN FILING LICENSE GRANTED

** 03/04/2009

Foreign Priority claimed ☐ yes ☐ no	STATE OR COUNTRY SINGAPORE	SHEETS DRAWING 15	TOTAL CLAIMS 21	INDEPENDENT CLAIMS 3
35 USC 119 (a-d) conditions met ☐ yes ☐ no ☐ Met after Allowance				
Verified and Acknowledged	Examiner's Signature	Initials		

ADDRESS

AIR MAIL

16162

TITLE

METHOD OF FORMING A HIGH VOLTAGE DEVICE

FILING FEE RECEIVED 1194	FEES: Authority has been given in Paper No. _____ to charge/credit DEPOSIT ACCOUNT No. _____ for following:	☐ All Fees ☐ 1.16 Fees (Filing) ☐ 1.17 Fees (Processing Ext. of time) ☐ 1.18 Fees (Issue) ☐ Other _____ ☐ Credit
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APPLICATION NO.	ISSUE DATE	PATENT NO.	ATTORNEY DOCKET NO.	CONFIRMATION NO.
12/390,509	11/08/2011	8053319	CSM P 2009 NAT 03 US 0	5260

16162 7590 10/19/2011
Horizon IP Pte Ltd.
7500A Beach Road, #04-306/308
The Plaza
Singapore, 199591
SINGAPORE

ISSUE NOTIFICATION

The projected patent number and issue date are specified above.

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (application filed on or after May 29, 2000)

The Patent Term Adjustment is 233 day(s). Any patent to issue from the above-identified application will include an indication of the adjustment on the front page.

If a Continued Prosecution Application (CPA) was filed in the above-identified application, the filing date that determines Patent Term Adjustment is the filing date of the most recent CPA.

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APPLICATION NO. 12/500,620

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number			
	Filing Date			
	First Named Inventor	Guowei ZHANG		
	Art Unit			
	Examiner Name	Not Assigned Yet		
	Attorney Docket Number	CSM P 2009 NAT 03 US 1		

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**INFORMATION DISCLOSURE
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(Not for submission under 37 CFR 1.99)

Application Number		
Filing Date		
First Named Inventor	Guowei ZHANG	
Art Unit		
Examiner Name	Not Assigned Yet	
Attorney Docket Number	CSM P 2009 NAT 03 US 1	

	1	LEONARD RUBIN ET AL., Narrow n+/p+ isolation in retrograde well implants, Solid State Technology Magazine, 01 July 2003, pages 119-125, Volume 46 Issue 7, Solid State Technology, USA.	☐
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(Not for submission under 37 CFR 1.99)

Application Number	
Filing Date	
First Named Inventor	Guowei ZHANG
Art Unit	
Examiner Name	Not Assigned Yet
Attorney Docket Number	CSM P 2009 NAT 03 US 1

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Signature	/dexter chin/	Date (YYYY-MM-DD)	2009-07-10
Name/Print	Dexter CHIN	Registration Number	38,842

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Application Number:

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Title of Invention:

HIGH VOLTAGE DEVICE

First Named Inventor/Applicant Name:

Guowei ZHANG

Filer:

Dexter Kam Chin

Attorney Docket Number:

CSM P 2009 NAT 03 US 1

Filed as Large Entity

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Utility Search Fee	1111	1	540	540
Utility Examination Fee	1311	1	220	220

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Application Number:	12500620
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Title of Invention:	HIGH VOLTAGE DEVICE
First Named Inventor/Applicant Name:	Guowei ZHANG
Customer Number:	31366
Filer:	Dexter Kam Chin
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Attorney Docket Number:	CSM P 2009 NAT 03 US 1
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			e62ab04b8a5d2eaedf68865f045572e5952876d2		
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4	Application Data Sheet	CSMP2009NAT03US1-ADS.pdf	990027	no	4
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5	Power of Attorney	CSMP2009NAT03US1-Decl_POA_copy.pdf	84891	no	3
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6	Information Disclosure Statement (IDS) Filed (SB/08)	CSMP2009NAT03US1-IDS.pdf	758623	no	4
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HIGH VOLTAGE DEVICE

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application is a continuation-in-part application which claims benefit of co-pending U.S. patent application Ser. No. 12/390,509, filed on February 23, 2009. All disclosures are incorporated herewith by reference.

BACKGROUND

[0002] Lateral Double-Diffused (LD) transistors have been widely employed in high voltage applications. For high performance LD transistors, low drain-to-source on-resistance ($R_{ds_{on}}$) is desired to minimize its power dissipation when it is turned on, as well as high breakdown voltage to maximize its voltage capability. To achieve low $R_{ds_{on}}$, the channel of the high LD transistor should be as short as possible.

[0003] As technology evolves into era of deep sub-micron (e.g., beyond 0.35 μm) Very-Large-Scale Integration (VLSI), there is a desire for both high voltage (HV) LD transistors and low voltage (LV) transistors to be fabricated on the same substrate. Generally, the processes for forming LV devices, such as complementary metal-oxide-semiconductor (CMOS) processes, are used to form the HV devices. However, conventional CMOS processes are incompatible for forming HV transistors. For example, process overlay issues make it difficult for alignment of the HV channel well, which defines the channel length. Process variations in process overlay requires large process windows. This makes forming short channel lengths to achieve low $R_{ds_{on}}$ difficult if not impossible. Furthermore, the thin gate electrodes used prevent them serving as a hard mask, further exacerbating process control in forming the channel well.

[0004] From the foregoing discussion, it is desirable to provide reliable HV devices formed with short channel lengths to have low $R_{ds_{on}}$.

SUMMARY

[0005] A method of forming a device is presented. The method includes providing a substrate prepared with an active device region. The active device region includes gate stack layers of a gate stack including at least a gate electrode layer over a gate dielectric layer. A first mask is provided on the substrate corresponding to the gate. The substrate is patterned to at least remove portions of a top gate stack layer unprotected by the first mask. A second mask is also provided on the substrate with an opening exposing a portion of the first mask and the top gate stack layer. A channel well is formed by implanting ions through the opening and gate stack layers into the substrate.

[0006] In another embodiment, another method of forming a device is disclosed. The method includes providing a substrate prepared with an active device region. The active device region includes a doped drift well in a first portion of the active device region and gate stack layers of a gate stack. The gate stack layers include at least a gate electrode layer over a gate dielectric layer on the surface of the substrate. A first mask is provided on the substrate corresponding to the gate. The substrate is patterned to at least remove portions of a top gate stack layer unprotected by the first mask. A second mask is also provided on the substrate with an opening exposing a portion of the first mask and the top gate stack layer. A channel well is formed by implanting ions through the opening and gate stack layers into the substrate.

[0007] In yet another embodiment, a device is presented. The device includes a substrate prepared with an active device region. The active device region includes a gate stack with a gate electrode layer over a gate dielectric layer. At least the gate electrode layer is patterned using a hard mask. The device also includes a doped channel well disposed in the substrate adjacent to the first edge of the gate. The first edge of the gate overlaps the channel well with a channel edge of the channel well beneath the gate. The first edge of the gate and channel edge define an

effective channel length of the device. A doped drift well adjacent to the second edge of the gate is also included.

[0008] These and other objects, along with advantages and features of the present invention herein disclosed, will become apparent through reference to the following description and the accompanying drawings. Furthermore, it is to be understood that the features of the various embodiments described herein are not mutually exclusive and can exist in various combinations and permutations.

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] In the drawings, like reference characters generally refer to the same parts throughout the different views. Also, the drawings are not necessarily to scale, emphasis instead generally being placed upon illustrating the principles of the invention. In the following description, various embodiments of the present invention are described with reference to the following drawings, in which:

[0010] Fig. 1 shows an embodiment of a device;

[0011] Figs. 2a-g show an embodiment of a process for forming a device;

[0012] Figs. 3a-g show an alternative embodiment of a process for forming a device;

[0013] Figs. 4a-g show another embodiment of a process for forming a device; and

[0014] Figs. 5a-g show another embodiment of a process for forming a device.

DETAILED DESCRIPTION

[0015] Embodiments generally relate to semiconductor devices or integrated circuits. More particularly, some embodiments relate to high power devices. For example, high power devices include lateral double-diffused (LD) transistors, such as metal oxide transistors (MOS). The high power devices can be employed as switching voltage regulators for power management applications. The LD transistors can be easily integrated into devices or ICs. The devices or ICs

can be incorporated into or used with, for example, consumer electronic products, particularly portable consumer products, such as cell phones, laptop computers and personal digital assistants (PDAs).

[0016] Fig. 1 shows a portion 100 of an embodiment of a device. As shown, the portion includes a substrate 101. The substrate, for example, comprises a silicon substrate. The substrate can be lightly doped with p-type dopants. N-type or other types of substrates, such as silicon germanium or silicon-on-insulator (SOI), are also useful.

[0017] The substrate may be prepared with first and second regions 110 and 160. In one embodiment, the first region comprises a low voltage (LV) device region while the second region comprises a high voltage (HV) device region. The regions may be other types of device regions or additional device regions may be provided. For example, a dual gate oxide (DGO) region may be provided for DGO devices. The DGO devices may be used for I/O circuitries. Alternatively, the device may include only a HV region.

[0018] The first region, in one embodiment, comprises first and second type active sub-regions 112 and 116. The first and second types are complementary types, forming a complementary type device. For example, the complementary type device comprises a complementary metal oxide semiconductor (CMOS) device. Forming non-complementary or other types of devices is also useful.

[0019] The first type active sub-region, for example, comprises a n-type active sub-region and the second type active sub-region comprises a p-type active region. The first type active sub-region comprises a second type doped well 122; the second type active sub-region comprises a first type doped well 124. For example, the first type active sub-region comprises a p-type doped well and the second type active sub-region comprises a n-type doped well. Other configurations of active sub-regions are also useful. P-type dopants can include boron (B),

indium (In) or a combination thereof while n-type dopants can include phosphorus (P), arsenic (As), antimony (Sb) or a combination thereof.

[0020] A first type transistor 140 is provided in the first type active sub-region and a second type transistor 150 is provided in the second type active sub-region. For example, a n-type transistor is provided in the n-type active sub-region with a p-type doped well while a p-type transistor is provided in the p-type active sub-region with a n-type doped well. The transistors, in one embodiment, are MOS field effect transistors (MOSFETs). Other types of transistors may also be useful.

[0021] A transistor, for example, comprises a gate having a gate electrode 144 over a gate dielectric 142. The gate electrode, for example, comprises polysilicon. Other types of gate electrode materials, such as polysilicide or metal silicide, are also useful. Polysilicide or metal silicide gates can be formed by reacting the desired metal with polysilicon. Various types of metal, such as cobalt, titanium or nickel, can be used. Other types of gate electrode materials may also be useful. Furthermore, the gate electrode may be doped with dopants. Doping the gate electrode may depend on the technology. The gate electrode, for example, may be doped with the same dopant type of the transistor type. Doping the gate electrode with other dopant types is also useful. The gate dielectric can be silicon oxide. Other types of gate electrode or gate dielectric materials can also be useful. The material of the gate electrode and gate dielectric may depend on the technology. For example, cobalt silicide gate electrodes may be used for 0.18 μm and 0.13 μm technologies while nickel silicide may be used for 0.09 μm technologies. Other configurations of gate electrode and gate dielectric materials are also useful.

[0022] Sidewalls of the gate can be provided with sidewall spacers 148. The sidewall spacers, for example, comprise a dielectric material, such as silicon oxide, silicon nitride or a combination thereof. Other types of materials can be used for the sidewall spacers.

[0023] Adjacent to the gate are diffusion regions 146. The diffusion regions serve as, for example, source and drain of the transistor. Extension diffusion regions 147 can be provided. The extension diffusion regions, for example, are shallow diffusion regions which extend under the spacers. The diffusion regions (including the extension diffusion regions) of the first type transistor are doped with the first type dopants and the diffusion regions of the second type transistor are doped with the second type dopants. For example p-type diffusion regions are provided for p-type transistors and n-type diffusion regions are provided for n-type transistors. The depth and dopant concentration of the diffusion regions may depend, for example, on the application, such as voltage requirements.

[0024] The HV region includes a HV active region 165. In one embodiment, the HV active region comprises a first type active region. For example, the HV active region comprises a n-type active region. Providing a p-type HV active region is also useful. Furthermore, it is understood that the HV region is depicted with only one active region for purpose of simplification and that the HV region can include numerous HV active regions or sub-regions. For example, the HV active sub-regions can be complementary types of HV sub-regions for complementary types of HV devices.

[0025] Isolation regions 180, such as shallow trench isolation (STI) regions are provided to isolate the LV and HV regions as well as active sub-regions within the regions. Other types of isolation regions are also useful.

[0026] In one embodiment, the first type HV active region comprises at least first and second portions. The first portion comprises a second type HV channel (or body) well 135 and the second portion comprises a first type HV drift well 136. The HV drift well, for example, defines the type of HV device. For example, a p-type HV active region would have a n-type HV channel well and a p-type HV drift well. In one embodiment, a third or middle portion separates the

channel and drift wells. In other embodiments, the first and second portions are contiguous portions. For example, the HV channel and HV drift wells contact each other.

[0027] A deep well 132 may be provided in the HV active region, encompassing the HV channel and drift wells. Generally, the deep well comprises dopants of the first type or same as the HV device type. The deep well may be used to isolate the drift well from the substrate. For example, a p-type device will have a n-type deep well. In some cases, the deep well can be optional or the deep well can comprise dopants of the second or opposite type of the HV device type. For example, in the case of a n-type HV device, the deep well can be a p-type or n-type deep well. The different types of deep well selected may depend on the type of substrate. Generally, p-type doped substrates are used for fabricating devices. When p-type substrates are used, the n-type deep well may be optional. This is because the n-type drift well is already different from the p-type substrate. For a n-type substrate, a p-type deep well may be used to isolate the channel well from the substrate. Other configurations of deep wells may also be useful.

[0028] As shown, a first type HV device is provided in the first type HV active region. The first type HV device, in one embodiment, comprises a LD device, such as a LDMOS device. The HV device comprises a gate with first and second edges 179a-b or sidewalls.

[0029] The gate includes a HV gate electrode 174 over a HV gate dielectric 172. The HV gate electrode, for example, comprises polysilicon. Other types of HV gate electrode materials, such as polysilicide or metal silicide, are also useful. Various types of metal, such as cobalt, titanium or nickel, can be used. Other types of HV gate electrode materials may also be useful. Furthermore, the gate electrode may be doped with dopants. The HV gate electrode, for example, may be doped with the same dopant type of the transistor type. Doping the HV gate electrode with other dopant types is also useful. The HV gate dielectric can be silicon oxide. Other types of HV gate dielectric materials can also be useful.

[0030] In one embodiment, the material of the HV gate electrode and HV gate dielectric may depend on the CMOS process and/or technology. In one embodiment, the material of the HV gate electrode and HV gate dielectric may depend on the CMOS process for LV devices and/or technology. For example, cobalt silicide gate electrodes may be used for 0.18 μm and 0.13 μm technologies while nickel silicide may be used for 0.09 μm technologies. Other configurations of gate electrode and gate dielectric materials are also useful.

[0031] Sidewalls of the gate can be provided with sidewall spacers 178. The sidewall spacers, for example, comprise a dielectric material, such as silicon oxide, silicon nitride or a combination thereof. Other types of materials can be used for the sidewall spacers.

[0032] In one embodiment, the various gate elements of the HV gate, such as gate electrode, gate dielectric and spacers are of the same type of material as the LV devices in the LV region. This facilitates process compatibility for forming HV and LV devices.

[0033] The HV gate overlaps the channel and drift wells in the first and second portions of the HV active region. A drift isolation region 180a may be provided within the drift well 136 on the high voltage side of the HV device. The drift isolation region 180a can improve breakdown voltage of the HV device due to high voltage. For example, the drift isolation region is particularly useful to sustain high voltages for applications beyond about 10V. As shown, the drift isolation region 180a is disposed in the drift well 136 under the gate. Disposing the drift well at other positions, such as partially underlying the gate may also be useful.

[0034] A first type drain diffusion region 176b is provided between the drift isolation region 180a and isolation region 180 which isolates the HV active region from other sub-active regions in the HV region. A first type source diffusion region 176a is provided in the channel well 135 of the first type HV active region. A source extension region 177 can be provided which extends under the spacer on the channel side of the HV gate. The first type source and drain diffusion regions, for example, comprise a p-type diffusion regions for a p-type HV active region.

Providing n-type diffusion regions for a n-type HV active region is also useful. The dopant depth and concentration of the diffusion regions may be selected depending on the application. In one embodiment, the dopant depth and concentration of diffusion regions may be the same as those of LV devices. Other dopant depths and concentrations are also useful.

[0035] A second type body contact region 171 may be provided in the first portion of the HV active region. As shown, the second type body contact region 171 is between the source diffusion region 176a and the isolation region 180. The second type body contact region comprises second type dopants. In one embodiment, dopant depth and concentration of the body contact region may be the same as the diffusion regions of the LV devices. Other dopant depths and concentrations are also useful. The body contact region 171 provides electrical coupling to the channel well 135.

[0036] An edge 133 of the channel well 135 extends below the gate on the source side of the gate. The edge 133, for example, can be referred to as the channel edge of the channel well. The channel edge of the channel well 135 and the first edge 179a of the gate on the source side define a channel of the HV transistor having an effective channel length L.

[0037] The effective channel length, in one embodiment, is controlled by ion implantation which is self-aligned to the gate edge. For example, dopant ions are implanted using an implant mask through the gate electrode or a hard mask and the gate electrode. In one embodiment, the implant mask is also used to pattern the channel edge of the HV gate. The effective channel length may be controlled by channel or body implant conditions selected to satisfy threshold voltage and breakdown voltage requirements. For example, shorter effective channel length may be achieved by decreasing tilt angle, implant energy or dosage and vice-versa.

[0038] With the channel well being self-aligned to the gate edge of the implant, process overlay issues are avoided. Very short effective channel lengths with very low $R_{ds(on)}$ can be achieved, enabling high current to pass through a small area. This is particularly useful for

technologies beyond 0.35 μm , such as, for example, 0.25 μm , 0.18 μm and 0.13 μm technologies.

[0039] Smaller area requirements for power management applications with reduced power dissipation increases performance while reducing costs.

[0040] In one embodiment, the effective channel length L is less than 0.4 μm . In other embodiments, the effective channel length is less than 0.3 μm . In yet another embodiment, the effective channel length is less than 0.25 μm . Other effective channel lengths may also be useful, for example, depending on the application.

[0041] Figs. 2a-g show cross-sectional views of an embodiment of a process for forming a device or IC 100. Referring to Fig. 2a, a substrate 101 is provided. The substrate can comprise a silicon substrate, such as a lightly p-type doped substrate. Other types of substrates, including n-type doped substrates, silicon germanium or silicon-on-insulator (SOI), are also useful.

[0042] As shown, first and second device regions 110 and 160 are defined on the substrate. The first region, for example, comprises a LV device region while the second region comprises a HV device region. Other types of device regions or additional device regions may be provided. Alternatively, the device may include only a HV region.

[0043] The LV device region 110 can be defined with first and second type active sub-regions 112 and 116. In one embodiment, the first and second type active sub-regions are complementary type active sub-regions to form a complementary device such as a CMOS device. Forming non-complementary or other types of devices is also useful.

[0044] As for the HV device region 160, a HV active region 165 is defined therein. In one embodiment, the HV active region 165 can be a first or second type HV active region. Although only one HV active region is depicted, it is understood that the HV device region may include additional HV active regions or sub-regions. The HV active sub-regions can be complementary HV active sub-regions.

[0045] The substrate is also prepared with isolation regions 180 to separate the device regions as well as other active device sub-regions. In one embodiment, the isolation regions 180 comprise STIs. Various conventional processes can be employed to form the STI regions. For example, the substrate can be etched using conventional etch and mask techniques to form trenches which are then filled with dielectric materials such as silicon oxide. Chemical mechanical polishing (CMP) can be performed to remove the excess dielectric materials and provide a planar substrate top surface. Other processes or materials can also be used to form the STIs.

[0046] In Fig. 2b, the doped wells for the active regions of the device regions are formed. In one embodiment, a first type doped well 124 is formed in the second type LV active sub-region 116 and a second type doped well 122 is formed in the first type LV active sub-region 112. For example, a n-type doped well is formed in the p-type active sub-region and a p-type doped well is formed in the n-type active sub-region.

[0047] In the HV active region 165, it includes at least first and second portions. For example, the first portion corresponds to a drain portion and the second portion corresponds to a source portion. In some embodiments, a third or middle portion is provided to separate the first and second portions. Alternatively, the first and second portions are contiguous portions without a middle portion.

[0048] In one embodiment, a doped well 136 is formed in the drain portion of the HV active region. The doped well, for example, is a drift well of the HV device. The drift well comprises the same type as the HV active region. In the case of a first type HV active region, the drift well comprises a first type. For example, a n-type drift well is provided for a n-type HV active region.

[0049] A drift STI region 180a may be optionally provided in the drift well 136. The drift STI region is disposed in the drift well to improve reliability of high voltage devices. For

example, the drift STI region improves breakdown reliability of the HV device. The drift STI region can be formed at the same time as the other STI regions.

[0050] A deep HV well 132 may be provided in the HV active region 165. As shown, the deep HV well 132 is formed in the whole HV active region 165 and encompasses the first and second portions of the HV active region 165. The deep HV well comprises an opposite type as the drift well and the HV active region. In the case of a first type HV active region, the deep well comprises a second type. For example, a p-type deep well is provided for a n-type HV active region.

[0051] The depth and concentration of dopants of the various wells may depend on, for example, application. For example, higher voltage applications may require lower doping concentration of dopants and greater depth of the wells. In one embodiment, the depth of the deep well 132 is about 3 μm with a dopant concentration of about $5 \times 10^{16} \text{cm}^{-3}$. For the drift well 136, it may have a depth of about 1.5 μm with a dopant concentration of about $5 \times 10^{17} \text{cm}^{-3}$. Other depths and dopant concentrations are also useful for the wells.

[0052] As described, the STI regions are formed prior to forming the various doped wells. Other process schemes, such as forming the STI regions after the various doped wells are formed may also be useful.

[0053] The doped wells can be formed by ion implantation. P-type dopants can include boron (B), BF_2 (boron compounded with fluorine), indium (In) or a combination thereof while n-type dopants can include phosphorus (P), arsenic (As), antimony (Sb) or a combination thereof. Generally, the first and second types of the active regions are formed selectively in separate processes. For example, the n-type regions can be implanted with n-type dopants while an implant mask prevents doping of the p-type regions.

[0054] In one embodiment, the wells of the HV region are formed separately from the LV regions. For example, the LV wells can be formed first followed by the HV wells. Forming the

HV wells prior to the LV wells is also useful. The different types of LV wells can be formed in separate processes and the different types of HV wells can also be formed in separate processes. For example, the first type and second type wells in the LV region are formed in separate implant processes. Similarly, the different types of wells in the HV region are formed in separate implant processes.

[0055] A doped well can be formed by a single implant process or multiple implant processes. In multiple implant processes, dopants can be implanted at different energy levels to achieve the desired doping profiles. In the case of a single implant process, a high temperature drive-in process is performed to achieve the desired doping profile.

[0056] A single implant process may be employed to, for example, form deep wells. For a n-type deep well, phosphorus may be implanted with a dopant concentration of about $6 \times 10^{12} \text{ cm}^{-2}$ at an energy level of about 2000 KeV followed by a two-hour drive-in at a temperature of about 1100 °C. Single implant processes may also be used to form other types of wells. Forming wells using multiple implant processes may also be useful. For example, LV wells may be formed by multiple implant processes.

[0057] In Fig. 2c, various layers of the gate are formed on the substrate. In one embodiment, a gate dielectric layer 242 is formed over the substrate. The gate dielectric layer, for example, comprises silicon oxide. Other types of dielectric materials, such as silicon oxynitride, may also be useful. Alternatively, high-k, low-k or composite of dielectric materials may be used. The thickness of the gate dielectric layer 242 may be about 30 Å for 1.8V gate voltage transistors. Other thicknesses may also be useful. The thickness, for example, may depend on the gate voltage applications. For example, higher gate voltages may require thicker gate dielectric layers. In one embodiment, the gate dielectric layer is formed by thermal oxidation. Other techniques, such as chemical vapor deposition (CVD), can also be used to form the gate dielectric layer.

[0058] A gate electrode layer 244 is deposited on the gate dielectric layer 242. The gate electrode layer comprises, in one embodiment, polysilicon (poly). The gate electrode layer 244 can be formed as an amorphous or non-amorphous layer. For amorphous deposited layer, subsequent processing can be performed to crystallize it. Other types of gate electrode materials are also useful. For example, the polysilicon can be subsequently processed to form polysilicide or metal gates. The thickness of the gate electrode layer may be about 2000 Å or less. In another embodiment, the thickness of the gate electrode layer is about 2500 Å or less. In yet another embodiment, the thickness of the gate electrode layer can be about 4000 Å or less. For example, in the case of polysilicide gates, the polysilicon may be about 2000 Å while the tungsten is about 2000 Å. Other thicknesses are also useful. Various techniques can be used to form the gate electrode layer. For example, polysilicon can be deposited by CVD while metal can be deposited by sputtering. Other techniques, depending on the material, may also be useful.

[0059] Referring to Fig. 2d, a mask layer 288 is formed on the substrate, covering the gate electrode layer. In one embodiment, the mask layer comprises photoresist. The mask is patterned as desired. To pattern the mask layer, photolithography can be employed. For example, the mask can be selectively exposed to an exposure source through a lithographic mask. Depending on whether a positive or negative resist is used, the exposed or unexposed portions are removed by development. To enhance lithographic resolution, an ARC layer (not shown) may be provided beneath the mask layer.

[0060] The mask layer 288 is patterned to form an opening 289 to expose a portion of the gate electrode layer 244. In one embodiment, the opening corresponds to the opening for the channel well implant. For example, the mask layer serves as the channel well implant mask. The ARC layer may be patterned with the mask layer. Subsequently patterning the ARC layer may also be useful.

[0061] In Fig. 2e, the substrate is implanted with dopants to form a channel well 135. In one embodiment, dopants of the opposite type as the type of the HV active region are implanted. For example, p-type dopants are implanted into the p-type deep well of a n-type HV active region. In one embodiment, the dopants are implanted at an angle to form a channel from the gate edge to the edge of the channel well. The implant angle θ can be in a range of about 1 to 45°. Other implant angles may also be useful. The implant conditions can be varied depending on application requirements. For example, the angle, dose and energy can be selected to achieve the desired effective channel length.

[0062] In one embodiment, the channel well is formed by multiple implants. For example, the channel well can be formed by at least two tilt and rotate implants. In one embodiment, the implant comprises a quad implant. A quad implant comprises 4 tilted angled implants, each rotated by a rotation angle. For example, a p-type channel well can be formed with a quad implant at a tilt angle of about 30° with rotation angle of about 45° with a boron dose of about $2 \times 10^{13} \text{cm}^{-2}$ at an energy level of about 130 KeV. Other tilt angles, rotation angles, doses and energy levels are also useful. For example, a quad implant can comprise a tilt angle of about 7° with rotation angle of about 45° with a boron dose of about $2 \times 10^{13} \text{cm}^{-2}$ at an energy level of about 150 KeV.

[0063] After forming the channel doped well 135, exposed portions of the gate electrode layer 244 are removed, as shown in Fig. 2f. In one embodiment, an anisotropic etch, such as a reactive ion etch (RIE) is performed to remove exposed portions of the gate electrode layer. The removal of the exposed portions of the gate electrode layer, in one embodiment, forms a first edge of a HV gate. The first edge, in one embodiment, corresponds to the source side of the HV gate. As shown, the gate dielectric layer 242 serves as an etch stop for etching the gate electrode layer. Leaving the gate dielectric layer can be advantageous since it can serve as an implant mask for subsequent ion implantation to protect the substrate from implant damage.

Alternatively, the exposed portion of the gate dielectric layer can be removed. Removing the gate dielectric layer 242 to expose the substrate may also be useful. After etching the gate electrode layer, the mask layer 288 is removed.

[0064] As described, the implant mask, in one embodiment, serves as a channel well implant mask as well as a mask for patterning the first edge (e.g., source side) of the HV gate. By using the same mask to form the channel well and source side of the gate, the channel is self-aligned to the gate edge. This enables the channel length of the HV device to be well controlled to produce very short effective channel lengths to reduce $R_{ds_{on}}$ performance. In one embodiment, the effective channel length L is less than $0.4\ \mu\text{m}$. In other embodiments, the effective channel length is less than $0.3\ \mu\text{m}$. In yet another embodiment, the effective channel length is less than $0.25\ \mu\text{m}$.

[0065] Referring to Fig. 2g, another mask layer 288b is formed on the substrate and patterned. The mask layer is patterned to expose portions of the gate electrode layer 244 to be removed, protecting portions corresponding to gates of the LV devices. For example, the mask serves as a gate mask. Additionally, the mask layer 288b protects the source portion of the HV region and the gate of the HV device. The exposed portions of the gate electrode layer 244 are removed to form gates of both the LV and HV devices.

[0066] After the gates are formed, the process continues to form the device, as illustrated in Fig. 1. The process, for example, includes forming extension regions 147 by ion implantation, followed by forming spacers 148 and 178 and diffusion regions 146 and 176. The different types of extension and diffusion regions can be formed in different processes. For example, n-type diffusion regions are formed in one implant process and p-type diffusion regions are formed in another process. Furthermore, the diffusion and extension regions of the different device regions can be formed separately. Silicide gates and contacts can be formed after the diffusion regions are formed. For example, metals, such as cobalt are deposited on the substrate 101 and reacted

to form silicide contacts and gates. The contacts and gates can be formed in the same or different processes. Unreacted metals are removed. A pre-metal and an inter-level dielectric layer can be formed in which contacts and interconnects are formed. Additional interconnect levels can be formed by, for example, dual damascene techniques. Final passivation, dicing, assembly and testing may be performed to complete the IC.

[0067] An alternative embodiment of a process of forming a device 100 is shown in Figs. 3a-g. Referring to Fig. 3a, a partially processed device is shown. The partially processed device is similar to that shown in Fig. 2c. A hard mask 388 is formed on the gate electrode layer 244. The hard mask, for example, comprises silicon oxide. In one embodiment, the hard mask 388 is formed by CVD. The thickness of the hard mask, for example, is about 40nm. Other materials, techniques or thicknesses may also be useful. For example, the hard mask may be formed from silicon nitride or other types of materials.

[0068] Referring to Fig. 3b, a soft mask 287, such as photoresist, is formed on the hard mask 388. The soft mask can be patterned by photolithographic techniques to form an opening 289 which can serve as a channel implant mask. An ARC layer can be provided between the soft mask 287 and the hard mask 388. The pattern of the soft mask is transferred to the hard mask by, for example, RIE.

[0069] In one embodiment, the soft mask layer is removed after patterning of the hard mask, as shown in Fig. 3c. The substrate is implanted with dopants to form a channel doped well 135. In one embodiment, dopants of the opposite type as the type of the HV active region are implanted. For example, p-type dopants are implanted into the p-type deep well of a n-type HV active region.

[0070] In one embodiment, the dopants are implanted at an angle to form a channel from the gate edge to the edge of the channel well. The implant angle θ can be in a range of about 1 to 45°. Other implant angles may also be useful. The implant conditions can be varied depending

on application requirements. For example, the angle, dose and energy can be selected to achieve the desired effective channel length.

[0071] In one embodiment, the channel well is formed by multiple implants. For example, the channel well can be formed by at least two tilt and rotate implants. In one embodiment, the implant comprises a quad implant. A quad implant comprises 4 tilted angled implants, each rotated by a rotation angle. For example, a p-type channel well can be formed with a quad implant at a tilt angle of about 30° with rotation angle of about 45° with a boron dose of about $2 \times 10^{13} \text{cm}^{-2}$ at an energy level of about 130 KeV. Other tilt angles, rotation angles, doses and energy levels are also useful. For example, a quad implant can comprise a tilt angle of about 7° with rotation angle of about 45° with a boron dose of about $2 \times 10^{13} \text{cm}^{-2}$ at an energy level of about 150 KeV.

[0072] As shown in Fig. 3d, another soft mask layer 287 is deposited on the substrate, covering the hard mask layer and exposed gate electrode layer. The soft mask is patterned by lithography. The soft mask protects portions of the hard mask layer 388 corresponding to the transistor gates. A new ARC layer can be provided between the soft mask 287 and the hard mask 388. In Fig. 3e, exposed portions of the hard mask are removed, exposing portions of the gate electrode layer. Patterning of the hard mask layer can be achieved by, for example, RIE. The patterned hard mask serves as a gate mask. The soft mask, as shown in Fig. 3f, is removed after patterning the hard mask.

[0073] Referring to Fig. 3g, the gate electrode layer 244 is patterned to form gates in the LV and HV region by, for example etch. Patterning of the gate electrode layer can be achieved using, for example, RIE. The patterning of the gate electrode layer also erodes the hard mask, reducing its thickness. The hard mask 388 can be removed by, for example, a clean step after etching the gate electrode layer. The process continues as previously described to complete the device.

[0074] Another embodiment of a process for forming a device 100 is shown in Figs. 4a-g. Referring to Fig. 4a, a partially processed device is shown. The partially processed device is similar to that shown in Fig. 3a.

[0075] Referring to Fig. 4b, a soft mask layer 387, such as a photoresist layer, is formed on the hard mask 388. An ARC layer can be provided between the soft mask and the hard mask. The soft mask layer is patterned to expose portions of the hard mask 388. The remaining portion of the soft mask corresponds to the HV gate in the HV region 160. Patterning of the soft mask layer can be achieved by photolithographic techniques. In one embodiment, the patterned soft mask layer also exposes the hard mask in the LV region 110.

[0076] As shown in Fig. 4c, the pattern of the soft mask is transferred to the hard mask by, for example, RIE. The RIE removes exposed portions of the hard mask to expose portions of the gate electrode layer 344 in the HV region which are to be removed. In one embodiment, the removal of the hard mask also exposes the gate electrode layer in the LV region. After patterning of the hard mask, the soft mask layer is removed.

[0077] In Fig. 4d, another soft mask layer 487, such as a photoresist layer, is formed over the substrate. An ARC layer may be provided beneath the soft mask layer. The soft mask layer can be patterned by photolithographic techniques to expose portions of the gate electrode layer 344 in the LV region which are to be removed, protecting portions corresponding to gates of the LV devices. For example, the patterned soft mask serves as a gate mask for the LV region. The soft mask layer is removed from the HV region.

[0078] The gate electrode layer is patterned in Fig. 4e. In one embodiment, the gate electrode layer in the LV region is patterned using the soft mask while the hard mask is used in the HV region. The use of the soft mask advantageously facilitates forming gates of critical dimensions in the LV region.

[0079] Referring to Fig. 4f, another soft mask layer 487b, such as photoresist, is formed over the substrate. An ARC layer may be provided beneath the soft mask. The soft mask layer can be patterned by photolithographic techniques to form an opening 489. The patterned soft mask layer serves as a channel implant mask.

[0080] As shown in Fig. 4g, the substrate is implanted with dopants to form a channel doped well 135. In one embodiment, dopants of the opposite type as the type of the HV active region are implanted. For example, p-type dopants are implanted into the p-type deep well of a n-type HV active region.

[0081] In one embodiment, the dopants are implanted at an angle to form a channel from the gate edge to the edge of the channel well. The implant angle θ can be in a range of about 1 to 45°. Other implant angles may also be useful. The implant conditions can be varied depending on application requirements. For example, the angle, dose and energy can be selected to achieve the desired effective channel length.

[0082] In one embodiment, the channel well is formed by multiple implants. For example, the channel well can be formed by at least two tilt and rotate implants. In one embodiment, the implant comprises a quad implant. A quad implant comprises 4 tilted angled implants, each rotated by a rotation angle. For example, a p-type channel well can be formed with a quad implant at a tilt angle of about 30° with rotation angle of about 45° with a boron dose of about $2 \times 10^{13} \text{cm}^{-2}$ at an energy level of about 130 KeV. Other tilt angles, rotation angles, doses and energy levels are also useful. For example, a quad implant can comprise a tilt angle of about 45° with rotation angle of about 45° with a boron dose of about $2 \times 10^{13} \text{cm}^{-2}$ at an energy level of about 150 KeV.

[0083] In one embodiment, the implant mask and hard mask are removed after forming the channel well. For example, the implant mask is first removed followed by the removal of the

hard mask. In some embodiments, the hard mask is remained. The process continues as previously described to complete the device after the formation of the gates.

[0084] An alternative embodiment of a process for forming a device 100 is shown in Figs. 5a-g. Referring to Fig. 5a, a partially processed device is shown. The partially processed device is similar to that shown in Fig. 4c.

[0085] As shown in Fig. 5b, a soft mask layer 587, such as a photoresist layer, is formed over the substrate. An ARC layer may be provided beneath the gate soft mask layer. The soft mask layer can be patterned by photolithographic techniques to expose the HV region, protecting the LV devices from being processed. As illustrated by Fig. 5c, the gate electrode in the HV region is patterned by, for example, RIE, to form the gate using the hard mask as the gate mask. After patterning of the gate electrode in the HV region, the soft mask layer is removed.

[0086] Referring to Fig. 5d, another soft mask layer 587b, such as photoresist, is formed over the substrate. An ARC may be provided beneath the soft mask layer. The soft mask layer can be patterned by photolithographic techniques to form an opening 589. The patterned soft mask layer serves as a channel implant mask.

[0087] As shown in Fig. 5e, the substrate is implanted with dopants to form a channel doped well 135. In one embodiment, dopants of the opposite type as the type of the HV active region are implanted. For example, p-type dopants are implanted into the p-type deep well of a n-type HV active region.

[0088] In one embodiment, the dopants are implanted at an angle to form a channel from the gate edge to the edge of the channel well. The implant angle θ can be in a range of about 1 to 45°. Other implant angles may also be useful. The implant conditions can be varied depending on application requirements. For example, the angle, dose and energy can be selected to achieve the desired effective channel length.

[0089] In one embodiment, the channel well is formed by multiple implants. For example, the channel well can be formed by at least two tilt and rotate implants. In one embodiment, the implant comprises a quad implant. A quad implant comprises 4 tilted angled implants, each rotated by a rotation angle. For example, a p-type channel well can be formed with a quad implant at a tilt angle of about 30° with rotation angle of about 45° with a boron dose of about $2 \times 10^{13} \text{cm}^{-2}$ at an energy level of about 130 KeV. Other tilt angles, rotation angles, doses and energy levels are also useful. For example, a quad implant can comprise a tilt angle of about 45° with rotation angle of about 45° with a boron dose of about $2 \times 10^{13} \text{cm}^{-2}$ at an energy level of about 150 KeV.

[0090] Referring to Fig. 5f, another layer of soft mask layer 587c, such as photoresist, is formed over the substrate. An ARC layer may be provided beneath the soft mask layer. The soft mask layer is patterned by photolithographic techniques to expose portions of the gate electrode layer 344 in the LV region to be removed. Additionally, the soft mask layer remains in the HV region to protect it from being processed. The exposed portions of the gate electrode layer 344 are removed to form the gates of the LV device, as illustrated in Fig. 5g.

[0091] Similar to the embodiment of Figs. 4a-g, the LV gates are patterned using a soft mask while a hard mask is used to pattern the HV gate. After forming the LV gates, the soft mask is removed followed by removal of the hard mask over the HV gate. In some embodiments, the hard mask can remain. The process continues to complete the device as described previously.

[0092] The invention may be embodied in other specific forms without departing from the spirit or essential characteristics thereof. The foregoing embodiments, therefore, are to be considered in all respects illustrative rather than limiting the invention described herein. Scope of the invention is thus indicated by the appended claims, rather than by the foregoing description, and all changes that come within the meaning and range of equivalency of the claims are intended to be embraced therein.

[0093] What is claimed is:

CLAIMS

1. A method of forming a device comprising:
 - providing a substrate prepared with an active device region, wherein the active device region includes gate stack layers of a gate stack comprising at least a gate electrode layer over a gate dielectric layer;
 - providing a first mask on the substrate corresponding to the gate;
 - patterning the substrate to at least remove portions of a top gate stack layer unprotected by the first mask;
 - providing a second mask on the substrate with an opening exposing a portion of the first mask and the top gate stack layer; and
 - implanting ions through the opening and gate stack layers into the substrate to form a channel well.
2. The method of claim 1 wherein the top gate stack layer comprises the gate electrode layer of the gate stack.
3. The method of claim 2 wherein the gate electrode layer comprises silicon, including amorphous or polysilicon.
4. The method of claim 2 wherein the first mask comprises a hard mask.
5. The method of claim 4 wherein a first edge of the gate corresponds to a channel edge of the gate of the device and a second edge of the gate corresponds to the drain edge of the gate of the device.

6. The method of claim 1 wherein the first edge of the gate corresponds to a channel edge of the gate of the device and the second edge of the gate corresponds to the drain edge of the gate of the device.
7. The method of claim 1 wherein patterning the substrate leaves the gate dielectric layer remaining on the substrate.
8. The method of claim 1 wherein patterning the gate stack layers leaves the gate dielectric layer remaining on the substrate.
9. The method of claim 1 wherein implanting ions comprises a tilt angle implant.
10. The method of claim 1 wherein a tilt angle of the implant is about $1 - 45^\circ$.
11. The method of claim 1 wherein implanting ions comprises multiple tilt angle implants which are rotated about a plane of the substrate.
12. The method of claim 11 wherein a tilt angle of the implant is about $1 - 45^\circ$.
13. The method of claim 1 wherein implanting ions comprises a quad tilt angle implant which is rotated about a plane of the substrate.
14. The method of claim 1 wherein the second mask comprises photoresist.

15. A method of forming a device comprising:

providing a substrate prepared with an active device region which includes a doped drift well in a first portion of the active device region and gate stack layers of a gate stack comprising at least a gate electrode layer over a gate dielectric layer on the surface of the substrate;

providing a first mask on the substrate corresponding to the gate;

patterning the substrate to at least remove portions of a top gate stack layer unprotected by the first mask;

providing a second mask on the substrate with an opening exposing a portion of the first mask and the top gate stack layer; and

implanting ions through the opening and gate stack layers into the substrate to form a channel well.

16. The method of claim 15 wherein the substrate comprises:

a drift isolation region in the doped drift well; and

a doped deep well in the substrate of the active device region encompassing the doped drift well.

17. The method of claim 14 wherein the first mask comprises a hard mask.

18. The method of claim 17 wherein a first edge of the gate corresponds to a channel edge of the gate of the device and a second edge of the gate corresponds to the drain edge of the gate of the device.

19. A device comprising:

a substrate prepared with an active device region, wherein the active device region includes a gate stack with a gate electrode layer over a gate dielectric layer, wherein at least the gate electrode layer is patterned using a hard mask;

a doped channel well disposed in the substrate adjacent to the first edge of the gate, wherein the first edge of the gate overlaps the channel well with a channel edge of the channel well beneath the gate, the first edge of the gate and channel edge defines an effective channel length of the device; and

a doped drift well adjacent to the second edge of the gate.

20. The device of claim 16 wherein the effective channel length is less than about 0.4 μm .

HIGH VOLTAGE DEVICE

ABSTRACT

A method of forming a device is presented. The method includes providing a substrate prepared with an active device region. The active device region includes gate stack layers of a gate stack including at least a gate electrode layer over a gate dielectric layer. A first mask is provided on the substrate corresponding to the gate. The substrate is patterned to at least remove portions of a top gate stack layer unprotected by the first mask. A second mask is also provided on the substrate with an opening exposing a portion of the first mask and the top gate stack layer. A channel well is formed by implanting ions through the opening and gate stack layers into the substrate.

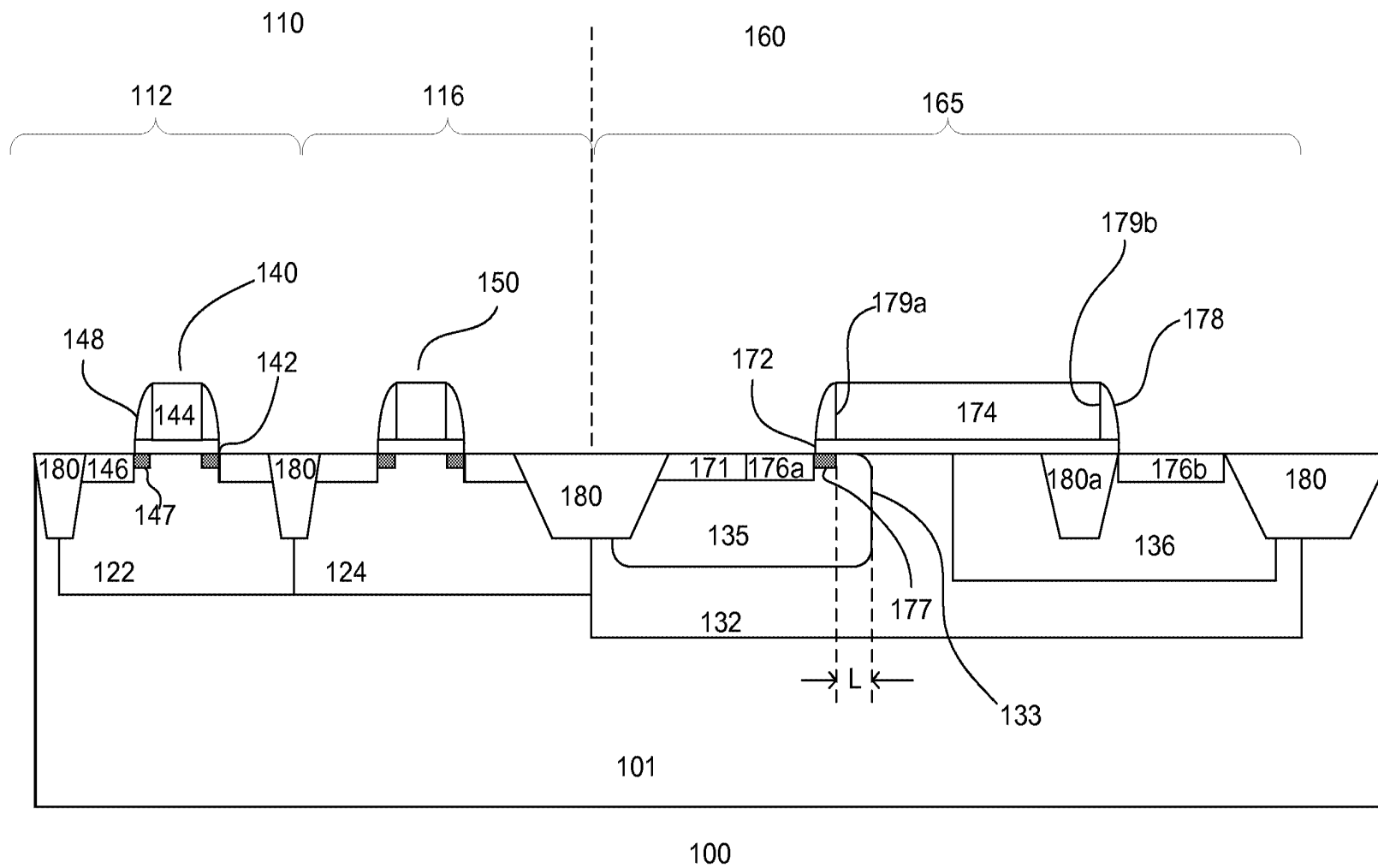


Fig. 1

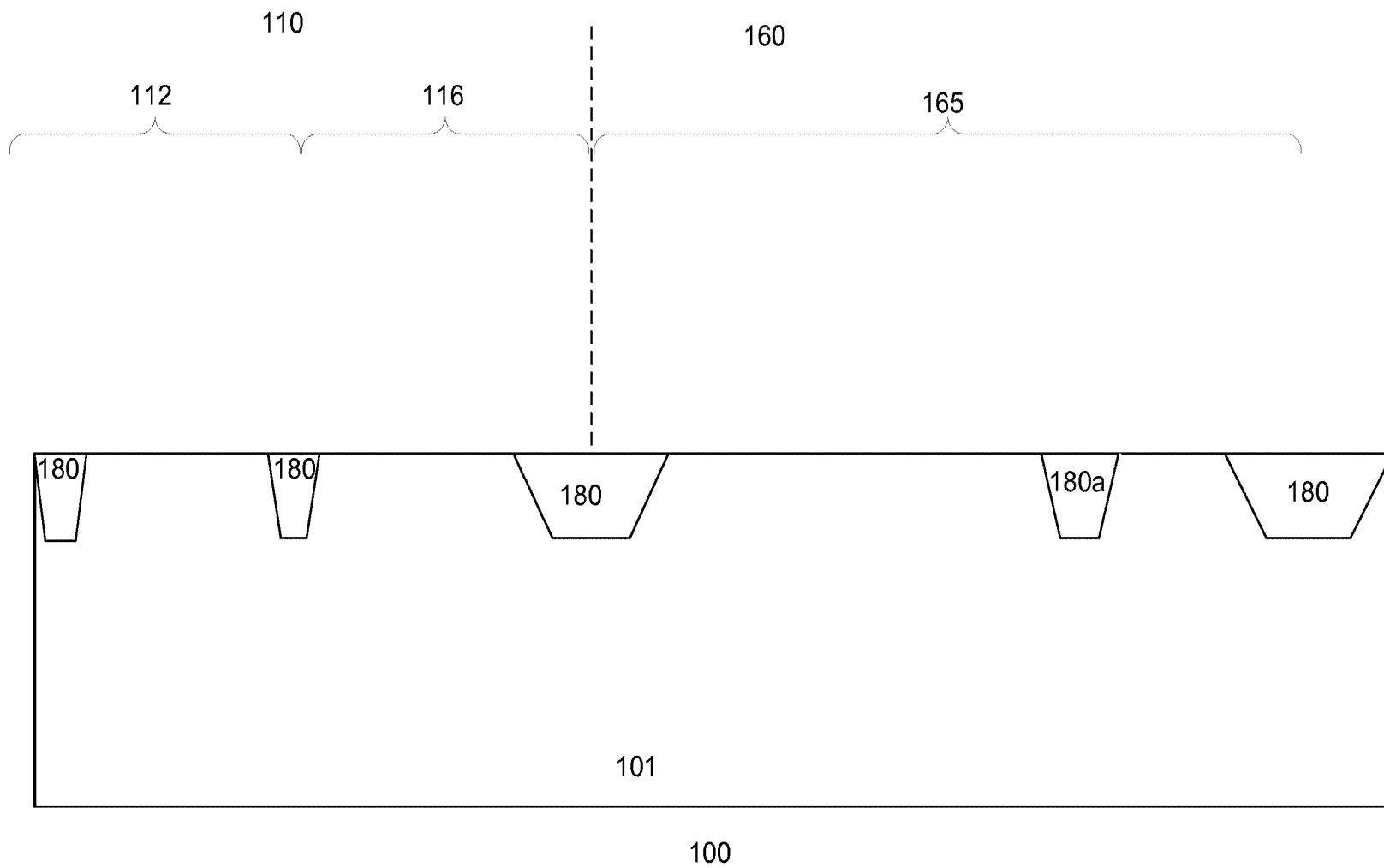


Fig. 2a

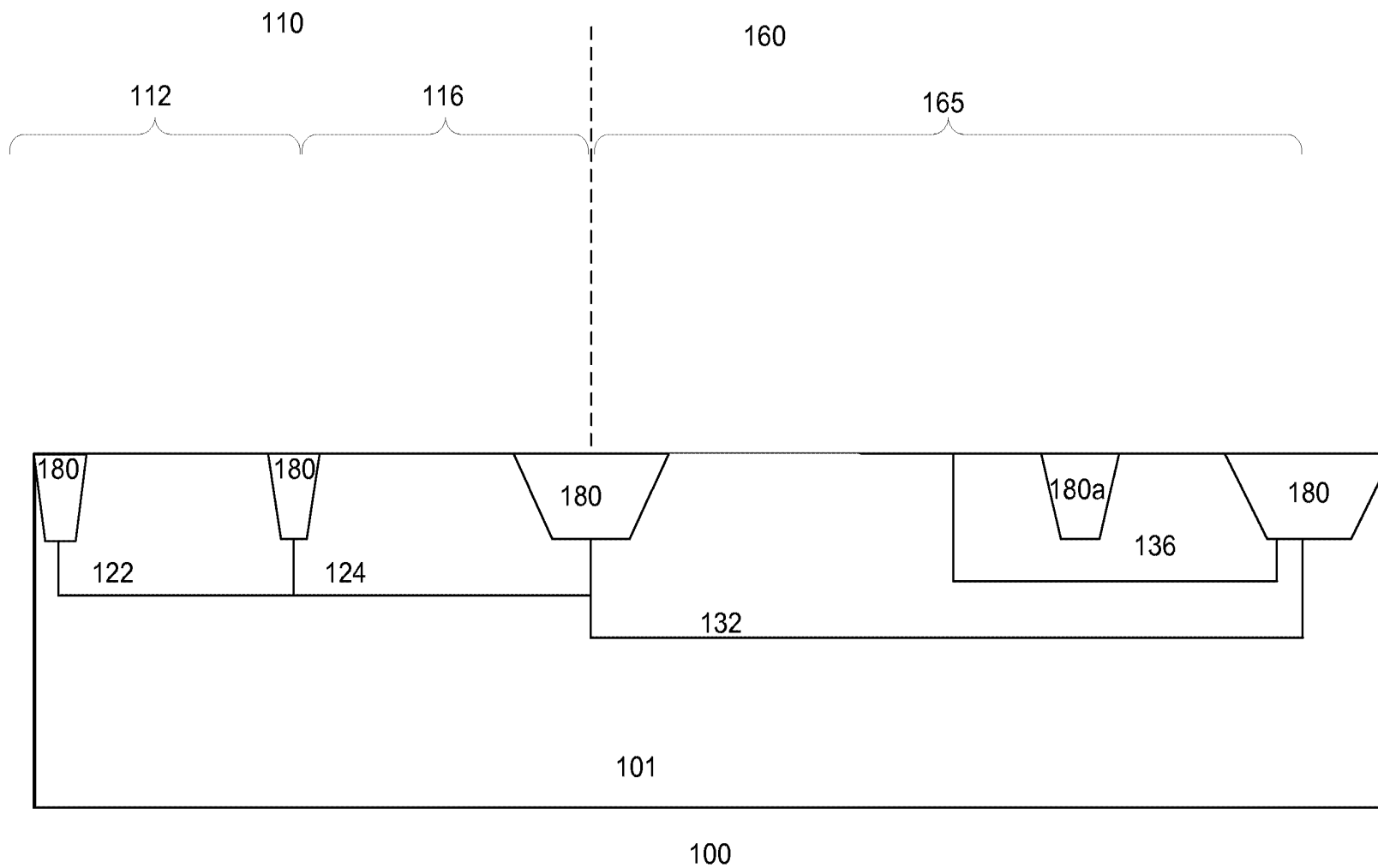


Fig. 2b

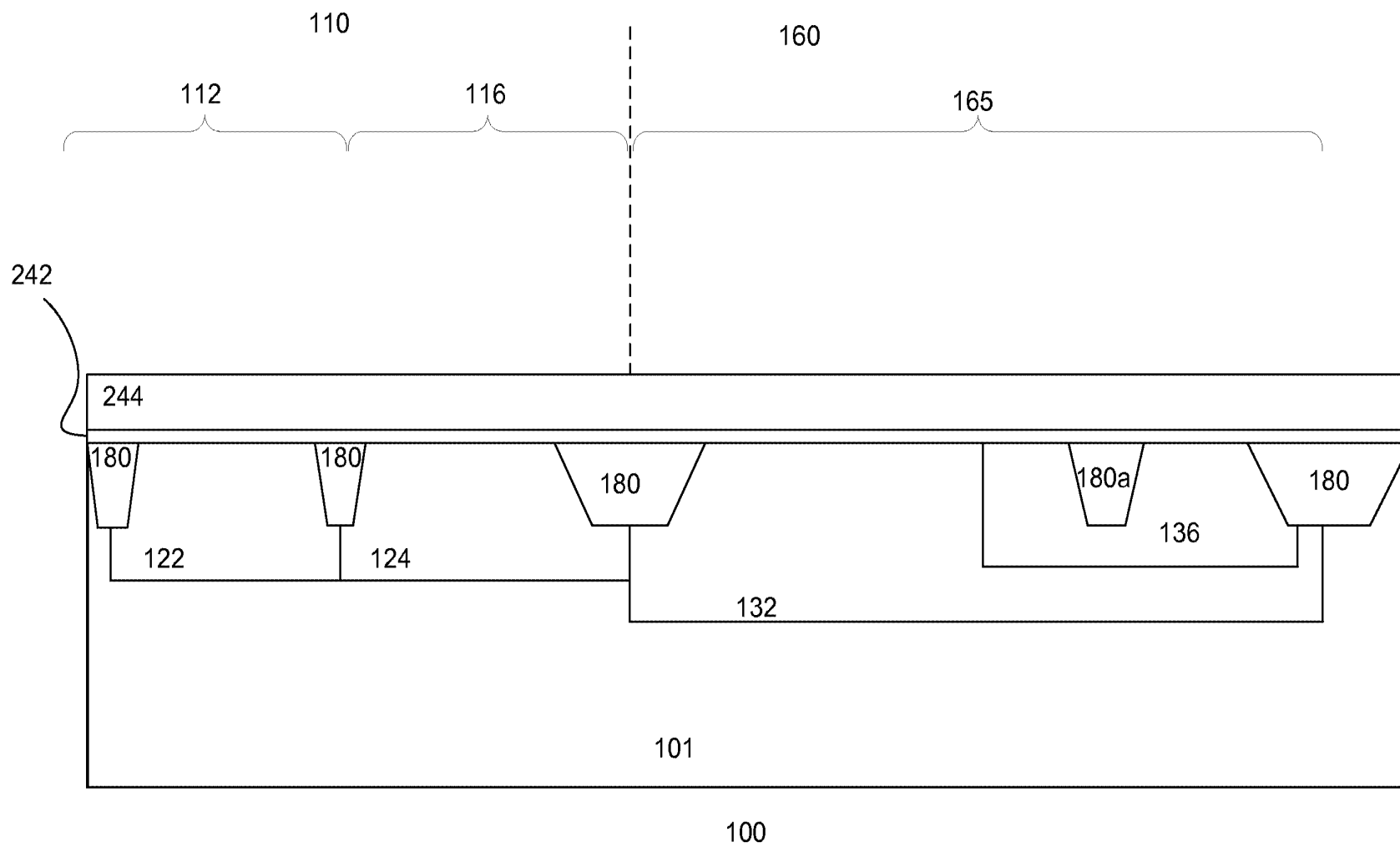


Fig. 2c

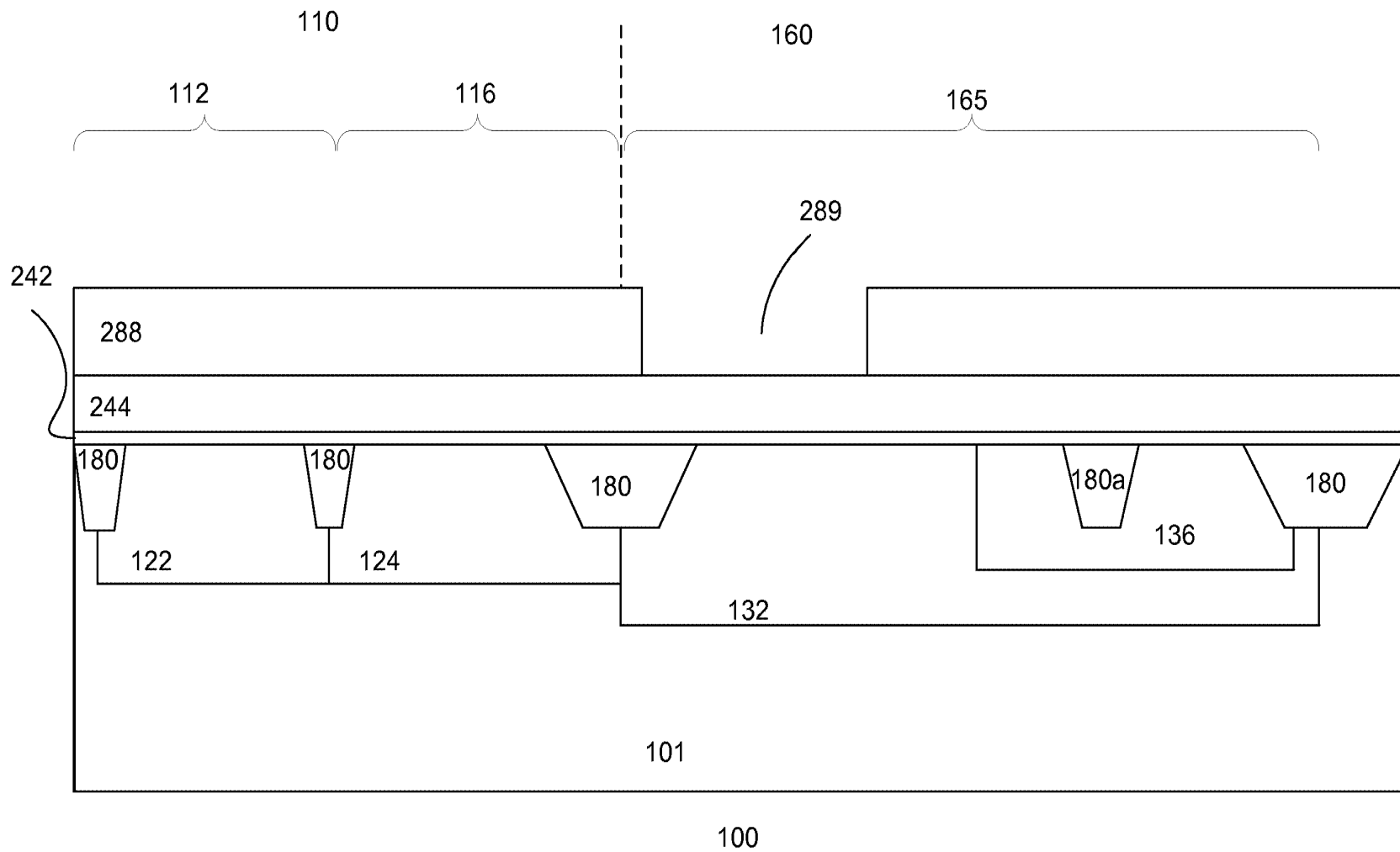


Fig. 2d

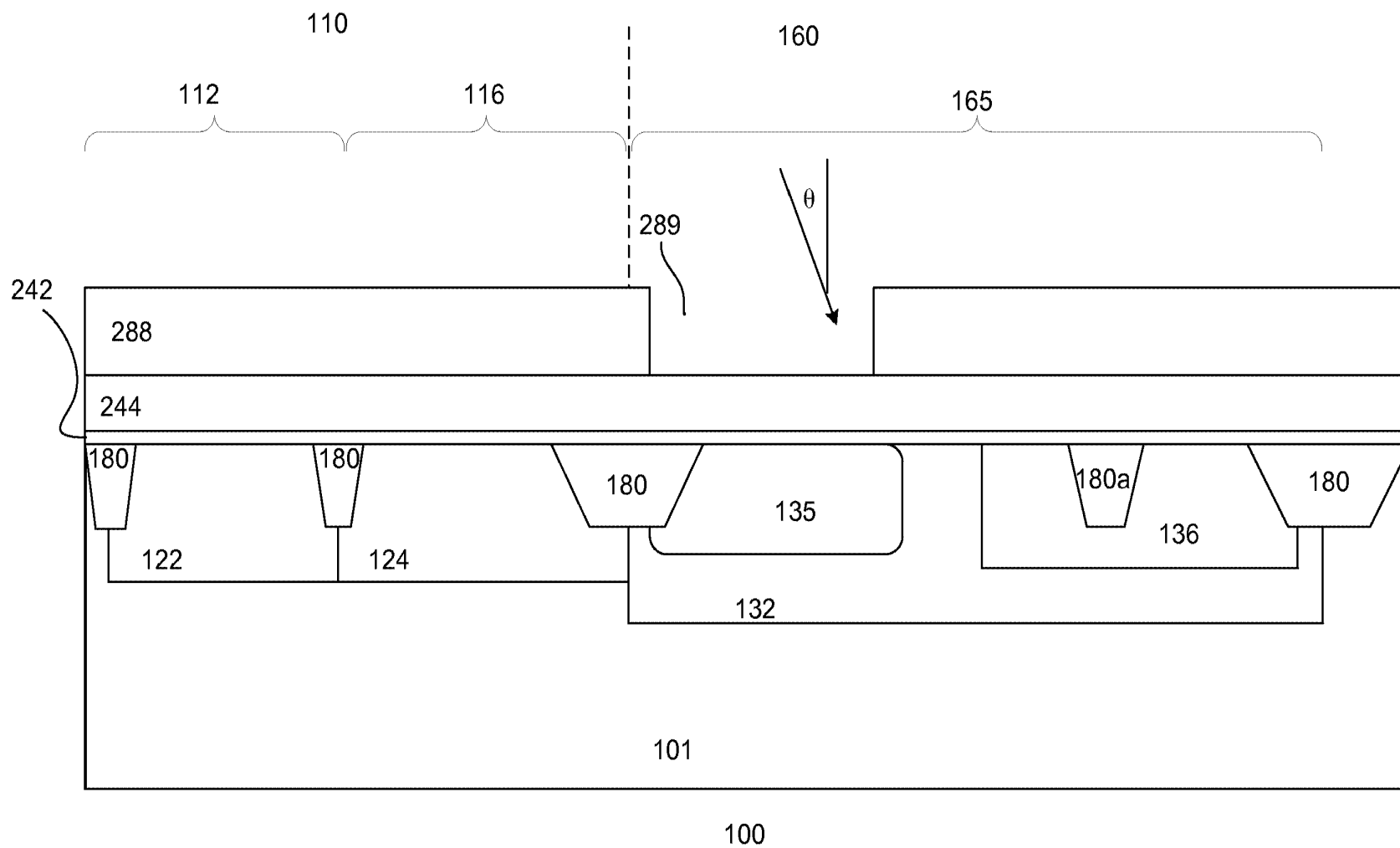


Fig. 2e

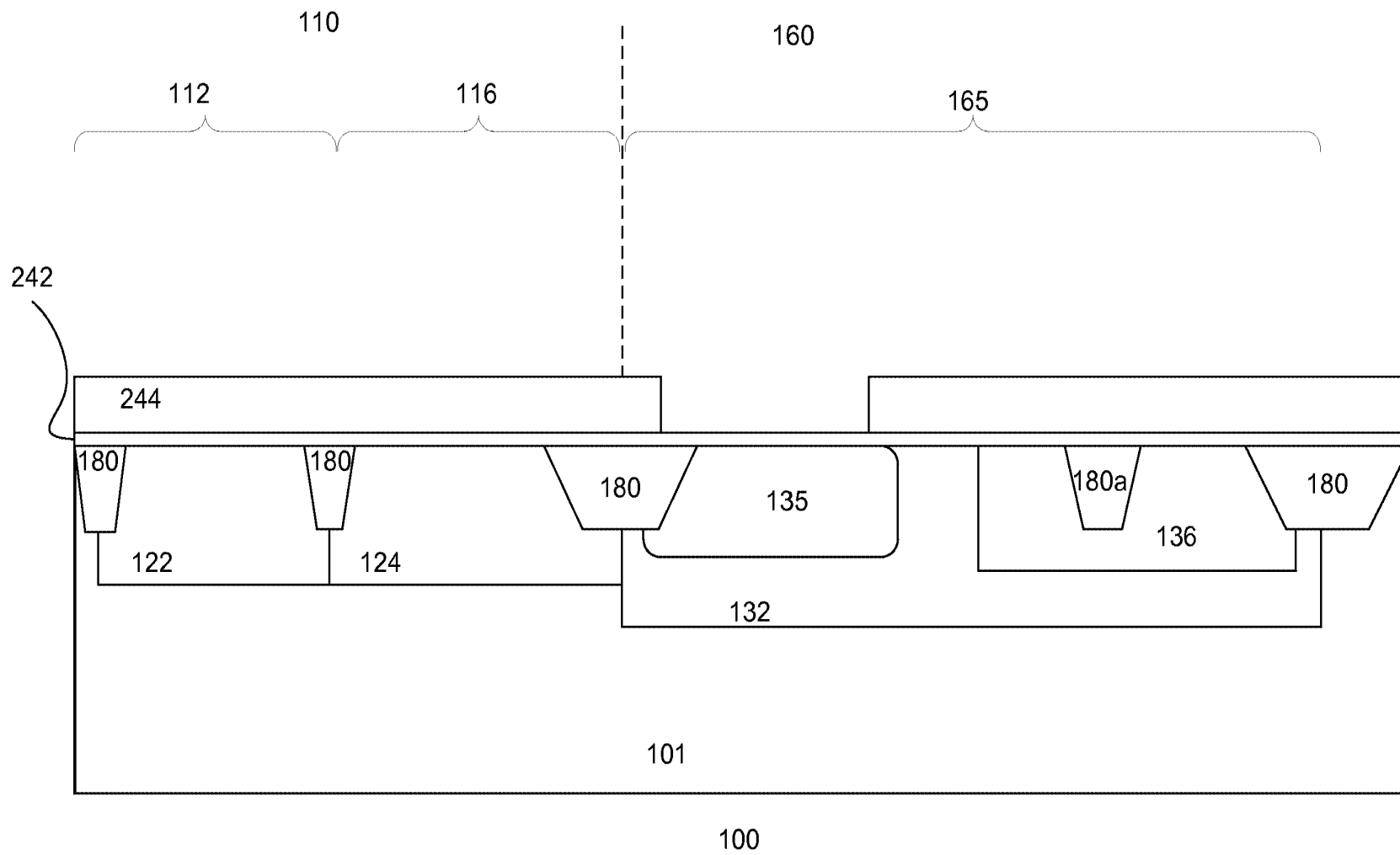


Fig. 2f

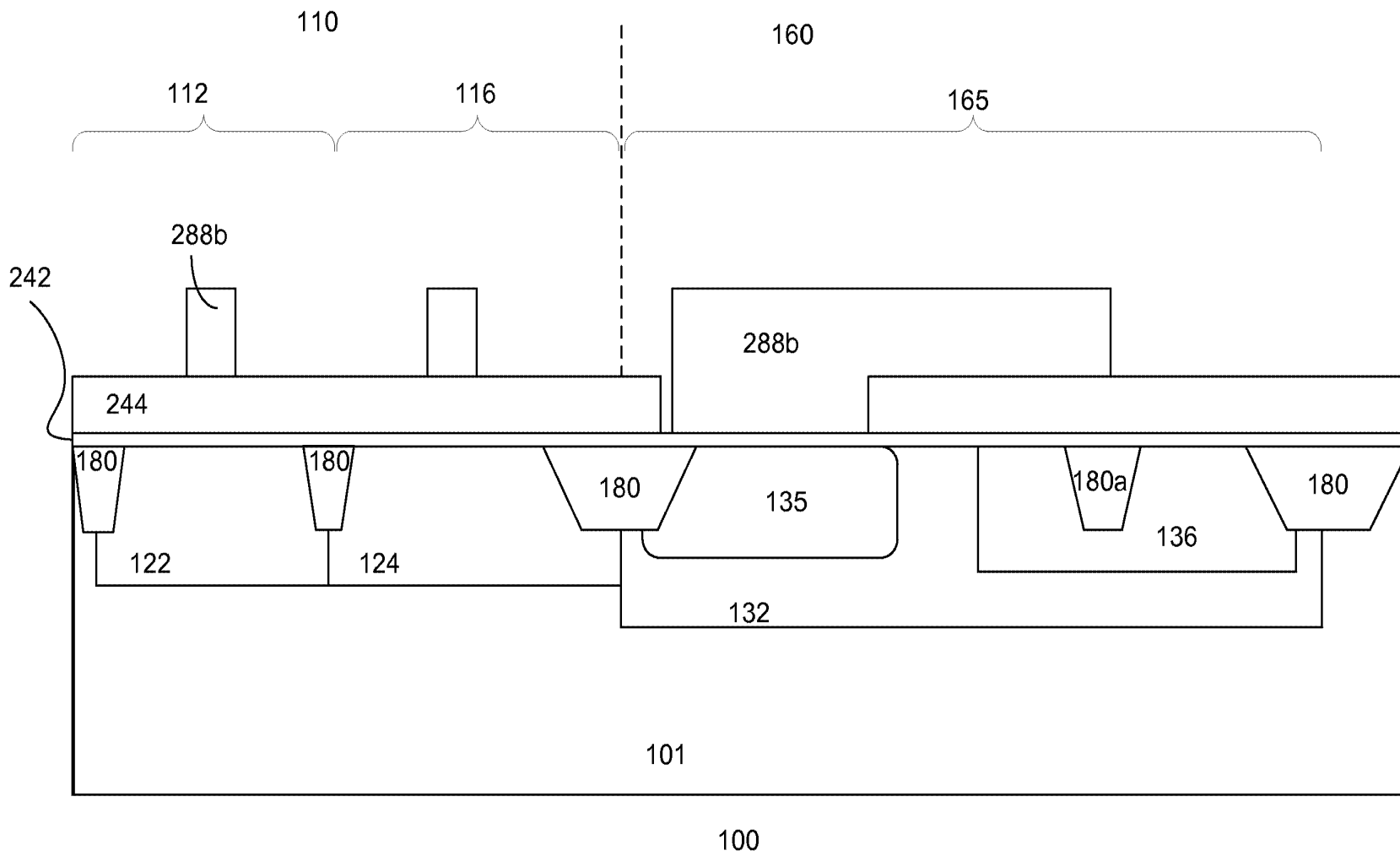


Fig. 2g

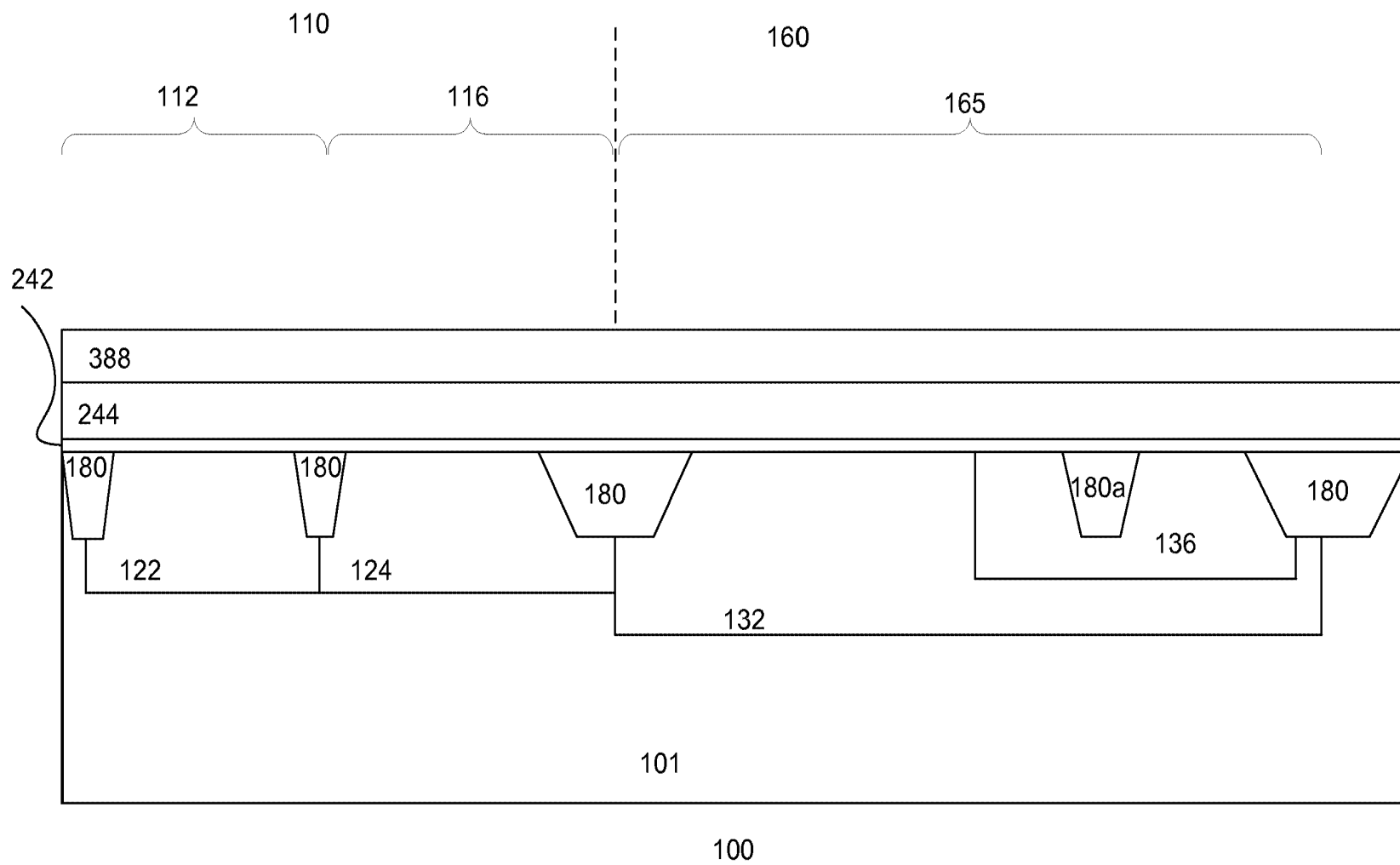


Fig. 3a

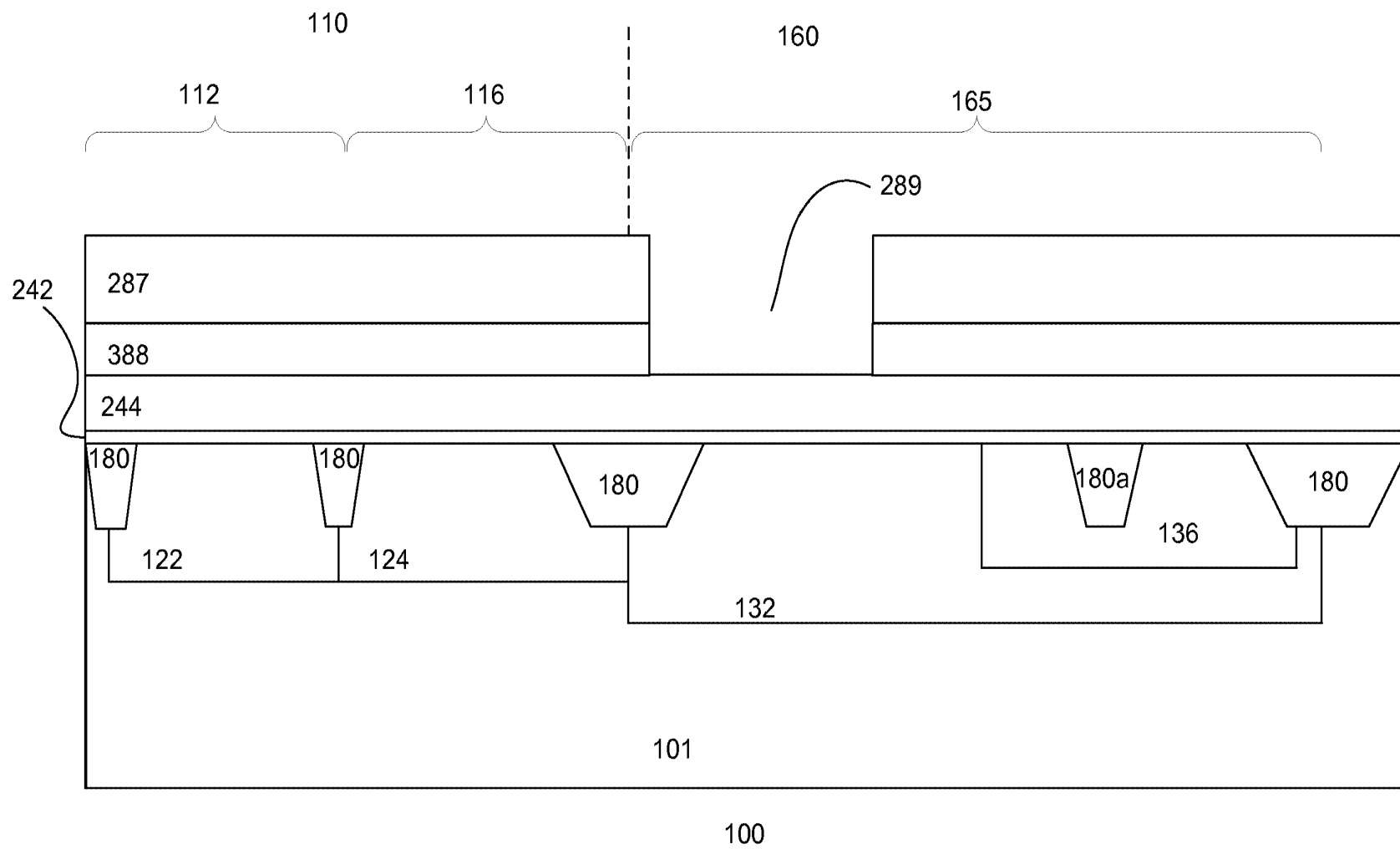


Fig. 3b

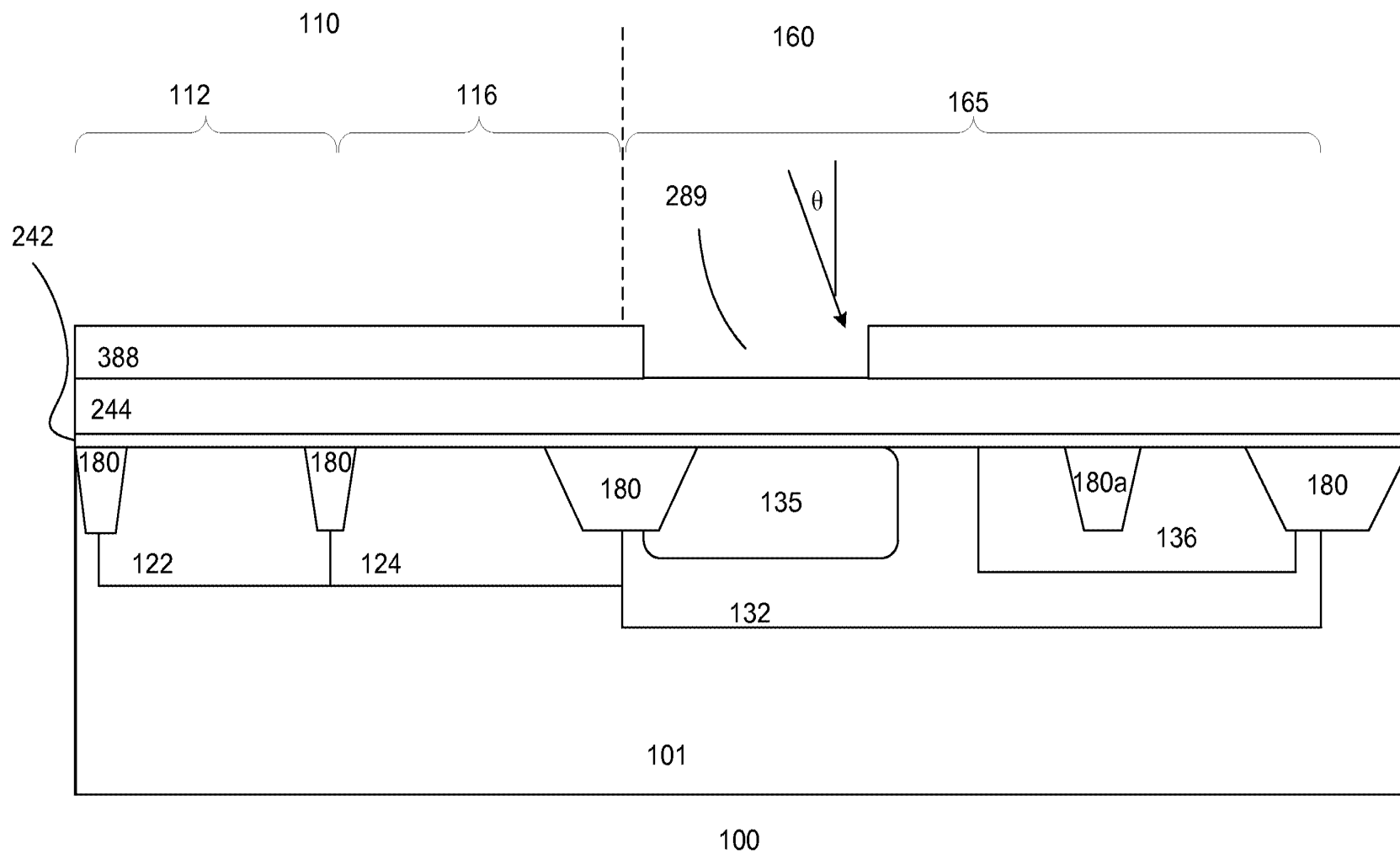
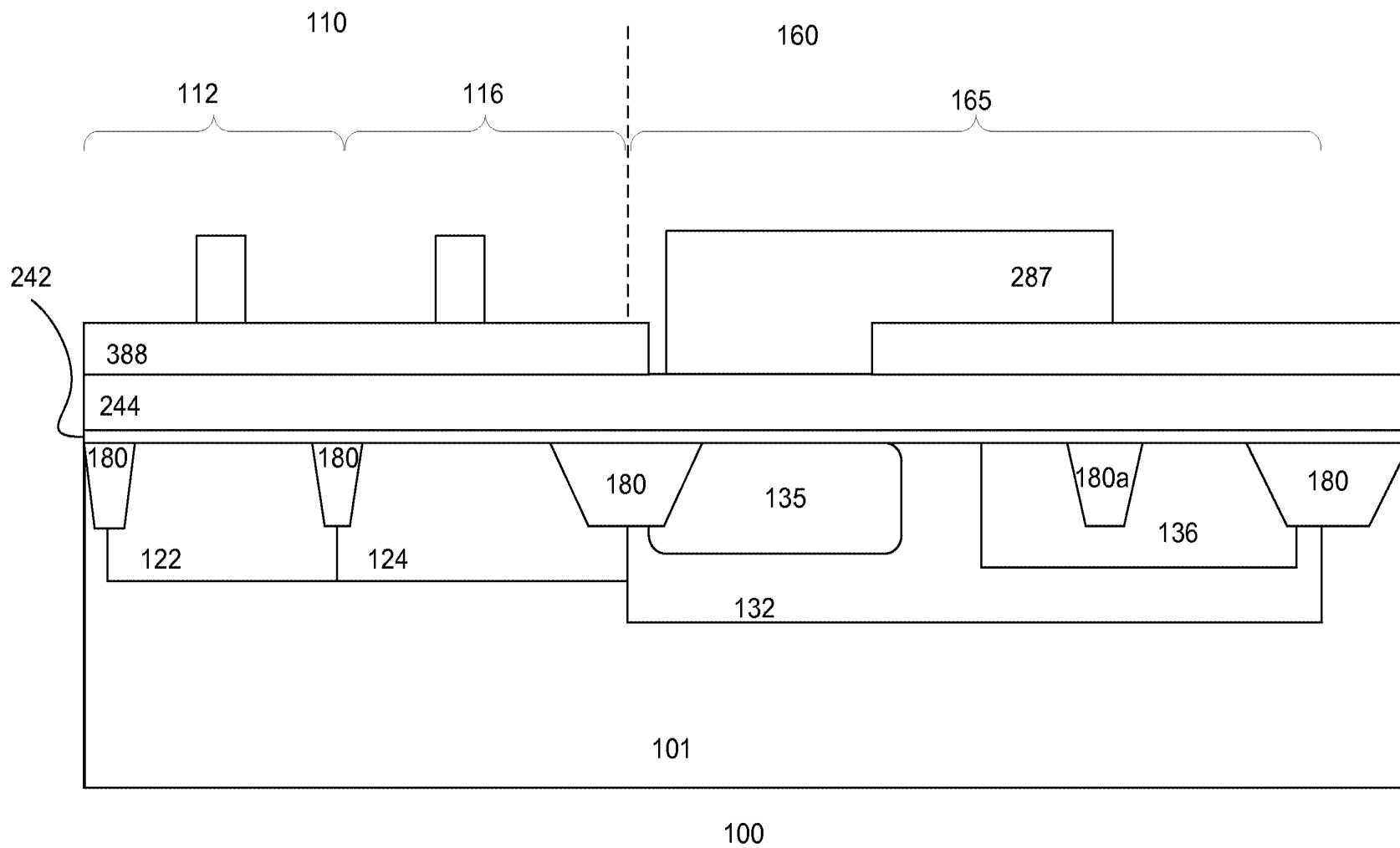


Fig. 3c



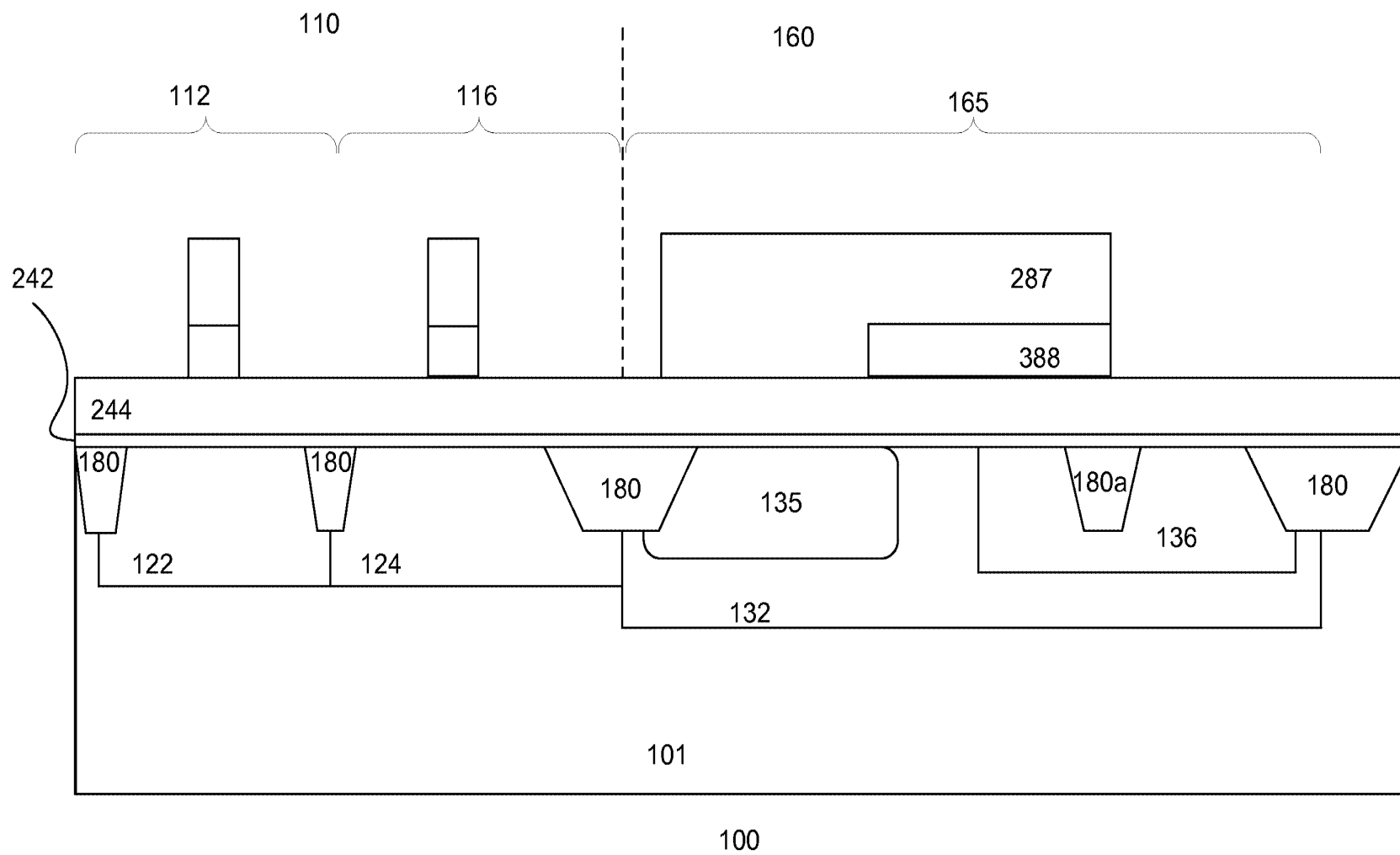


Fig. 3e

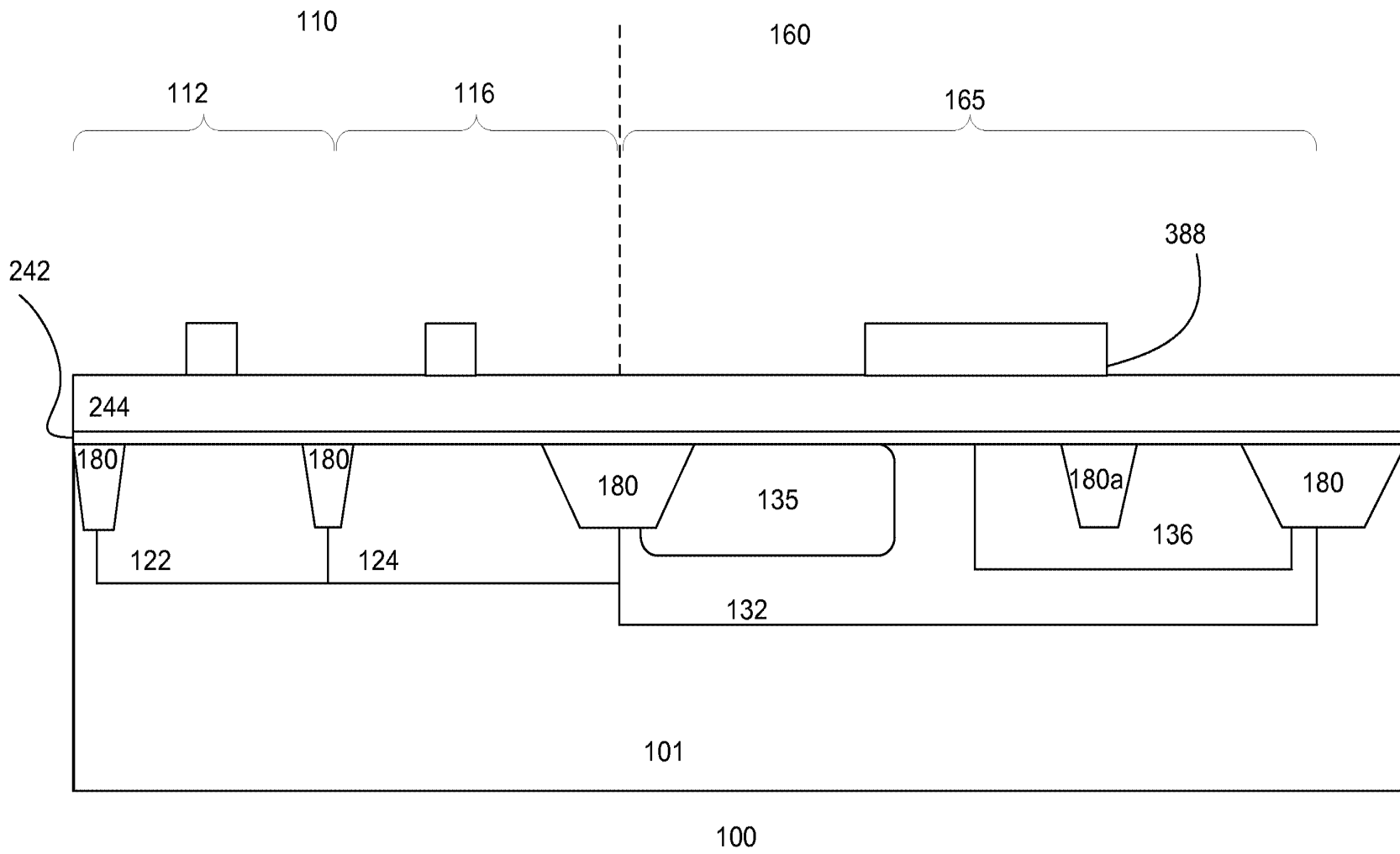


Fig. 3f

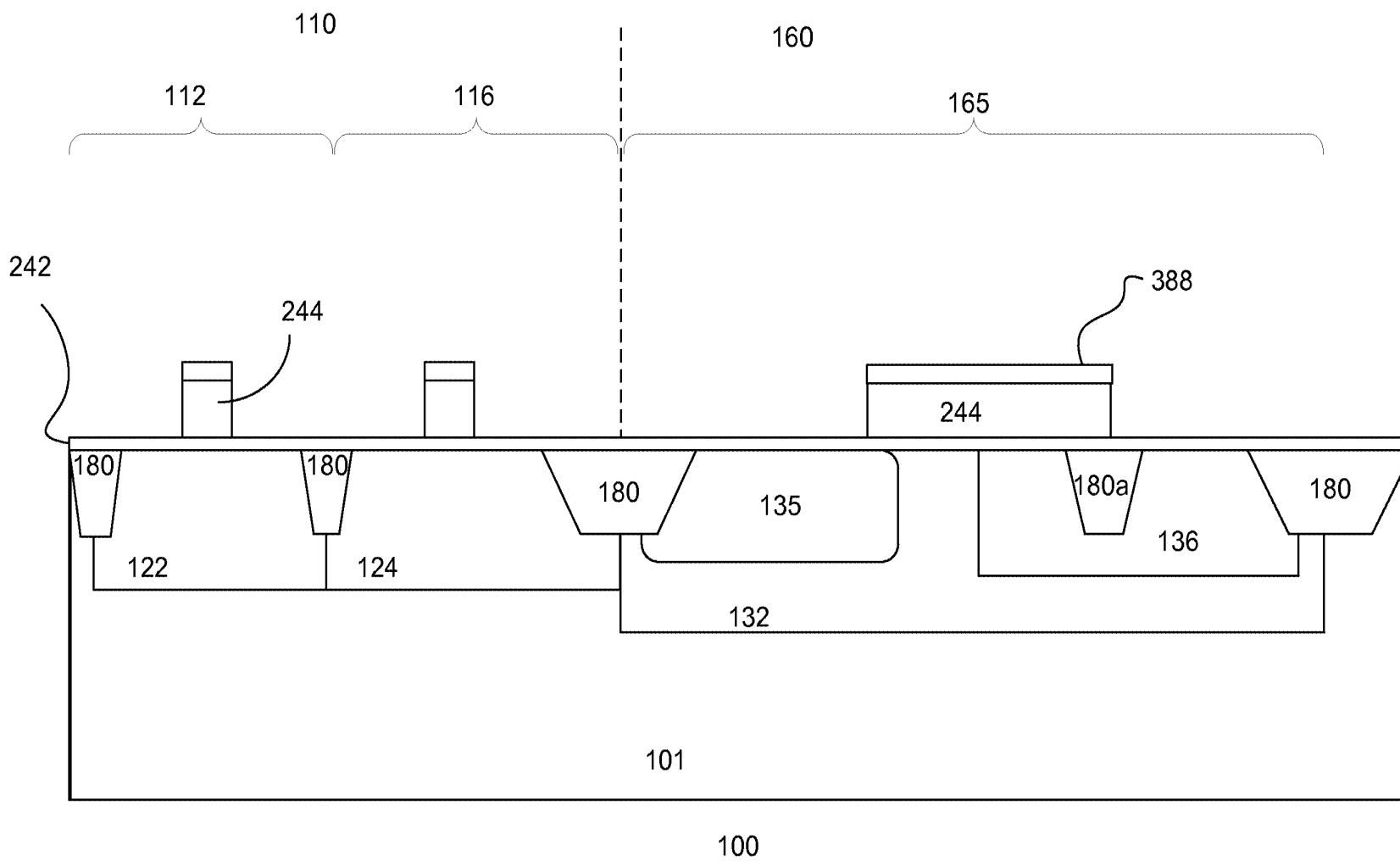


Fig. 3g

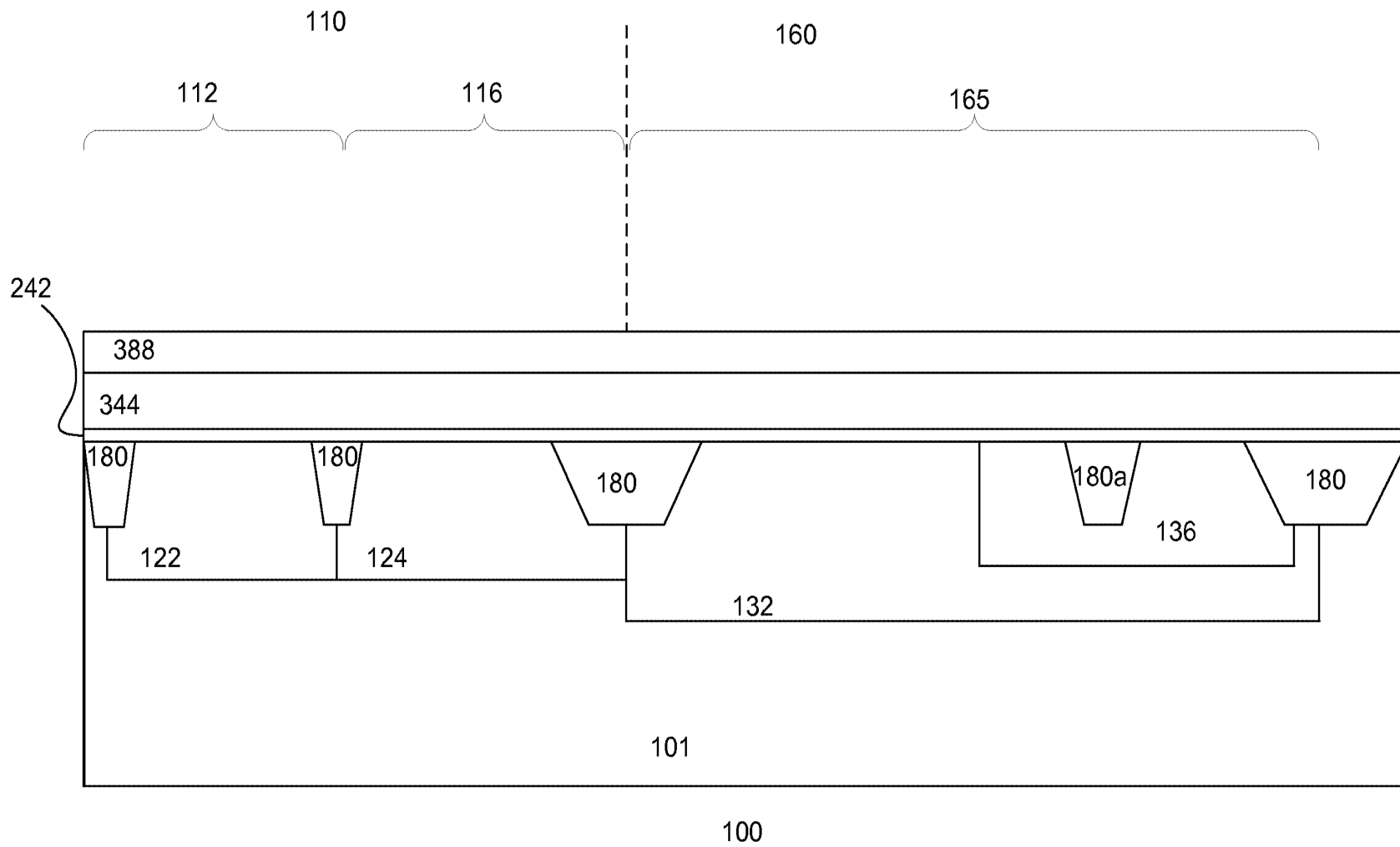


Fig. 4a

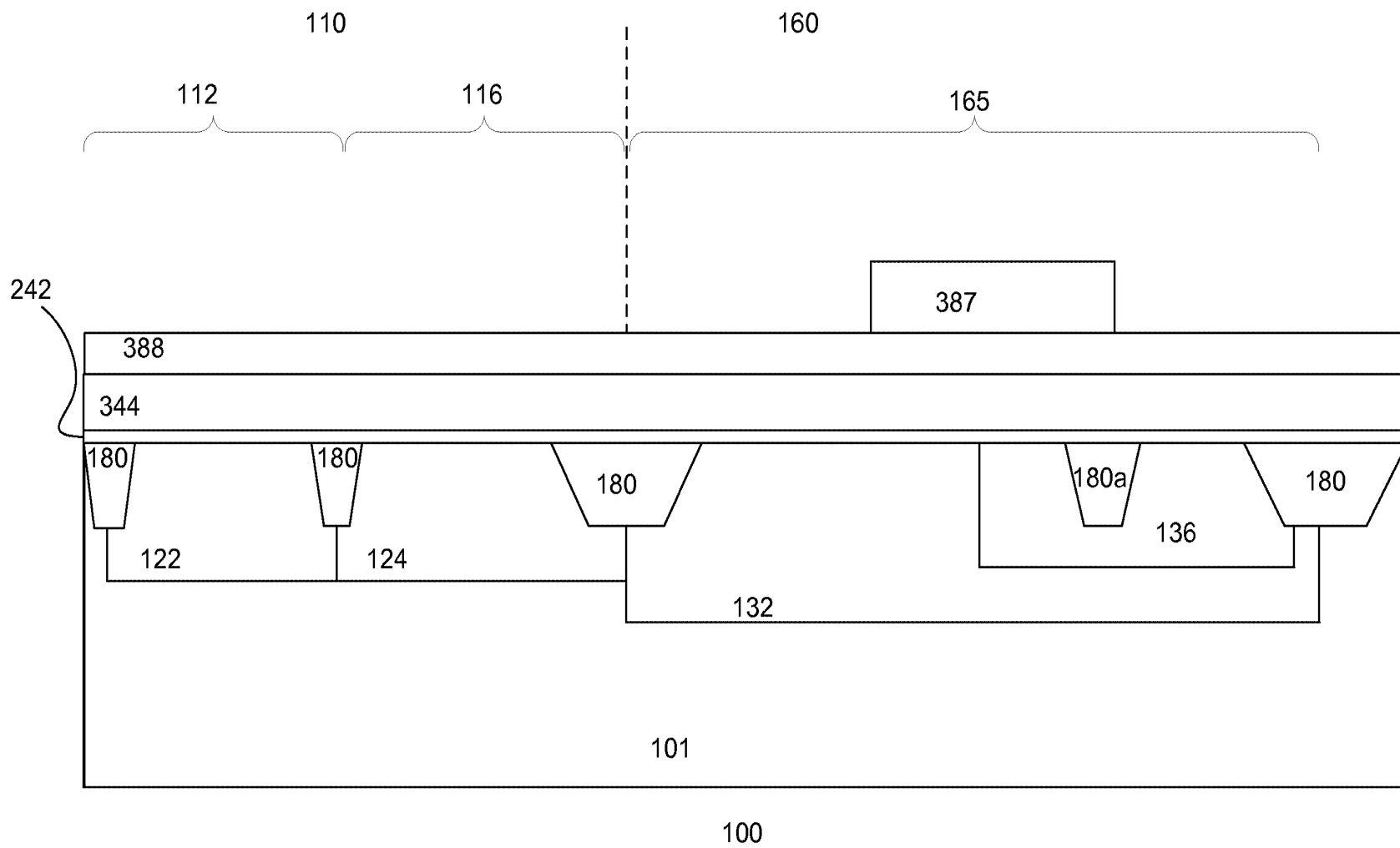


Fig. 4b

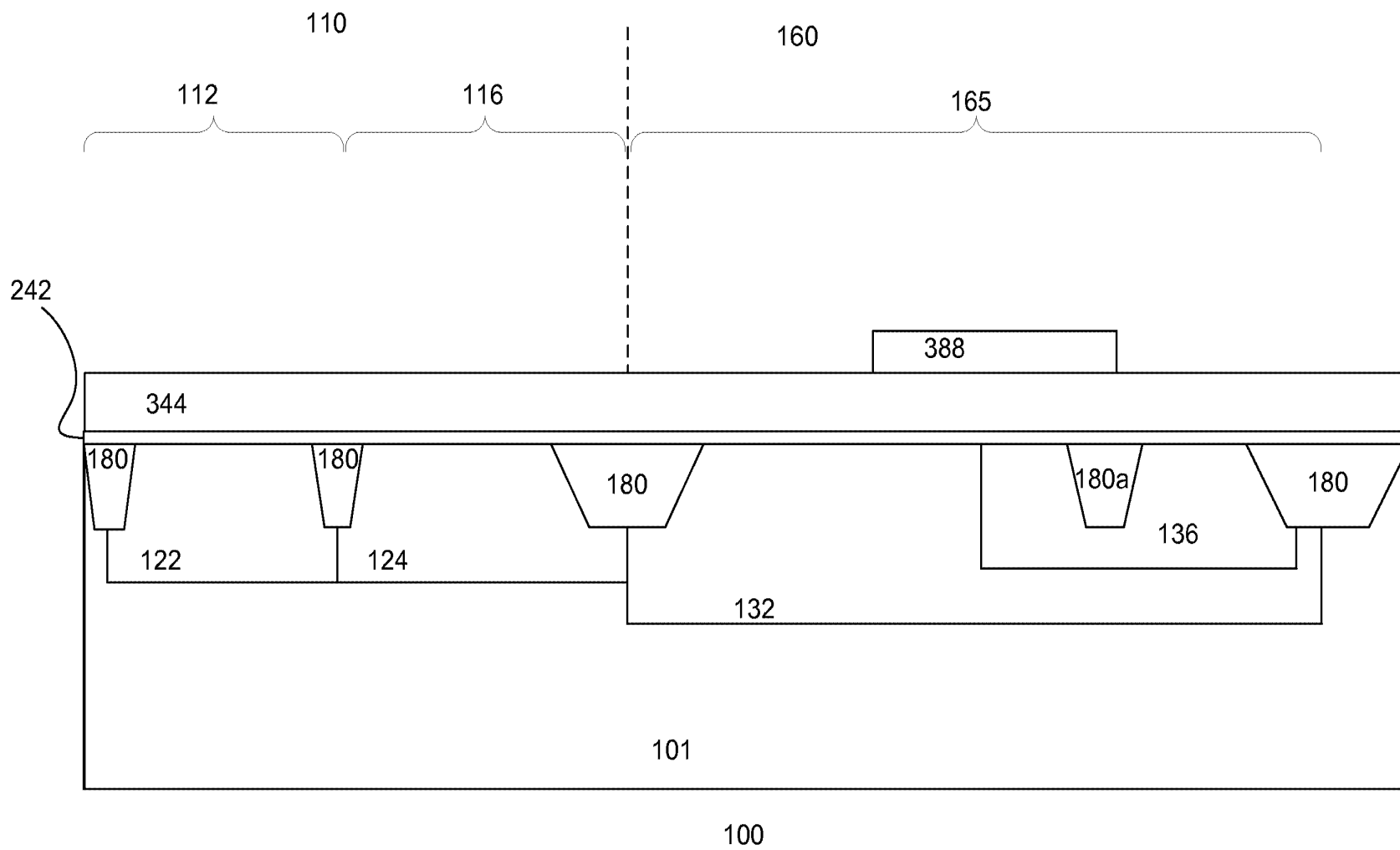


Fig. 4c

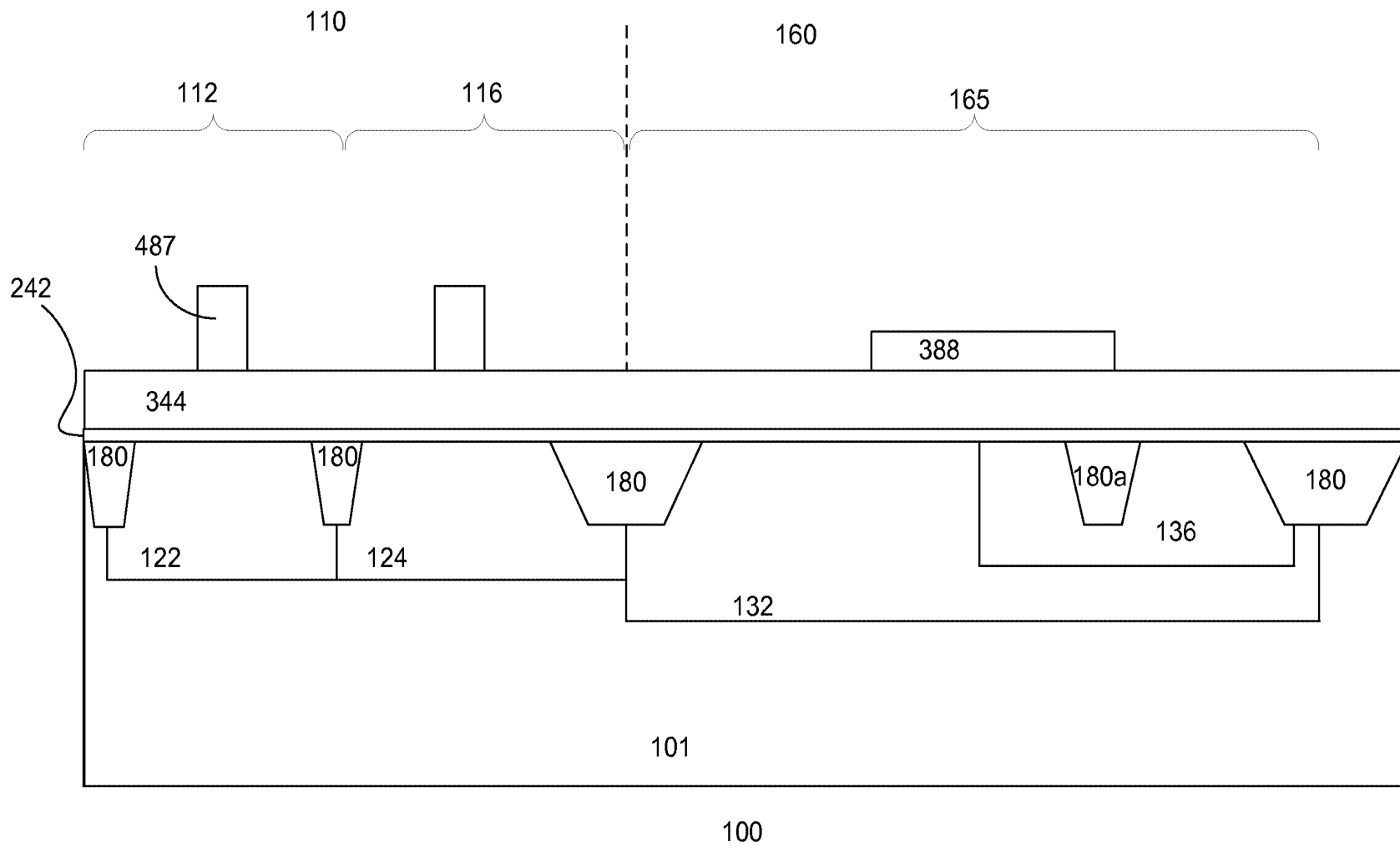


Fig. 4d

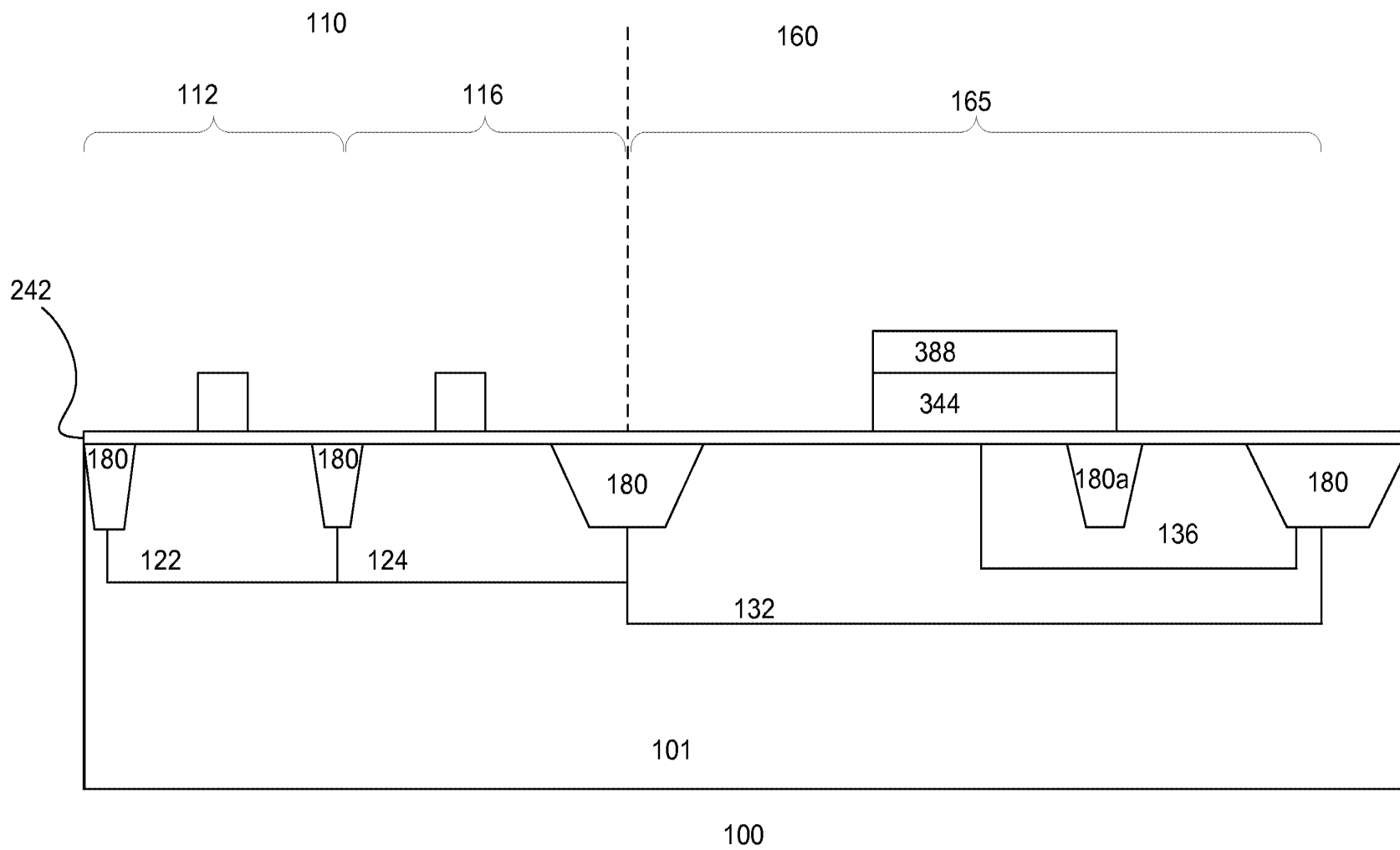


Fig. 4e

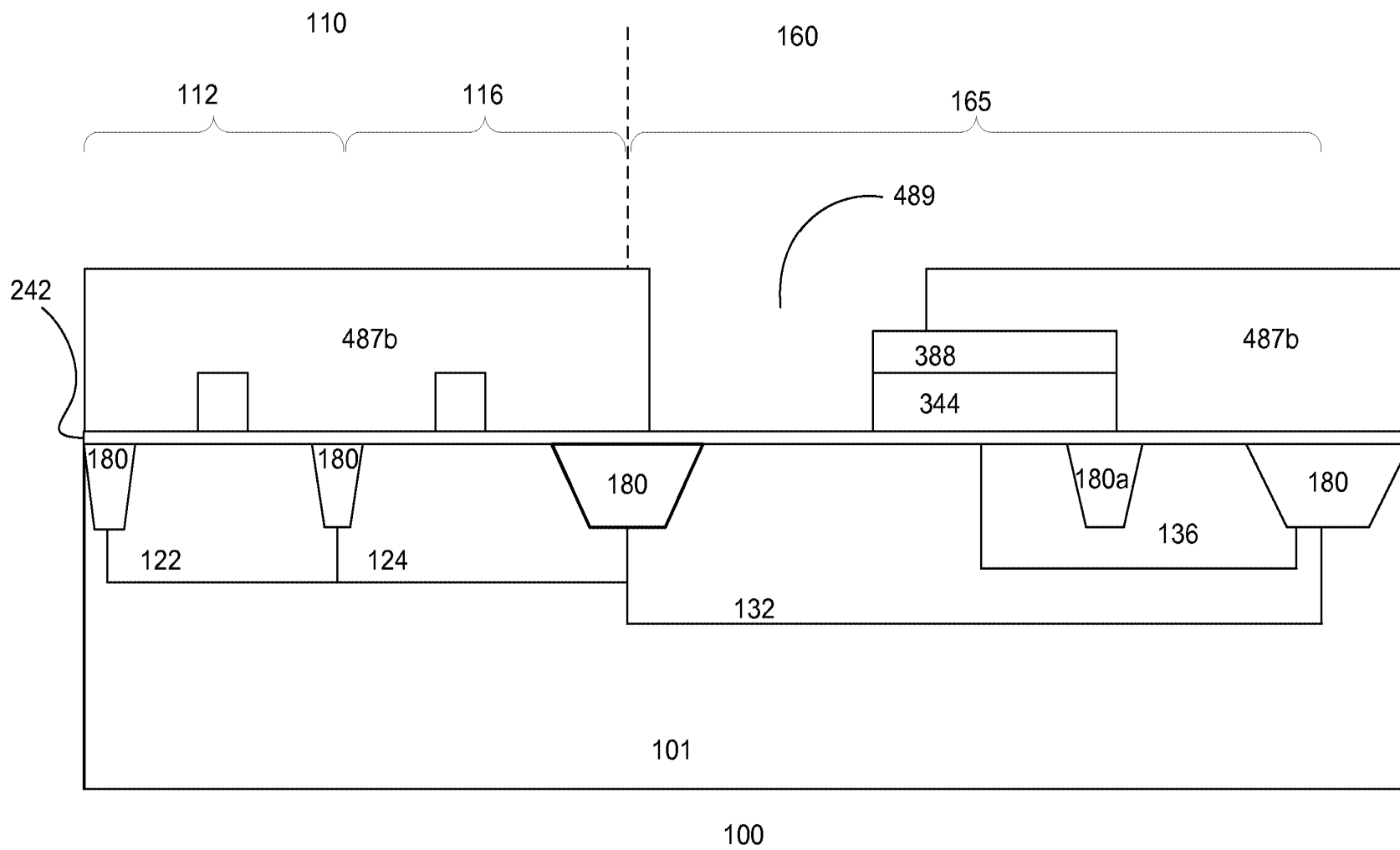


Fig. 4f

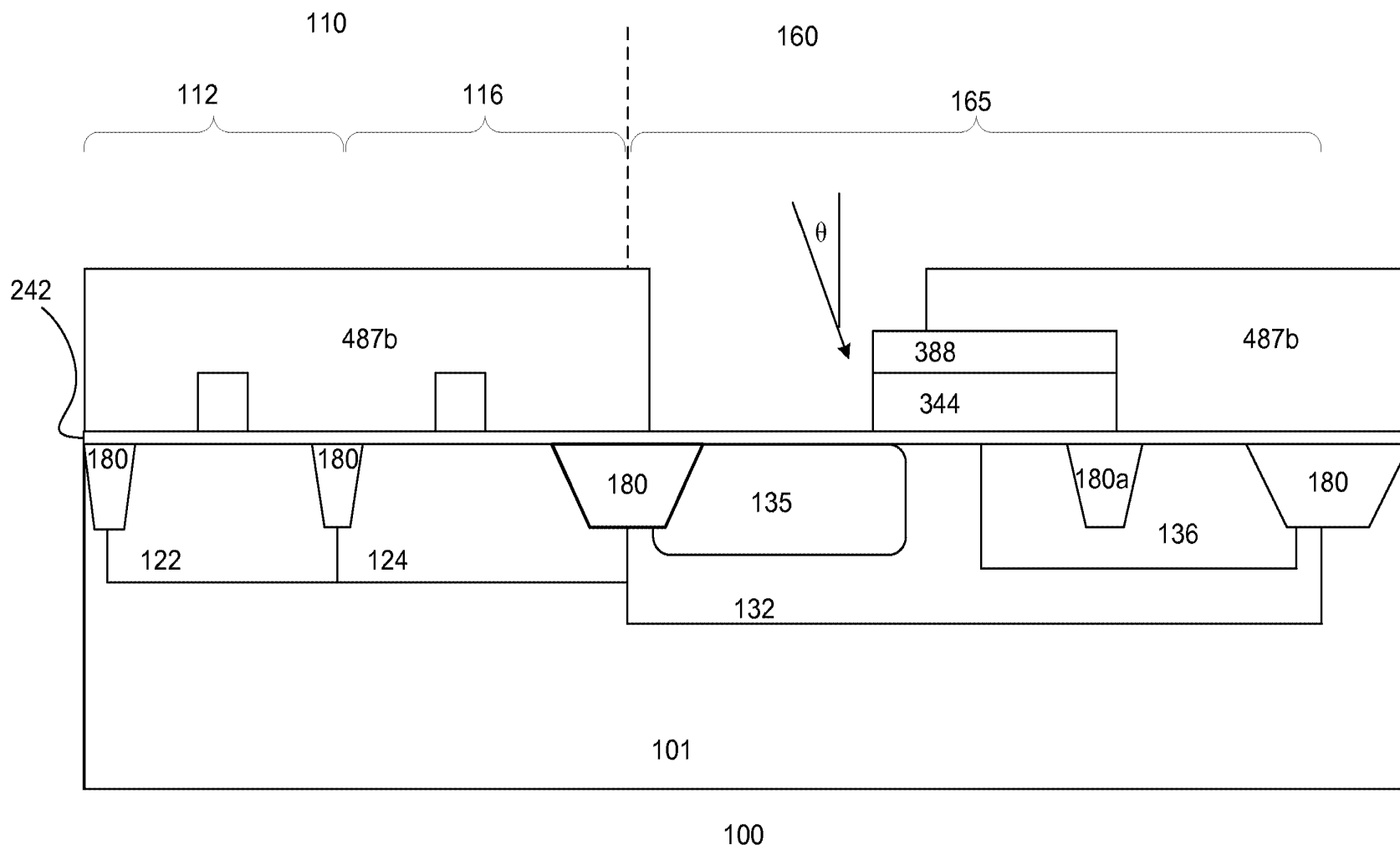


Fig. 4g

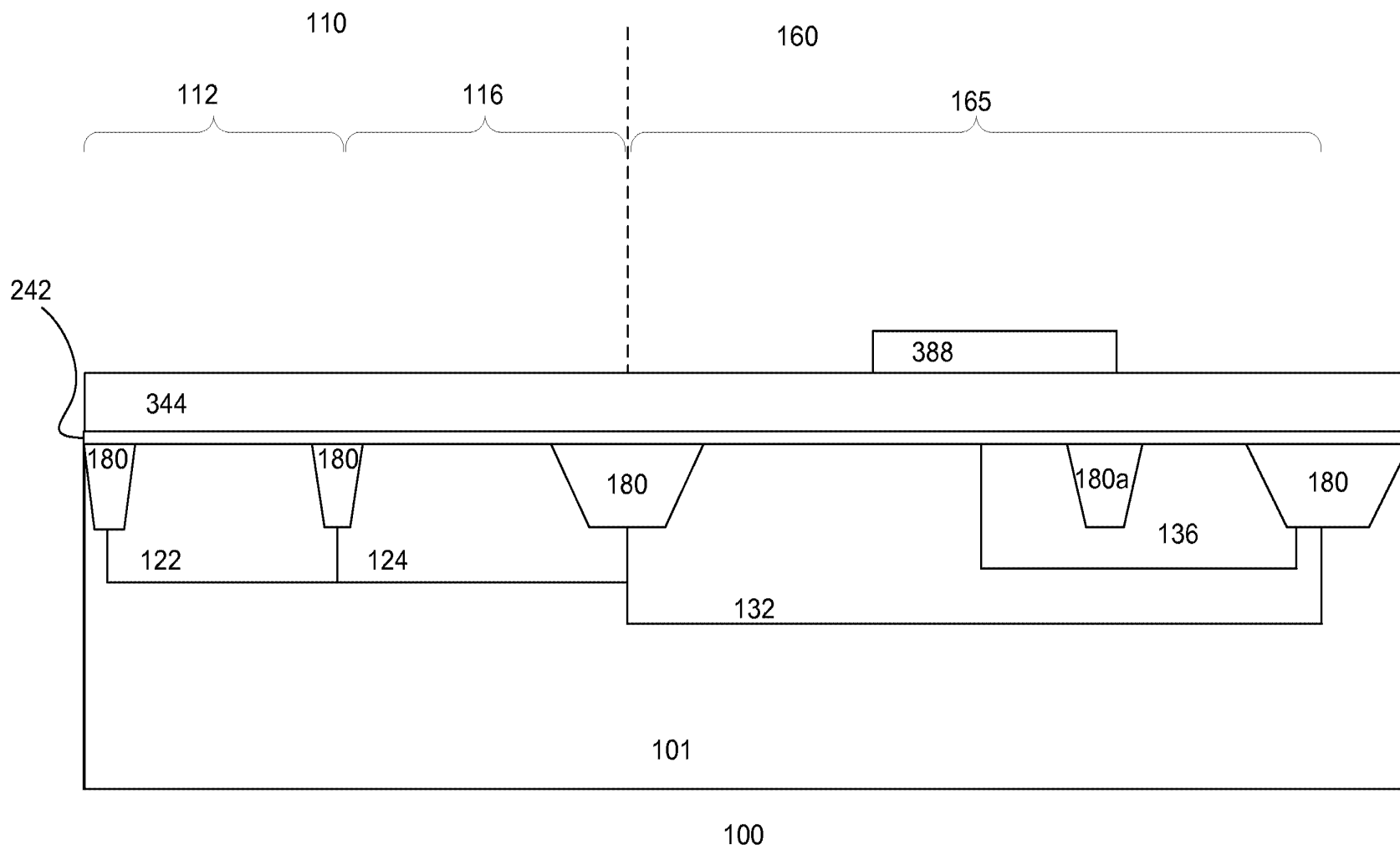


Fig. 5a

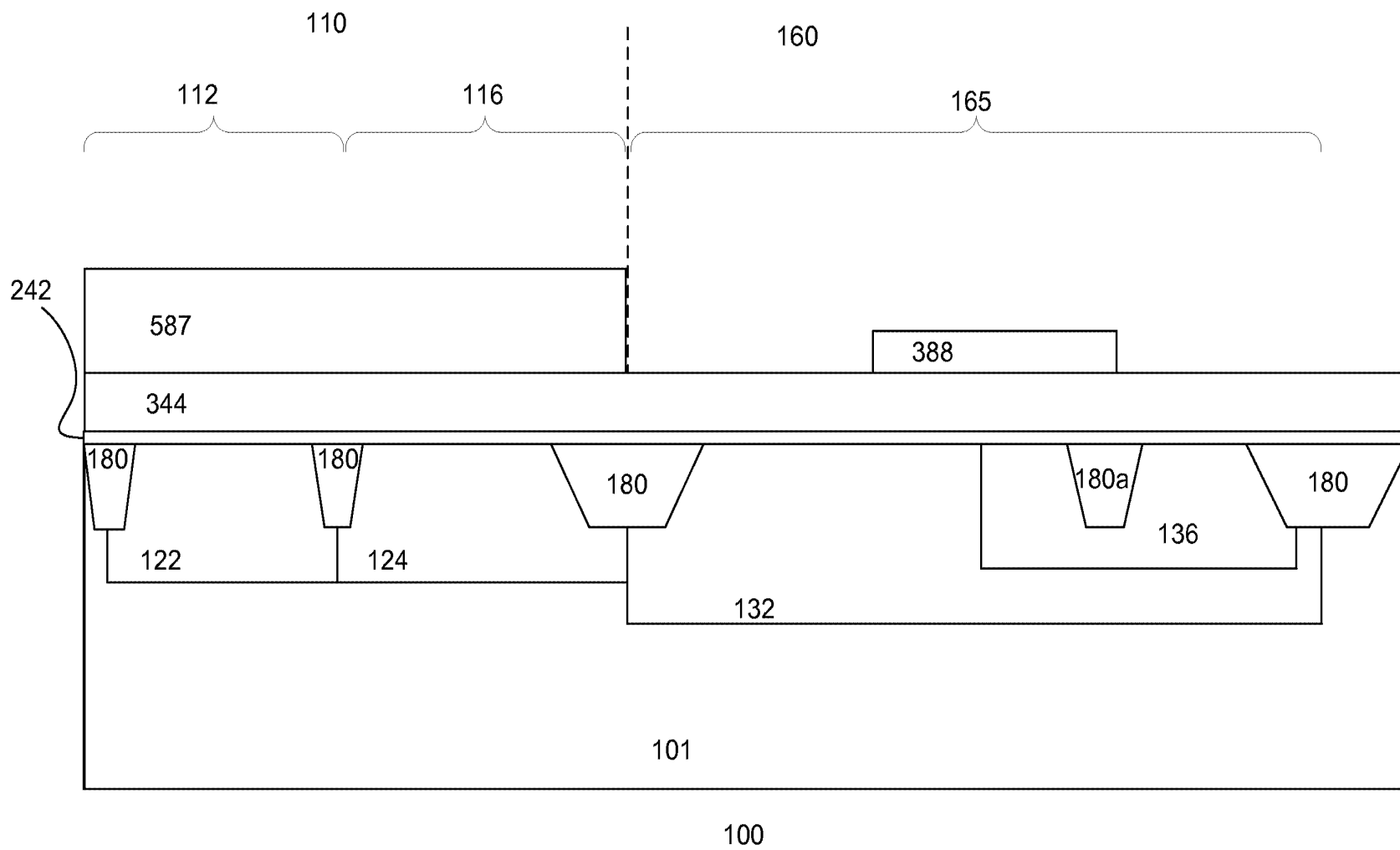


Fig. 5b

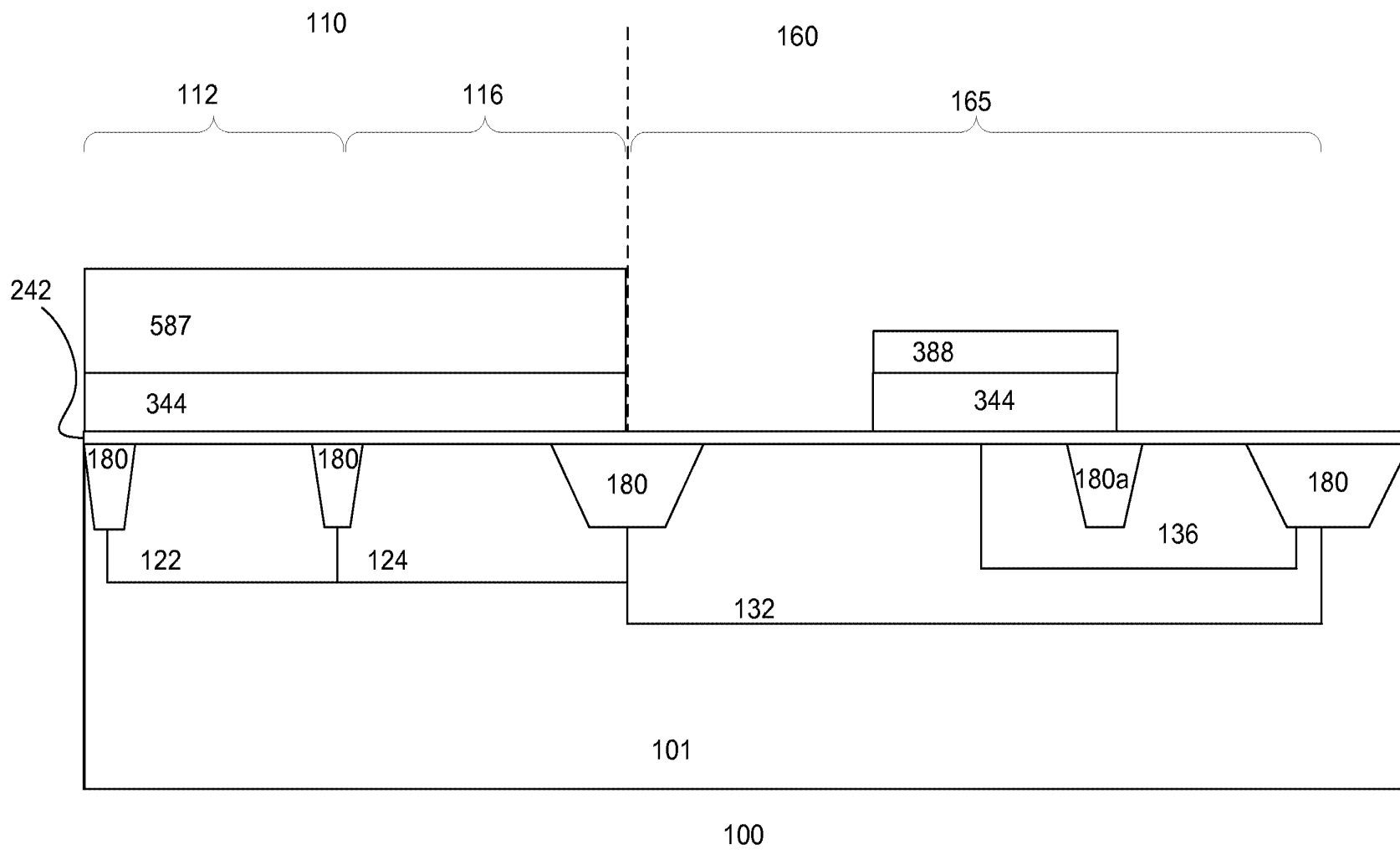


Fig. 5c

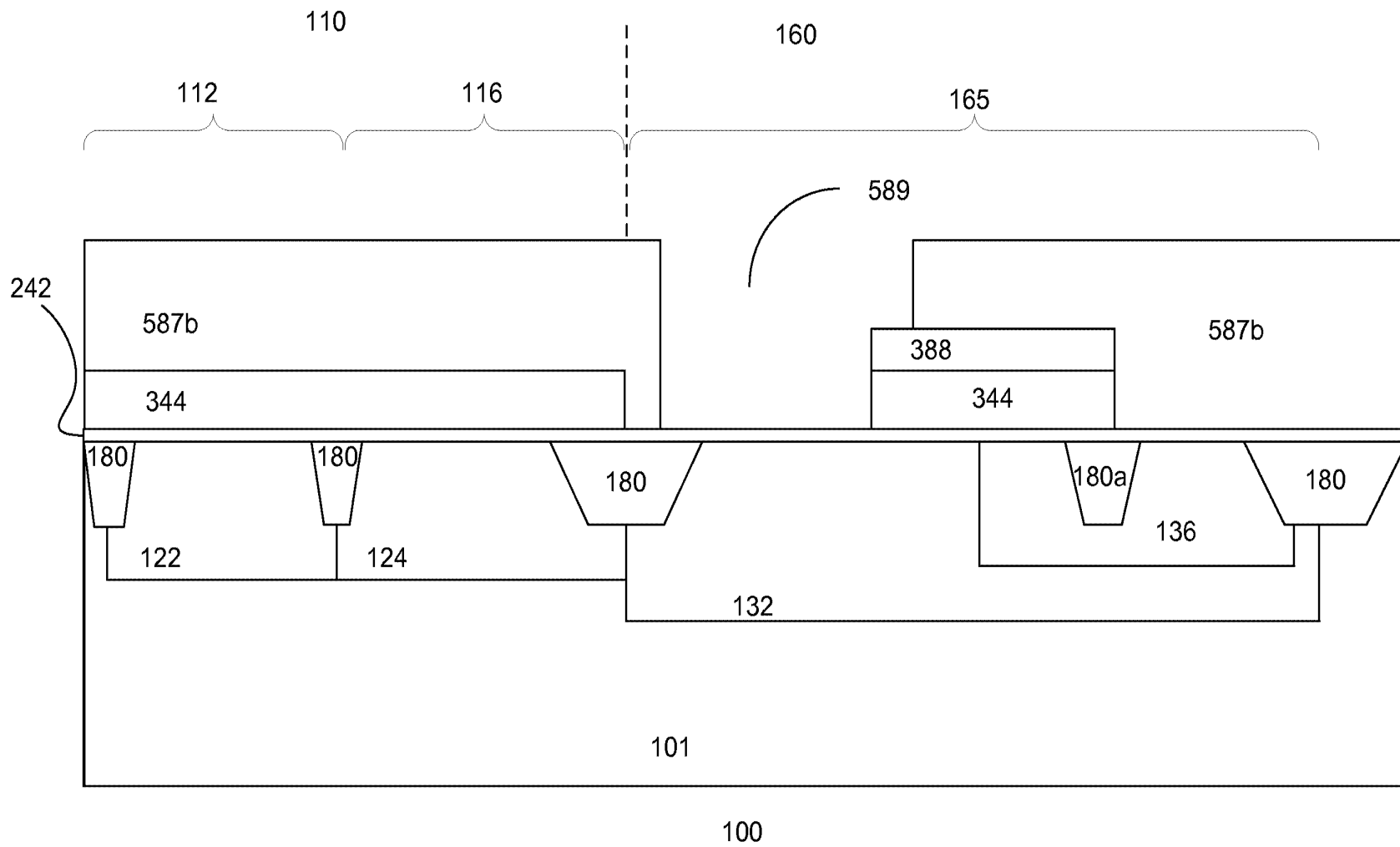


Fig. 5d

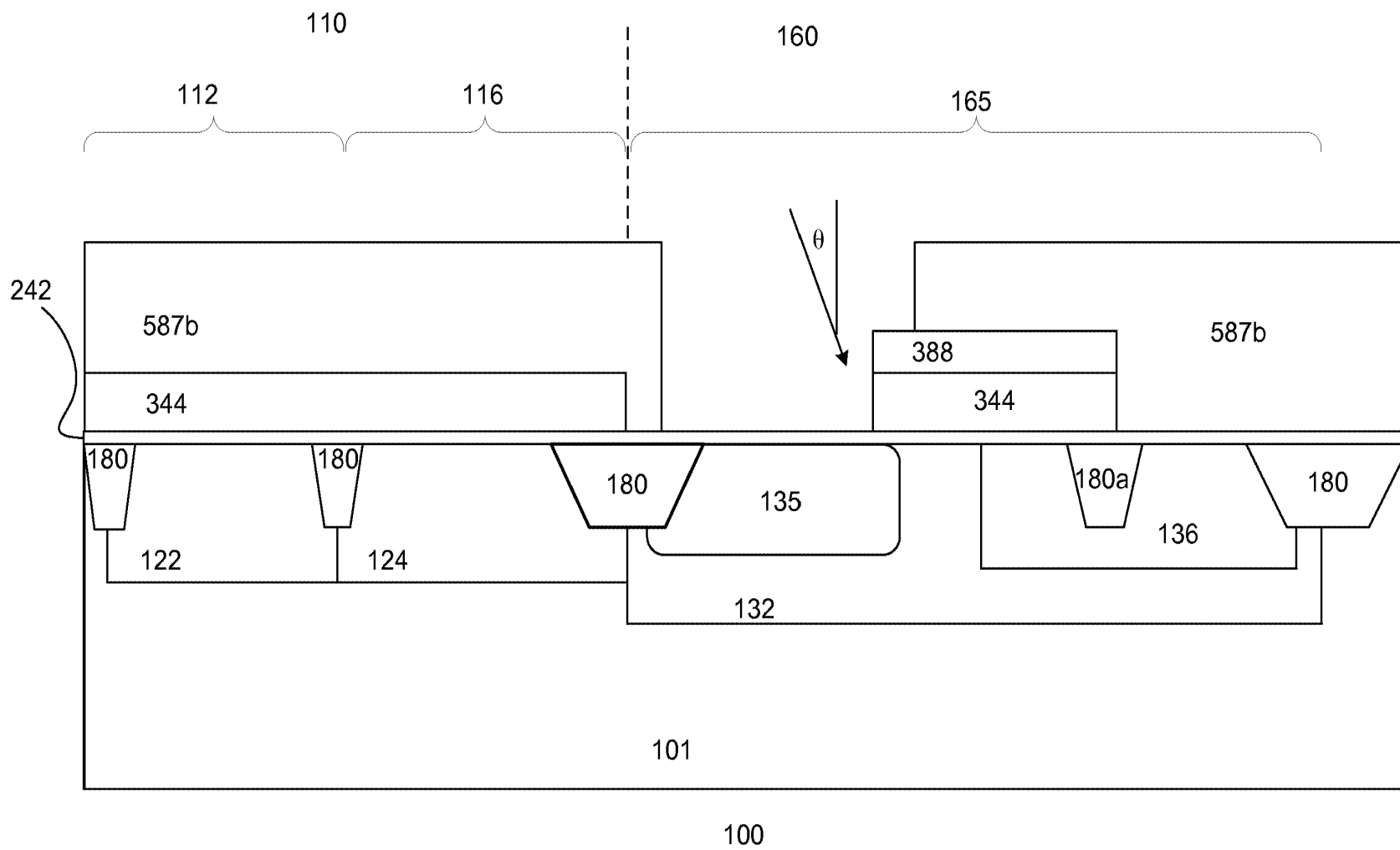


Fig. 5e

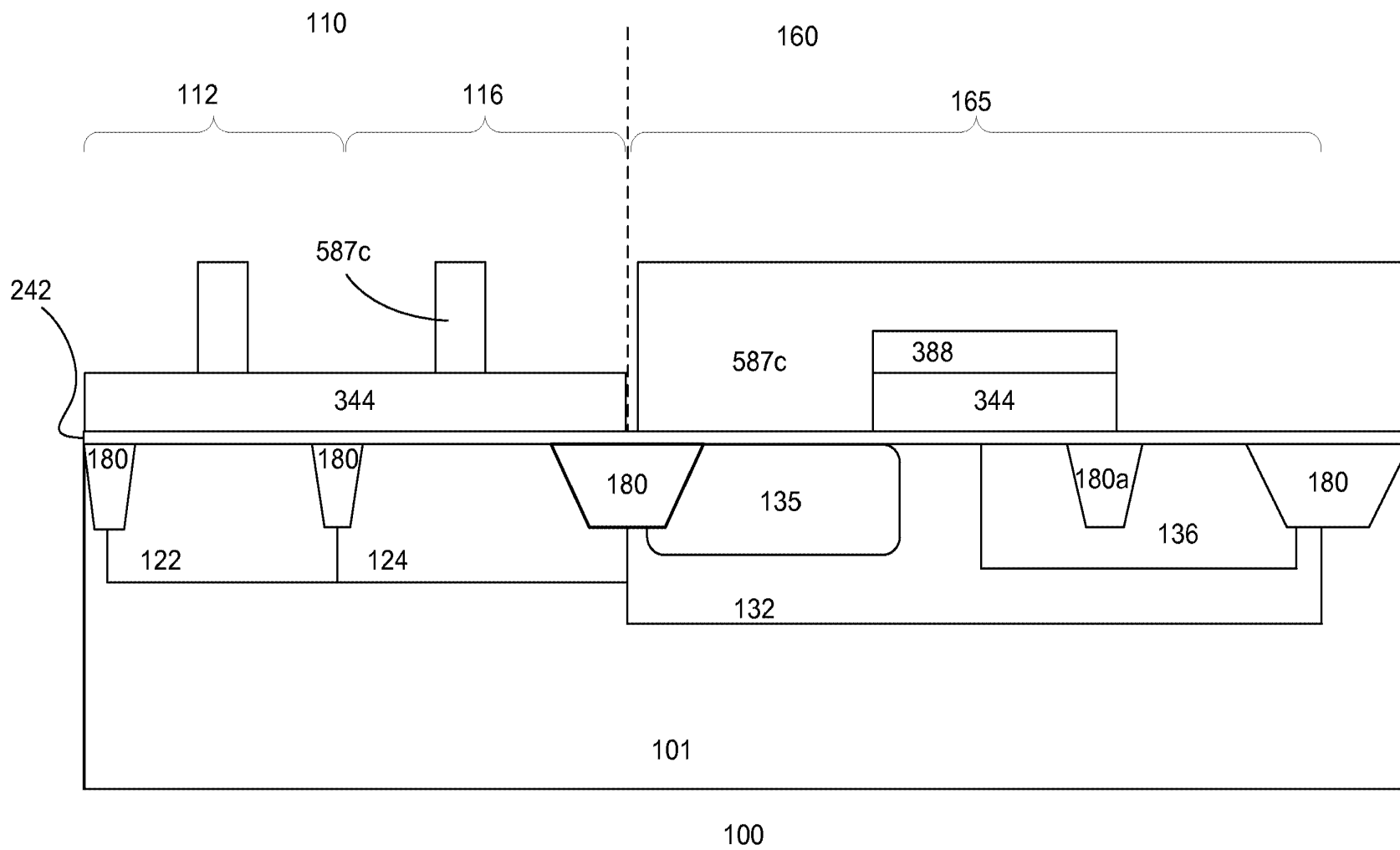


Fig. 5f

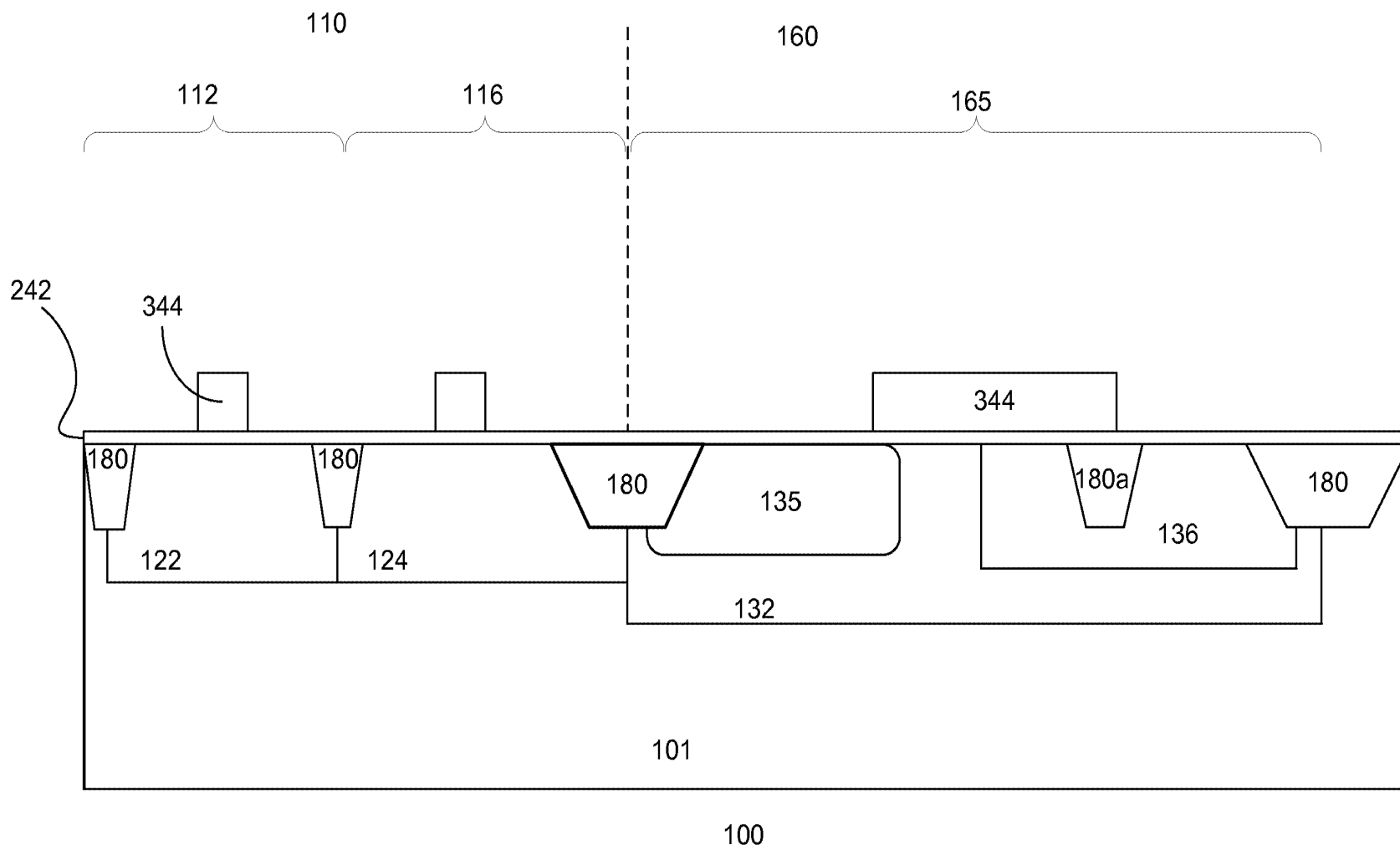


Fig. 5g

DECLARATION AND POWER OF ATTORNEY

As a below named inventor, I hereby declare that:

My residence, post office address, and citizenship are as stated below next to my name,

I believe I am the original, first and sole inventor (if only one name is listed below) or an original, first and joint inventor (if plural names are listed below) of the subject matter claimed and for which a patent is sought on the invention entitled:

HIGH VOLTAGE DEVICE

the specification of which

☒ is attached hereto

☐ was filed on ___ as Application Serial No. ___ and was amended on ___ (if applicable).

I hereby state that I have reviewed and understand the contents of the above-identified specification, including the claims, as amended by any amendment referred to above.

I do not know and do not believe the same was ever known or used in the United States of America before this invention thereof or more than one year prior to this application, and that the same was not in public use or on sale in the United States of America more than one year prior to this application. I acknowledge the duty to disclose information which is known to me to be material to patentability in accordance with title 37, Code of Federal Regulations, section 1.56.

I hereby claim foreign priority benefits under Title 35, United States Code, Section 119(a)-(d) or Section 365(b) of any foreign application(s) for patent or inventor's certificate, or Section 365(a) of any PCT international application which designated at least one country other than the United States, listed below and have also identified below any foreign application for patent or inventor's certificate having a filing date before that of the application on which priority is claimed:

Prior Foreign Application(s):

<u>Number</u>	<u>Country</u>	<u>Day/Month/Year filed</u>	<u>Priority Claimed</u>	
			<u>Yes</u>	<u>No</u>
				☒

I hereby claim the benefit under 35 USC 119(e) of any United States provisional application(s) listed below:

Prior Provisional Application(s):

<u>Application Number</u>	<u>Filing Date</u>

I hereby claim the benefit under Title 35, United States Code, Section 120 of any United States application(s), or Section 365(c) of any PCT international application designating the United States, listed below and, insofar as the subject matter of each of the claims of this application is not disclosed in the prior United States or PCT international application in the manner provided by the first paragraph of Title 35, United States Code, Section 112, I acknowledge the duty to disclose material information as defined in Title 37, Code of Federal

Regulations, Section 1.56 which occurred between the filing date of the prior application and the national or PCT international filing date of this application:

Prior U.S. Application(s):

<u>Serial No.</u>	<u>Filing Date</u>	<u>Status: Patented, Pending, Abandoned</u>
12/390,509	02/23/2009	Pending

I hereby declare that all statements made herein of my own knowledge are true and that all statements made on information and belief are believed to be true; and further that these statements were made with the knowledge that willful false statements and the like so made are punishable by fine or imprisonment, or both, under Section 1001 of Title 18 of the United States Code and that such willful false statements may jeopardize the validity of the application or any patent issued thereon.

I hereby revoke any previous Powers of Attorney and appoint the following attorney(s) and/or agent(s), each said individual being a member or associate of **Horizon IP Pte. Ltd.** and associated with:

Customer Number: 31366

for so long as they remain with such law offices or company with full power of substitution and revocation, to prosecute this application and to transact in connection therewith all business in the Patent and Trademark Office and before competent International Authorities; said appointment to be to the exclusion of myself and my other attorney(s).

Correspondence should be addressed to the address associated with:

Customer Number: 31366

.....

Inventor's signature: _____

Date: 03/07/2009

Full name of sole or first inventor:

Guo Wei ZHANG

Citizenship:

SG

Residence Address:

Block 426, Woodlands Street 41, #10-206,
Singapore 730426

P.O. Address:

Chartered Semiconductor Manufacturing, Ltd.
60 Woodlands Industrial Park D Street 2
Singapore 738406

Inventor's signature: 

Date: 3/07/09

Full name of second inventor:

Purakh Raj VERMA

Citizenship:

SG

Residence Address:

29 Rosewood Drive, #04-23, Casablanca,
Singapore 737291

P.O. Address:

Chartered Semiconductor Manufacturing, Ltd.
60 Woodlands Industrial Park D Street 2
Singapore 738406

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	CSM P 2009 NAT 03 US 1
		Application Number	
Title of Invention	HIGH VOLTAGE DEVICE		
The application data sheet is part of the provisional or nonprovisional application for which it is being submitted. The following form contains the bibliographic data arranged in a format specified by the United States Patent and Trademark Office as outlined in 37 CFR 1.76. This document may be completed electronically and submitted to the Office in electronic format using the Electronic Filing System (EFS) or the document may be printed and included in a paper filed application.			

Secrecy Order 37 CFR 5.2

- ☐ Portions or all of the application associated with this Application Data Sheet may fall under a Secrecy Order pursuant to 37 CFR 5.2 (Paper filers only. Applications that fall under Secrecy Order may not be filed electronically.)

Applicant Information:

Applicant 1					Remove
Applicant Authority		☒ Inventor		☐ Legal Representative under 35 U.S.C. 117	☐ Party of Interest under 35 U.S.C. 118
Prefix	Given Name	Middle Name	Family Name		Suffix
	Guowei		ZHANG		
Residence Information (Select One) ☐ US Residency ☒ Non US Residency ☐ Active US Military Service					
City	Singapore		Country Of Residenceⁱ	SG	
Citizenship under 37 CFR 1.41(b)ⁱ		SG			
Mailing Address of Applicant:					
Address 1		60 Woodlands Industrial Park D Street 2			
Address 2					
City	Singapore		State/Province		
Postal Code	738406		Countryⁱ	SG	
Applicant 2					Remove
Applicant Authority		☒ Inventor		☐ Legal Representative under 35 U.S.C. 117	☐ Party of Interest under 35 U.S.C. 118
Prefix	Given Name	Middle Name	Family Name		Suffix
	Purakh Raj		VERMA		
Residence Information (Select One) ☐ US Residency ☒ Non US Residency ☐ Active US Military Service					
City	Singapore		Country Of Residenceⁱ	SG	
Citizenship under 37 CFR 1.41(b)ⁱ		SG			
Mailing Address of Applicant:					
Address 1		60 Woodlands Industrial Park D Street 2			
Address 2					
City	Singapore		State/Province		
Postal Code	738406		Countryⁱ	SG	
All Inventors Must Be Listed - Additional Inventor Information blocks may be generated within this form by selecting the Add button.					

Correspondence Information:

Enter either Customer Number or complete the Correspondence Information section below.
For further information see 37 CFR 1.33(a).

- ☐ An Address is being provided for the correspondence Information of this application.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	CSM P 2009 NAT 03 US 1	
		Application Number		
Title of Invention	HIGH VOLTAGE DEVICE			
Customer Number	31366			
Email Address	dexter.chin@horizonip.com.sg	Add Email		Remove Email

Application Information:

Title of the Invention	HIGH VOLTAGE DEVICE			
Attorney Docket Number	CSM P 2009 NAT 03 US 1	Small Entity Status Claimed ☐		
Application Type	Nonprovisional			
Subject Matter	Utility			
Suggested Class (if any)		Sub Class (if any)		
Suggested Technology Center (if any)				
Total Number of Drawing Sheets (if any)	29	Suggested Figure for Publication (if any)		

Publication Information:

☐	Request Early Publication (Fee required at time of Request 37 CFR 1.219)
☐	Request Not to Publish. I hereby request that the attached application not be published under 35 U.S.C. 122(b) and certify that the invention disclosed in the attached application has not and will not be the subject of an application filed in another country, or under a multilateral international agreement, that requires publication at eighteen months after filing.

Representative Information:

Representative information should be provided for all practitioners having a power of attorney in the application. Providing this information in the Application Data Sheet does not constitute a power of attorney in the application (see 37 CFR 1.32). Enter either Customer Number or complete the Representative Name section below. If both sections are completed the Customer Number will be used for the Representative Information during processing.			
Please Select One:	☒ Customer Number	☐ US Patent Practitioner	☐ Limited Recognition (37 CFR 11.9)
Customer Number	31366		

Domestic Benefit/National Stage Information:

This section allows for the applicant to either claim benefit under 35 U.S.C. 119(e), 120, 121, or 365(c) or indicate National Stage entry from a PCT application. Providing this information in the application data sheet constitutes the specific reference required by 35 U.S.C. 119(e) or 120, and 37 CFR 1.78(a)(2) or CFR 1.78(a)(4), and need not otherwise be made part of the specification.			
Prior Application Status	Pending	Remove	
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)
	Continuation in part of	12390509	2009-02-23
Additional Domestic Benefit/National Stage Data may be generated within this form by selecting the Add button.			Add

Foreign Priority Information:

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	CSM P 2009 NAT 03 US 1
		Application Number	
Title of Invention	HIGH VOLTAGE DEVICE		

This section allows for the applicant to claim benefit of foreign priority and to identify any prior foreign application for which priority is not claimed. Providing this information in the application data sheet constitutes the claim for priority as required by 35 U.S.C. 119(b) and 37 CFR 1.55(a).

			Remove
Application Number	Country ⁱ	Parent Filing Date (YYYY-MM-DD)	Priority Claimed
			☐ Yes ☒ No
Additional Foreign Priority Data may be generated within this form by selecting the Add button.			Add

Assignee Information:

Providing this information in the application data sheet does not substitute for compliance with any requirement of part 3 of Title 37 of the CFR to have an assignment recorded in the Office.

Assignee 1			Remove
If the Assignee is an Organization check here. ☒			
Organization Name	Chartered Semiconductor Manufacturing, Ltd.		
Mailing Address Information:			
Address 1	60 Woodlands Industrial Park D Street 2		
Address 2			
City	Singapore	State/Province	
Country ⁱ	SG	Postal Code	738406
Phone Number	+65 6362 2838	Fax Number	+65 6360 4970
Email Address			
Additional Assignee Data may be generated within this form by selecting the Add button.			Add

Signature:

A signature of the applicant or representative is required in accordance with 37 CFR 1.33 and 10.18. Please see 37 CFR 1.4(d) for the form of the signature.

Signature	/dexter chin/		Date (YYYY-MM-DD)	2009-07-10
First Name	Dexter	Last Name	CHIN	Registration Number
				38842

This collection of information is required by 37 CFR 1.76. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 23 minutes to complete, including gathering, preparing, and submitting the completed application data sheet form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

DECLARATION AND POWER OF ATTORNEY

As a below named inventor, I hereby declare that:

My residence, post office address, and citizenship are as stated below next to my name,

I believe I am the original, first and sole inventor (if only one name is listed below) or an original, first and joint inventor (if plural names are listed below) of the subject matter claimed and for which a patent is sought on the invention entitled:

HIGH VOLTAGE DEVICE

the specification of which

☒ is attached hereto

☐ was filed on ___ as Application Serial No. ___ and was amended on ___ (if applicable).

I hereby state that I have reviewed and understand the contents of the above-identified specification, including the claims, as amended by any amendment referred to above.

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Prior Foreign Application(s):

<u>Number</u>	<u>Country</u>	<u>Day/Month/Year filed</u>	<u>Priority Claimed</u>	
			<u>Yes</u>	<u>No</u>
				☒

I hereby claim the benefit under 35 USC 119(e) of any United States provisional application(s) listed below:

Prior Provisional Application(s):

<u>Application Number</u>	<u>Filing Date</u>

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Regulations, Section 1.56 which occurred between the filing date of the prior application and the national or PCT international filing date of this application:

Prior U.S. Application(s):

<u>Serial No.</u>	<u>Filing Date</u>	<u>Status: Patented, Pending, Abandoned</u>
12/390,509	02/23/2009	Pending

I hereby declare that all statements made herein of my own knowledge are true and that all statements made on information and belief are believed to be true; and further that these statements were made with the knowledge that willful false statements and the like so made are punishable by fine or imprisonment, or both, under Section 1001 of Title 18 of the United States Code and that such willful false statements may jeopardize the validity of the application or any patent issued thereon.

I hereby revoke any previous Powers of Attorney and appoint the following attorney(s) and/or agent(s), each said individual being a member or associate of **Horizon IP Pte. Ltd.** and associated with:

Customer Number: 31366

for so long as they remain with such law offices or company with full power of substitution and revocation, to prosecute this application and to transact in connection therewith all business in the Patent and Trademark Office and before competent International Authorities; said appointment to be to the exclusion of myself and my other attorney(s).

Correspondence should be addressed to the address associated with:

Customer Number: 31366

.....

Inventor's signature: _____

Date: 03/07/2009

Full name of sole or first inventor:

Guo Wei ZHANG

Citizenship:

SG

Residence Address:

Block 426, Woodlands Street 41, #10-206,
Singapore 730426

P.O. Address:

Chartered Semiconductor Manufacturing, Ltd.
60 Woodlands Industrial Park D Street 2
Singapore 738406

Inventor's signature: 

Date: 3/07/09

Full name of second inventor:

Purakh Raj VERMA

Citizenship:

SG

Residence Address:

29 Rosewood Drive, #04-23, Casablanca,
Singapore 737291

P.O. Address:

Chartered Semiconductor Manufacturing, Ltd.
60 Woodlands Industrial Park D Street 2
Singapore 738406

Filing Date: 07/10/09

Approved for use through 7/31/2006. OMB 0651-0032

U.S. Patent and Trademark Office; U.S. DEPARTMENT OF COMMERCE

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PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875					Application or Docket Number 12/500,620	
APPLICATION AS FILED – PART I						
(Column 1)		(Column 2)		SMALL ENTITY		OR
OTHER THAN SMALL ENTITY						
FOR	NUMBER FILED	NUMBER EXTRA	RATE (\$)	FEE (\$)	RATE (\$)	FEE (\$)
BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A		N/A	330
SEARCH FEE (37 CFR 1.16(k), (l), or (m))	N/A	N/A	N/A		N/A	540
EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A		N/A	220
TOTAL CLAIMS (37 CFR 1.16(i))	20	minus 20 =	x\$26		x\$52	
INDEPENDENT CLAIMS (37 CFR 1.16(h))	3	minus 3 =	x\$110		x\$220	
APPLICATION SIZE FEE (37 CFR 1.16(s))		If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$270 (\$135 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR				
MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))			195		390	
			TOTAL		TOTAL	1090
* If the difference in column 1 is less than zero, enter "0" in column 2.						
APPLICATION AS AMENDED – PART II						
(Column 1)		(Column 2)		(Column 3)		
AMENDMENT A	CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)	RATE (\$)
	Total (37 CFR 1.16(i))	Minus **	=	x =		x =
	Independent (37 CFR 1.16(h))	Minus ***	=	x =		x =
	Application Size Fee (37 CFR 1.16(s))					
	FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))			N/A		N/A
			TOTAL ADD'T FEE		TOTAL ADD'T FEE	
(Column 1)		(Column 2)		(Column 3)		
AMENDMENT B	CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)	RATE (\$)
	Total (37 CFR 1.16(i))	Minus **	=	x =		x =
	Independent (37 CFR 1.16(h))	Minus ***	=	x =		x =
	Application Size Fee (37 CFR 1.16(s))					
	FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))			N/A		N/A
			TOTAL ADD'T FEE		TOTAL ADD'T FEE	
* If the entry in column 1 is less than the entry in column 2, write "0" in column 3. ** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20". *** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3". The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.						

This collection of information is required by 37 CFR 1.16. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.

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www.uspto.gov

APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
12/500,620	07/10/2009	2812	1090	CSM P 2009 NAT 03 US 1	20	3

CONFIRMATION NO. 2701

31366
HORIZON IP PTE LTD
7500A Beach Road
#04-306/308 The Plaza
SINGAPORE 199591,
SINGAPORE

FILING RECEIPT



OC000000037003815

Date Mailed: 07/27/2009

Receipt is acknowledged of this non-provisional patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF APPLICANT, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection. Please verify the accuracy of the data presented on this receipt. **If an error is noted on this Filing Receipt, please submit a written request for a Filing Receipt Correction. Please provide a copy of this Filing Receipt with the changes noted thereon. If you received a "Notice to File Missing Parts" for this application, please submit any corrections to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections**

Applicant(s)

Guo Wei ZHANG, Singapore, SINGAPORE;
Purakh Raj VERMA, Singapore, SINGAPORE;

Assignment For Published Patent Application

CHARTERED SEMICONDUCTOR MANUFACTURING, LTD., Singapore, SINGAPORE

Power of Attorney: The patent practitioners associated with Customer Number 31366

Domestic Priority data as claimed by applicant

This application is a CIP of 12/390,509 02/23/2009

Foreign Applications

If Required, Foreign Filing License Granted: 07/20/2009

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 12/500,620**

Projected Publication Date: 08/26/2010

Non-Publication Request: No

Early Publication Request: No

Title

HIGH VOLTAGE DEVICE

Preliminary Class

438

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

Almost every country has its own patent law, and a person desiring a patent in a particular country must make an application for patent in that country in accordance with its particular laws. Since the laws of many countries differ in various respects from the patent law of the United States, applicants are advised to seek guidance from specific foreign countries to ensure that patent rights are not lost prematurely.

Applicants also are advised that in the case of inventions made in the United States, the Director of the USPTO must issue a license before applicants can apply for a patent in a foreign country. The filing of a U.S. patent application serves as a request for a foreign filing license. The application's filing receipt contains further information and guidance as to the status of applicant's license for foreign filing.

Applicants may wish to consult the USPTO booklet, "General Information Concerning Patents" (specifically, the section entitled "Treaties and Foreign Patents") for more information on timeframes and deadlines for filing foreign patent applications. The guide is available either by contacting the USPTO Contact Center at 800-786-9199, or it can be viewed on the USPTO website at <http://www.uspto.gov/web/offices/pac/doc/general/index.html>.

For information on preventing theft of your intellectual property (patents, trademarks and copyrights), you may wish to consult the U.S. Government website, <http://www.stopfakes.gov>. Part of a Department of Commerce initiative, this website includes self-help "toolkits" giving innovators guidance on how to protect intellectual property in specific countries such as China, Korea and Mexico. For questions regarding patent enforcement issues, applicants may call the U.S. Government hotline at 1-866-999-HALT (1-866-999-4158).

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set forth in 37 CFR 5.15. The scope and limitations of this license are set forth in 37 CFR 5.15(a) unless an earlier license has been issued under 37 CFR 5.15(b). The license is subject to revocation upon written notification. The date indicated is the effective date of the license, unless an earlier license of similar scope has been granted under 37 CFR 5.13 or 5.14.

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NOT GRANTED

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APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
12/500,620	07/10/2009	Guo Wei ZHANG	CSM P 2009 NAT 03 US 1

CONFIRMATION NO. 2701

PUBLICATION NOTICE



OC000000043223793

31366
HORIZON IP PTE LTD
7500A Beach Road
#04-306/308 The Plaza
SINGAPORE 199591,
SINGAPORE

Title:HIGH VOLTAGE DEVICE

Publication No.US-2010-0213543-A1

Publication Date:08/26/2010

NOTICE OF PUBLICATION OF APPLICATION

The above-identified application will be electronically published as a patent application publication pursuant to 37 CFR 1.211, et seq. The patent application publication number and publication date are set forth above.

The publication may be accessed through the USPTO's publically available Searchable Databases via the Internet at www.uspto.gov. The direct link to access the publication is currently <http://www.uspto.gov/patft/>.

The publication process established by the Office does not provide for mailing a copy of the publication to applicant. A copy of the publication may be obtained from the Office upon payment of the appropriate fee set forth in 37 CFR 1.19(a)(1). Orders for copies of patent application publications are handled by the USPTO's Office of Public Records. The Office of Public Records can be reached by telephone at (703) 308-9726 or (800) 972-6382, by facsimile at (703) 305-8759, by mail addressed to the United States Patent and Trademark Office, Office of Public Records, Alexandria, VA 22313-1450 or via the Internet.

In addition, information on the status of the application, including the mailing date of Office actions and the dates of receipt of correspondence filed in the Office, may also be accessed via the Internet through the Patent Electronic Business Center at www.uspto.gov using the public side of the Patent Application Information and Retrieval (PAIR) system. The direct link to access this status information is currently <http://pair.uspto.gov/>. Prior to publication, such status information is confidential and may only be obtained by applicant using the private side of PAIR.

Further assistance in electronically accessing the publication, or about PAIR, is available by calling the Patent Electronic Business Center at 1-866-217-9197.

Office of Data Management, Application Assistance Unit (571) 272-4000, or (571) 272-4200, or 1-888-786-0101



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APPLICATION NUMBER	PATENT NUMBER	GROUP ART UNIT	FILE WRAPPER LOCATION
12/500,620		2892	



Correspondence Address/Fee Address Change

The following fields have been set to Customer Number 16162 on 07/19/2011

- Correspondence Address

The address of record for Customer Number 16162 is:

16162
Horizon IP Pte Ltd.
7500A Beach Road, #04-306/308
The Plaza
Singapore, 199591
SINGAPORE



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APPLICATION NUMBER	PATENT NUMBER	GROUP ART UNIT	FILE WRAPPER LOCATION
12/500,620		2892	



Correspondence Address/Fee Address Change

The following fields have been set to Customer Number 16162 on 07/20/2011

- Correspondence Address

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16162
Horizon IP Pte Ltd.
7500A Beach Road, #04-306/308
The Plaza
Singapore, 199591
SINGAPORE



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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
12/500,620	07/10/2009	Guo Wei ZHANG	CSM P 2009 NAT 03 US 1	2701
16162	7590	01/12/2012		
Horizon IP Pte Ltd. 7500A Beach Road, #04-306/308 The Plaza Singapore, 199591 SINGAPORE			EXAMINER HOANG, QUOC DINH	
			ART UNIT 2892	PAPER NUMBER
			NOTIFICATION DATE 01/12/2012	DELIVERY MODE ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

oa.notification@horizonip.com.sg
dexter.chin@horizonip.com.sg
info@horizonip.com.sg

Office Action Summary	Application No.	Applicant(s)	
	12/500,620	ZHANG ET AL.	
	Examiner	Art Unit	
	QUOC HOANG	2892	

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 1 MONTH(S) OR THIRTY (30) DAYS, WHICHEVER IS LONGER, FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

- 1) ☒ Responsive to communication(s) filed on 10 July 2009.
- 2a) ☐ This action is **FINAL**. 2b) ☒ This action is non-final.
- 3) ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on ____; the restriction requirement and election have been incorporated into this action.
- 4) ☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims

- 5) ☒ Claim(s) 1-20 is/are pending in the application.
- 5a) Of the above claim(s) ____ is/are withdrawn from consideration.
- 6) ☐ Claim(s) ____ is/are allowed.
- 7) ☐ Claim(s) ____ is/are rejected.
- 8) ☐ Claim(s) ____ is/are objected to.
- 9) ☒ Claim(s) 1-20 are subject to restriction and/or election requirement.

Application Papers

- 10) ☐ The specification is objected to by the Examiner.
- 11) ☐ The drawing(s) filed on ____ is/are: a) ☐ accepted or b) ☐ objected to by the Examiner.
Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).
Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).
- 12) ☐ The oath or declaration is objected to by the Examiner. Note the attached Office Action or form PTO-152.

Priority under 35 U.S.C. § 119

- 13) ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).
- a) ☐ All b) ☐ Some * c) ☐ None of:
1. ☐ Certified copies of the priority documents have been received.
2. ☐ Certified copies of the priority documents have been received in Application No. ____.
3. ☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

* See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

- | | |
|--|---|
| 1) ☐ Notice of References Cited (PTO-892) | 4) ☐ Interview Summary (PTO-413) |
| 2) ☐ Notice of Draftsperson's Patent Drawing Review (PTO-948) | Paper No(s)/Mail Date. ____. |
| 3) ☐ Information Disclosure Statement(s) (PTO/SB/08) | 5) ☐ Notice of Informal Patent Application |
| Paper No(s)/Mail Date ____. | 6) ☐ Other: ____. |

DETAILED ACTION

Election/Restrictions

1. Restriction to one of the following inventions is required under 35 U.S.C. 121:
 - I. Claims 19-20, drawn to a device, classified in class 257, subclass 213.
 - II. Claims 1-18, drawn to a method of manufacturing a device, classified in class 438, subclass 199.
2. The inventions are distinct, each from the other because of the following reasons:

Inventions II and I are related as process of making and product made. The inventions are distinct if either or both of the following can be shown: (1) that the process as claimed can be used to make other and materially different product or (2) that the product as claimed can be made by another and materially different process (MPEP § 806.05(f)). In the instant case unpatentability of Group I invention would not necessarily imply unpatentability of the process of the Group II invention, since the device of group I invention could be made by processes different from those of group II invention, for example, in claim 19, the gate electrode could be formed without using the first mask.
3. Because these inventions are distinct for the reasons given above and have acquired a separate status in the art as shown by their different classification, the fields of search are not co-extensive and separate examination would be required, restriction for examination purposes as indicated is proper.
4. Applicant is advised that the reply to this requirement to be complete must include an election of the invention to be examined even though the requirement be traversed (37 CFR 1.143).

Art Unit: 2892

5. Applicant is reminded that upon the cancellation of claims to a non-elected invention, the inventorship must be amended in compliance with 37 CFR 1.48(b) of one or more of the currently named inventors is no longer an inventor of at least one claim remaining in the application. Any amendment of the inventorship must be accompanied by a petition under 37 CFR 1.48(b) and by the fee required under 37 CFR 1.17(I).

Conclusion

6. Any inquiry concerning this communication or earlier communications from the examiner should be directed to Quoc Hoang whose telephone number is (571) 272-1780. The examiner can normally be reached on Monday-Friday from 8.00 AM to 5.00 PM.

If attempt to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Thao Le can be reached on (571) 272-1708. The fax phone numbers of the organization where this application or proceeding is assigned are (571) 273-8300 for regular communications and (571) 273-8300 for After Final communications.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free).

/Quoc D Hoang/

Primary Examiner, Art Unit 2892

Application/Control Number: 12/500,620

Page 4

Art Unit: 2892

<i>Index of Claims</i> 	Application/Control No. 12500620	Applicant(s)/Patent Under Reexamination ZHANG ET AL.
	Examiner QUOC HOANG	Art Unit 2892

✓	Rejected	-	Cancelled	N	Non-Elected	A	Appeal
=	Allowed	÷	Restricted	I	Interference	O	Objected

☐ Claims renumbered in the same order as presented by applicant ☐ CPA ☐ T.D. ☐ R.1.47										
CLAIM		DATE								
Final	Original	01/06/2012								
	1	÷								
	2	÷								
	3	÷								
	4	÷								
	5	÷								
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	20	÷								

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The Plaza
Singapore, 199591
SINGAPORE



**Courtesy Reminder for
Application Serial No: 12/500,620**

Attorney Docket No: CSM P 2009 NAT 03 US 1
Customer Number: 16162
Date of Electronic Notification: 01/12/2012

This is a courtesy reminder that new correspondence is available for this application. The official date of notification of the outgoing correspondence will be indicated on the form PTOL-90 accompanying the correspondence.

An email notification regarding the correspondence was sent to the following email address(es) associated with your customer number:

oa.notification@horizonip.com.sg
dexter.chin@horizonip.com.sg
info@horizonip.com.sg

Please verify that these email addresses are correct.

To view your correspondence online or update your email addresses, please visit us anytime at <https://sportal.uspto.gov/secure/myportal/privatepair>. If you have any questions, please email the Electronic Business Center (EBC) at EBC@uspto.gov or call 1-866-217-9197.

Appl. No. 12/500,620
Amdt dated February 10, 2012
Reply to Office Action dated January 12, 2012

PATENT

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Appl. No. : 12/500,620 Confirmation No. : 2701
Applicant : Guo Wei ZHANG et al.
Filed : 07/10/2009
TC / A.U. : 2892
Examiner : HOANG, QUOC DINH
Docket No. : CSM P 2009 NAT 03 US 1
Customer No. : 016162
Title : HIGH VOLTAGE DEVICE

CERTIFICATE OF TRANSMISSION

I hereby certify that this correspondence is being electronically transmitted to the United States Patent and Trademark Office, Alexandria, VA 22313-1450 on February 10, 2012.

Signed:

/nannan chen/

Nannan CHEN

Mail Stop AMENDMENT
Commissioner For Patents
P.O. Box 1450
Alexandria, VA 22313-1450

AMENDMENT

Sir:

In response to the Office Action mailed January 12, 2012, Applicants elect group II invention (claims 1-18) for prosecution on the merits and cancel group I invention (claims 19-20) from consideration without prejudice.

In view of the foregoing, Applicants believe that all claims now pending in this application are in condition for allowance. The issuance of a formal Notice of Allowance at an early date is respectfully requested.

Appl. No. 12/500,620
Amdt dated February 10, 2012
Reply to Office Action dated January 12, 2012

Should the Examiner believe that a telephone conference would expedite prosecution of this application, please telephone the undersigned attorney at his number set out below.

Date: February 10, 2012

Respectfully submitted,

/dexter chin/

Dexter CHIN
Attorney for Applicants
Reg. No.: 38,842

Horizon IP Pte Ltd
7500A Beach Road
#04-306/308 The Plaza
Singapore 199591
Tel.: (65) 9836 9908
Fax: (65) 6846 2005
Email: dexter.chin@horizonip.com.sg

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

TRANSMITTAL FORM <i>(to be used for all correspondence after initial filing)</i>	Application Number	12/500,620	
	Filing Date	07/10/2009	
	First Named Inventor	Guo Wei ZHANG	
	Art Unit	2892	
	Examiner Name	HOANG, QUOC DINH	
Total Number of Pages in This Submission	4	Attorney Docket Number	CSM P 2009 NAT 03 US 1

ENCLOSURES (Check all that apply)		
☐ Fee Transmittal Form ☐ Fee Attached ☒ Amendment/Reply ☐ After Final ☐ Affidavits/declaration(s) ☐ Extension of Time Request ☐ Express Abandonment Request ☐ Information Disclosure Statement ☐ Certified Copy of Priority Document(s) ☐ Reply to Missing Parts/ Incomplete Application ☐ Reply to Missing Parts under 37 CFR 1.52 or 1.53	☐ Drawing(s) ☐ Licensing-related Papers ☐ Petition ☐ Petition to Convert to a Provisional Application ☐ Power of Attorney, Revocation Change of Correspondence Address ☐ Terminal Disclaimer ☐ Request for Refund ☐ CD, Number of CD(s) _____ ☐ Landscape Table on CD	☐ After Allowance Communication to TC ☐ Appeal Communication to Board of Appeals and Interferences ☐ Appeal Communication to TC (Appeal Notice, Brief, Reply Brief) ☐ Proprietary Information ☐ Status Letter ☐ Other Enclosure(s) (please identify below):
Remarks		
SIGNATURE OF APPLICANT, ATTORNEY, OR AGENT		
Firm Name	Horizon IP Pte. Ltd.	
Signature	/dexter chin/	
Printed name	Dexter CHIN	
Date	02/10/2012	Reg. No. 38,842

CERTIFICATE OF TRANSMISSION/MAILING		
I hereby certify that this correspondence is being facsimile transmitted to the USPTO or deposited with the United States Postal Service with sufficient postage as first class mail in an envelope addressed to: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450 on the date shown below:		
Signature	/nannan chen/	
Typed or printed name	Nannan CHEN	Date 02/10/2012

This collection of information is required by 37 CFR 1.5. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to 2 hours to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

Privacy Act Statement

The **Privacy Act of 1974 (P.L. 93-579)** requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (*i.e.*, GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Electronic Acknowledgement Receipt

EFS ID:	12045324
Application Number:	12500620
International Application Number:	
Confirmation Number:	2701
Title of Invention:	HIGH VOLTAGE DEVICE
First Named Inventor/Applicant Name:	Guo Wei ZHANG
Customer Number:	16162
Filer:	Dexter Kam Chin
Filer Authorized By:	
Attorney Docket Number:	CSM P 2009 NAT 03 US 1
Receipt Date:	10-FEB-2012
Filing Date:	10-JUL-2009
Time Stamp:	00:38:01
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Response to Election / Restriction Filed	CSMP2009NAT03US1- ResponsetoER.pdf	76065 f41ee305ead579d35687d6a51b15e697c6552083	no	2

Warnings:

Information:

2	Miscellaneous Incoming Letter	CSMP2009NAT03US1-TF.pdf	277435	no	2
			e5933190f189af8fd0c70aea250a398aeeb8a7a0		
Warnings:					
Information:					
Total Files Size (in bytes):				353500	
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					



UNITED STATES PATENT AND TRADEMARK OFFICE

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United States Patent and Trademark Office
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P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

NOTICE OF ALLOWANCE AND FEE(S) DUE

16162 7590 03/19/2012
Horizon IP Pte Ltd.
7500A Beach Road, #04-306/308
The Plaza
Singapore, 199591
SINGAPORE

EXAMINER

HOANG, QUOC DINH

ART UNIT

PAPER NUMBER

2892

DATE MAILED: 03/19/2012

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
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12/500,620

07/10/2009

Guo Wei ZHANG

CSM P 2009 NAT 03 US 1

2701

TITLE OF INVENTION: HIGH VOLTAGE DEVICE

APPLN. TYPE	SMALL ENTITY	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	NO	\$1740	\$300	\$0	\$2040	06/19/2012

THE APPLICATION IDENTIFIED ABOVE HAS BEEN EXAMINED AND IS ALLOWED FOR ISSUANCE AS A PATENT. PROSECUTION ON THE MERITS IS CLOSED. THIS NOTICE OF ALLOWANCE IS NOT A GRANT OF PATENT RIGHTS. THIS APPLICATION IS SUBJECT TO WITHDRAWAL FROM ISSUE AT THE INITIATIVE OF THE OFFICE OR UPON PETITION BY THE APPLICANT. SEE 37 CFR 1.313 AND MPEP 1308.

THE ISSUE FEE AND PUBLICATION FEE (IF REQUIRED) MUST BE PAID WITHIN THREE MONTHS FROM THE MAILING DATE OF THIS NOTICE OR THIS APPLICATION SHALL BE REGARDED AS ABANDONED. THIS STATUTORY PERIOD CANNOT BE EXTENDED. SEE 35 U.S.C. 151. THE ISSUE FEE DUE INDICATED ABOVE DOES NOT REFLECT A CREDIT FOR ANY PREVIOUSLY PAID ISSUE FEE IN THIS APPLICATION. IF AN ISSUE FEE HAS PREVIOUSLY BEEN PAID IN THIS APPLICATION (AS SHOWN ABOVE), THE RETURN OF PART B OF THIS FORM WILL BE CONSIDERED A REQUEST TO REAPPLY THE PREVIOUSLY PAID ISSUE FEE TOWARD THE ISSUE FEE NOW DUE.

HOW TO REPLY TO THIS NOTICE:

I. Review the SMALL ENTITY status shown above.

If the SMALL ENTITY is shown as YES, verify your current SMALL ENTITY status:

A. If the status is the same, pay the TOTAL FEE(S) DUE shown above.

B. If the status above is to be removed, check box 5b on Part B - Fee(s) Transmittal and pay the PUBLICATION FEE (if required) and twice the amount of the ISSUE FEE shown above, or

If the SMALL ENTITY is shown as NO:

A. Pay TOTAL FEE(S) DUE shown above, or

B. If applicant claimed SMALL ENTITY status before, or is now claiming SMALL ENTITY status, check box 5a on Part B - Fee(s) Transmittal and pay the PUBLICATION FEE (if required) and 1/2 the ISSUE FEE shown above.

II. PART B - FEE(S) TRANSMITTAL, or its equivalent, must be completed and returned to the United States Patent and Trademark Office (USPTO) with your ISSUE FEE and PUBLICATION FEE (if required). If you are charging the fee(s) to your deposit account, section "4b" of Part B - Fee(s) Transmittal should be completed and an extra copy of the form should be submitted. If an equivalent of Part B is filed, a request to reapply a previously paid issue fee must be clearly made, and delays in processing may occur due to the difficulty in recognizing the paper as an equivalent of Part B.

III. All communications regarding this application must give the application number. Please direct all communications prior to issuance to Mail Stop ISSUE FEE unless advised to the contrary.

IMPORTANT REMINDER: Utility patents issuing on applications filed on or after Dec. 12, 1980 may require payment of maintenance fees. It is patentee's responsibility to ensure timely payment of maintenance fees when due.

PART B - FEE(S) TRANSMITTAL

**Complete and send this form, together with applicable fee(s), to: Mail Mail Stop ISSUE FEE
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450
or Fax (571)-273-2885**

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence including the Patent, advance orders and notification of maintenance fees will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications.

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

16162 7590 03/19/2012
Horizon IP Pte Ltd.
7500A Beach Road, #04-306/308
The Plaza
Singapore, 199591
SINGAPORE

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being facsimile transmitted to the USPTO (571) 273-2885, on the date indicated below.

(Depositor's name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
-----------------	-------------	----------------------	---------------------	------------------

12/500,620 07/10/2009 Guo Wei ZHANG CSM P 2009 NAT 03 US 1 2701

TITLE OF INVENTION: HIGH VOLTAGE DEVICE

APPLN. TYPE	SMALL ENTITY	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
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nonprovisional NO \$1740 \$300 \$0 \$2040 06/19/2012

EXAMINER	ART UNIT	CLASS-SUBCLASS
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HOANG, QUOC DINH 2892 257-335000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

- ☐ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached.
☐ "Fee Address" indication (or "Fee Address" Indication form PTO/SB/47; Rev 03-02 or more recent) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list

- (1) the names of up to 3 registered patent attorneys or agents OR, alternatively, 1 _____
(2) the name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed. 2 _____
3 _____

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document has been filed for recordation as set forth in 37 CFR 3.11. Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE (B) RESIDENCE: (CITY and STATE OR COUNTRY)

Please check the appropriate assignee category or categories (will not be printed on the patent) : ☐ Individual ☐ Corporation or other private group entity ☐ Government

4a. The following fee(s) are submitted:

- ☐ Issue Fee
☐ Publication Fee (No small entity discount permitted)
☐ Advance Order - # of Copies _____

4b. Payment of Fee(s); (Please first reapply any previously paid issue fee shown above)

- ☐ A check is enclosed.
☐ Payment by credit card. Form PTO-2038 is attached.
☐ The Director is hereby authorized to charge the required fee(s), any deficiency, or credit any overpayment, to Deposit Account Number _____ (enclose an extra copy of this form).

5. Change in Entity Status (from status indicated above)

- ☐ a. Applicant claims SMALL ENTITY status. See 37 CFR 1.27. ☐ b. Applicant is no longer claiming SMALL ENTITY status. See 37 CFR 1.27(g)(2).

NOTE: The Issue Fee and Publication Fee (if required) will not be accepted from anyone other than the applicant; a registered attorney or agent; or the assignee or other party in interest as shown by the records of the United States Patent and Trademark Office.

Authorized Signature _____ Date _____

Typed or printed name _____ Registration No. _____

This collection of information is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450.

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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
12/500,620	07/10/2009	Guo Wei ZHANG	CSM P 2009 NAT 03 US 1	2701

16162 7590 03/19/2012
Horizon IP Pte Ltd.
7500A Beach Road, #04-306/308
The Plaza
Singapore, 199591
SINGAPORE

EXAMINER

HOANG, QUOC DINH

ART UNIT	PAPER NUMBER
----------	--------------

2892

DATE MAILED: 03/19/2012

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (application filed on or after May 29, 2000)

The Patent Term Adjustment to date is 489 day(s). If the issue fee is paid on the date that is three months after the mailing date of this notice and the patent issues on the Tuesday before the date that is 28 weeks (six and a half months) after the mailing date of this notice, the Patent Term Adjustment will be 489 day(s).

If a Continued Prosecution Application (CPA) was filed in the above-identified application, the filing date that determines Patent Term Adjustment is the filing date of the most recent CPA.

Applicant will be able to obtain more detailed information by accessing the Patent Application Information Retrieval (PAIR) WEB site (<http://pair.uspto.gov>).

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Customer Service Center of the Office of Patent Publication at 1-(888)-786-0101 or (571)-272-4200.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Notice of Allowability	Application No.	Applicant(s)	
	12/500,620	ZHANG ET AL.	
	Examiner	Art Unit	
	QUOC HOANG	2892	

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. ☒ This communication is responsive to 01/10/2012.
2. ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on ____; the restriction requirement and election have been incorporated into this action.
3. ☒ The allowed claim(s) is/are 1-18.
4. ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).
 - a) ☐ All b) ☐ Some* c) ☐ None of the:
 1. ☐ Certified copies of the priority documents have been received.
 2. ☐ Certified copies of the priority documents have been received in Application No. ____.
 3. ☐ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).

* Certified copies not received: ____.

Applicant has THREE MONTHS FROM THE "MAILING DATE" of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.

THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.

5. ☐ A SUBSTITUTE OATH OR DECLARATION must be submitted. Note the attached EXAMINER'S AMENDMENT or NOTICE OF INFORMAL PATENT APPLICATION (PTO-152) which gives reason(s) why the oath or declaration is deficient.
 6. ☐ CORRECTED DRAWINGS (as "replacement sheets") must be submitted.
 - (a) ☐ including changes required by the Notice of Draftsperson's Patent Drawing Review (PTO-948) attached
 - 1) ☐ hereto or 2) ☐ to Paper No./Mail Date ____.
 - (b) ☐ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date ____.
- Identifying indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).**
7. ☐ DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

Attachment(s)

- | | |
|---|--|
| <ol style="list-style-type: none"> 1. ☒ Notice of References Cited (PTO-892) 2. ☐ Notice of Draftsperson's Patent Drawing Review (PTO-948) 3. ☒ Information Disclosure Statements (PTO/SB/08),
Paper No./Mail Date <u>07/09</u>. 4. ☐ Examiner's Comment Regarding Requirement for Deposit of Biological Material | <ol style="list-style-type: none"> 5. ☐ Notice of Informal Patent Application 6. ☐ Interview Summary (PTO-413),
Paper No./Mail Date ____. 7. ☐ Examiner's Amendment/Comment 8. ☒ Examiner's Statement of Reasons for Allowance 9. ☐ Other ____. |
|---|--|

/Quoc D Hoang/
Primary Examiner, Art Unit 2892

DETAILED ACTION

Election/Restrictions

1. Applicant's election without traverse of Group II (claims 1-18) and cancel group I (claims 19-20) in the reply filed on 02/10/2012 is acknowledged.

Allowable Subject Matter

2. Claims 1-18 are allowed.

The following is an examiner's statement of reasons for allowance: the prior art of record taken along or in combination, fails to teach or reasonably suggests patterning the substrate to at least remove portions of a top gate stack layer unprotected by the first mask; and providing a second mask on the substrate with an opening exposing a portion of the first mask and the top gate stack layer.

Yang discloses a method of forming a LDMOS device.

Any comments considered necessary by applicant must be submitted no later than the payment of the issue fee and, to avoid processing delays, should preferably accompany the issue fee. Such submissions should be clearly labeled "Comments on Statement of Reasons for Allowance."

Conclusion

3. Any inquiry concerning this communication or earlier communications from the examiner should be directed to Quoc Hoang whose telephone number is (571) 272-1780. The examiner can normally be reached on Monday-Friday from 8.00 AM to 5.00 PM.

Art Unit: 2892

If attempt to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Thao Le can be reached on (571) 272-1708. The fax phone numbers of the organization where this application or proceeding is assigned are (571) 273-8300 for regular communications and (571) 273-8300 for After Final communications.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free).

/Quoc D Hoang/

Primary Examiner, Art Unit 2892

Notice of References Cited	Application/Control No. 12/500,620	Applicant(s)/Patent Under Reexamination ZHANG ET AL.	
	Examiner QUOC HOANG	Art Unit 2892	Page 1 of 1

U.S. PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Name	Classification
*	A	US-7,776,700	08-2010	Yang et al.	438/286
	B	US-			
	C	US-			
	D	US-			
	E	US-			
	F	US-			
	G	US-			
	H	US-			
	I	US-			
	J	US-			
	K	US-			
	L	US-			
	M	US-			

FOREIGN PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Country	Name	Classification
	N					
	O					
	P					
	Q					
	R					
	S					
	T					

NON-PATENT DOCUMENTS

*		Include as applicable: Author, Title Date, Publisher, Edition or Volume, Pertinent Pages)
	U	
	V	
	W	
	X	

*A copy of this reference is not being furnished with this Office action. (See MPEP § 707.05(a).)
Dates in MM-YYYY format are publication dates. Classifications may be US or foreign.

Receipt date: 07/10/2009

Doc code: IDS

Doc description: Information Disclosure Statement (IDS) Filed

12500620 - GAI: 2892

Approved for use through 07/31/2009. OMB 0651-0031

U.S. Patent and Trademark Office; U.S. DEPARTMENT OF COMMERCE

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INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		
	Filing Date		
	First Named Inventor	Guowei ZHANG	
	Art Unit		
	Examiner Name	Not Assigned Yet	
	Attorney Docket Number	CSM P 2009 NAT 03 US 1	

U.S. PATENTS							Remove
Examiner Initial*	Cite No	Patent Number	Kind Code ¹	Issue Date	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines where Relevant Passages or Relevant Figures Appear	
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Receipt date: 07/10/2009 INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		12500620 - GAU: 2892
	Filing Date		
	First Named Inventor	Guowei ZHANG	
	Art Unit		
	Examiner Name	Not Assigned Yet	
	Attorney Docket Number	CSM P 2009 NAT 03 US 1	

1	LEONARD RUBIN ET AL., Narrow n+/p+ isolation in retrograde well implants, Solid State Technology Magazine, 01 July 2003, pages 119-125, Volume 46 Issue 7, Solid State Technology, USA.	☐
---	---	--------------------------

If you wish to add additional non-patent literature document citation information please click the Add button **Add**

EXAMINER SIGNATURE

Examiner Signature	/Quoc Hoang/	Date Considered	03/11/2012
--------------------	--------------	-----------------	------------

***EXAMINER:** Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through a citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.



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BIB DATA SHEET

CONFIRMATION NO. 2701

SERIAL NUMBER 12/500,620	FILING or 371(c) DATE 07/10/2009 RULE	CLASS 257	GROUP ART UNIT 2892	ATTORNEY DOCKET NO. CSM P 2009 NAT 03 US 1		
APPLICANTS Guo Wei ZHANG, Singapore, SINGAPORE; Purakh Raj VERMA, Singapore, SINGAPORE; ** CONTINUING DATA ***** This application is a CIP of 12/390,509 02/23/2009 PAT 8,053,319 ** FOREIGN APPLICATIONS ***** ** IF REQUIRED, FOREIGN FILING LICENSE GRANTED ** 07/20/2009						
Foreign Priority claimed ☐ Yes ☒ No 35 USC 119(a-d) conditions met ☐ Yes ☒ No Verified and /QUOC DINH HOANG/ Acknowledged Examiner's Signature		☐ Met after Allowance QH Initials	STATE OR COUNTRY SINGAPORE	SHEETS DRAWINGS 29	TOTAL CLAIMS 20	INDEPENDENT CLAIMS 3
ADDRESS Horizon IP Pte Ltd. 7500A Beach Road, #04-306/308 The Plaza Singapore, 199591 SINGAPORE						
TITLE HIGH VOLTAGE DEVICE						
FILING FEE RECEIVED 1090	FEES: Authority has been given in Paper No. _____ to charge/credit DEPOSIT ACCOUNT No. _____ for following:			☐ All Fees		
				☐ 1.16 Fees (Filing)		
				☐ 1.17 Fees (Processing Ext. of time)		
				☐ 1.18 Fees (Issue)		
				☐ Other _____		
			☐ Credit			

<i>Issue Classification</i> 	Application/Control No. 12500620	Applicant(s)/Patent Under Reexamination ZHANG ET AL.
	Examiner QUOC HOANG	Art Unit 2892

[illegible][illegible]

NONE		Total Claims Allowed:	
(Assistant Examiner)	(Date)	18	
/QUOC HOANG/ Primary Examiner.Art Unit 2892	03/11/2012	O.G. Print Claim(s)	O.G. Print Figure
(Primary Examiner)	(Date)	1	4g

<i>Index of Claims</i> 	Application/Control No. 12500620	Applicant(s)/Patent Under Reexamination ZHANG ET AL.
	Examiner QUOC HOANG	Art Unit 2892

✓	Rejected	-	Cancelled	N	Non-Elected	A	Appeal
=	Allowed	÷	Restricted	I	Interference	O	Objected

☐ Claims renumbered in the same order as presented by applicant				☐ CPA				☐ T.D.				☐ R.1.47			
CLAIM		DATE													
Final	Original	01/06/2012	03/11/2012												
1	1	÷	=												
2	2	÷	=												
3	3	÷	=												
4	4	÷	=												
5	5	÷	=												
6	6	÷	=												
7	7	÷	=												
8	8	÷	=												
9	9	÷	=												
10	10	÷	=												
11	11	÷	=												
12	12	÷	=												
13	13	÷	=												
14	14	÷	=												
17	15	÷	=												
18	16	÷	=												
15	17	÷	=												
16	18	÷	=												
	19	÷	-												
	20	÷	-												

EAST Search History

EAST Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L1	1	"12500620"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2012/03/10 11:38
L2	64	((guo near wei near zhang) (purakh near raj near verma)).in.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2012/03/10 11:58
L3	2	2 and ((first and second) near mask and implant\$3 and channel near well).clm.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2012/03/10 11:59
L4	3	2 and (mask and remov\$3 and implant\$3 and channel near well).clm.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2012/03/10 12:07
L5	1	("5589414").PN.	USPAT; USOCR	OR	OFF	2012/03/10 12:17
L6	44	gate near (insulating dielectric) and gate near electrode and (first and second) near mask and implant\$3 and channel near well and @ad<"20090223"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2012/03/10 12:21
L7	107	gate near (insulating dielectric) and gate near electrode and mask\$3 and pattern\$3 and remov\$3 and implant\$3 with channel near well and @ad<"20090223"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2012/03/10 12:36
L8	143	pattern\$3 with gate with (mask\$3 photoresist) and implant\$3 with well with (mask\$3 photoresist) with opening and @ad<"20090223"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2012/03/10 12:46
L9	9	("6606124").URPN.	USPAT	OR	OFF	2012/03/10 13:58
L10	38	overlap\$4 with (mask\$3 photoresist resist) with implant\$3 with well and opening and @ad<"20090223"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO;	OR	OFF	2012/03/10 14:16

			DERWENT; IBM_TDB			
L11	1	overlap\$4 with (mask\$3 photoresist resist) with (dop\$3 implant\$3) with channel near well and @ad< "20090223"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2012/03/10 16:15
L12	102	overlap\$4 with (mask\$3 photoresist resist) with (dop\$3 implant\$3) with well and @ad< "20090223"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2012/03/10 16:17
L13	10378	(mask\$3 photoresist resist) with (dop\$3 implant\$3) with well and @ad< "20090223"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2012/03/10 16:44
L14	150	stack\$3 with (mask\$3 photoresist resist) with (dop\$3 implant\$3) with well and @ad< "20090223"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2012/03/10 16:45
L15	411560	438/551,671,942.ccls. "257"/("E21.035" "E21.058" "E21.434" "E21.63").ccls.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2012/03/10 17:40
L16	355	(stack\$3 multi\$3 overlap\$3) with (mask\$3 photoresist resist) with (dop\$3 implant\$3) with well and @ad< "20090223"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2012/03/10 17:41
L17	114	16 and 15	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	OFF	2012/03/10 17:41

3/ 10/ 2012 5:55:57 PM

Search Notes 	Application/Control No. 12500620	Applicant(s)/Patent Under Reexamination ZHANG ET AL.
	Examiner QUOC HOANG	Art Unit 2892

SEARCHED			
Class	Subclass	Date	Examiner
438	551, 671, 942	3/11/2012	QH
257	E21.035, E21.058, E21.434, E21.63	3/11/2012	QH

SEARCH NOTES		
Search Notes	Date	Examiner
East (search history printout)	3/11/2012	QH
US_PGPub database	3/11/2012	QH

INTERFERENCE SEARCH			
Class	Subclass	Date	Examiner
438	551	3/11/2012	QH

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PART B - FEE(S) TRANSMITTAL

**Complete and send this form, together with applicable fee(s), to: Mail Mail Stop ISSUE FEE
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450
or Fax (571)-273-2885**

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence including the Patent, advance orders and notification of maintenance fees will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications.

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Horizon IP Pte Ltd.
7500A Beach Road, #04-306/308
The Plaza
Singapore, 199591
SINGAPORE

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Nannan CHEN	(Depositor's name)
/nannan chen/	(Signature)
06/19/2012	(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
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12/500,620

07/10/2009

Guo Wei ZHANG

CSM P 2009 NAT 03 US 1

2701

TITLE OF INVENTION: HIGH VOLTAGE DEVICE

APPLN. TYPE	SMALL ENTITY	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
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nonprovisional

NO

\$1740

\$300

\$0

\$2040

06/19/2012

EXAMINER	ART UNIT	CLASS-SUBCLASS
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HOANG, QUOC DINH

2892

257-335000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

☐ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached.

☐ "Fee Address" indication (or "Fee Address" Indication form PTO/SB/47; Rev 03-02 or more recent) attached. Use of a Customer Number is required.

2. For printing on the patent front page, list

(1) the names of up to 3 registered patent attorneys or agents OR, alternatively,

1 Horizon IP Pte Ltd

(2) the name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed.

2

3

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document has been filed for recordation as set forth in 37 CFR 3.11. Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE

GLOBALFOUNDRIES Singapore Pte. Ltd.

(B) RESIDENCE: (CITY and STATE OR COUNTRY)

Singapore

Please check the appropriate assignee category or categories (will not be printed on the patent): ☐ Individual ☒ Corporation or other private group entity ☐ Government

4a. The following fee(s) are submitted:

☒ Issue Fee

☒ Publication Fee (No small entity discount permitted)

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5. Change in Entity Status (from status indicated above)

☐ a. Applicant claims SMALL ENTITY status. See 37 CFR 1.27.

☐ b. Applicant is no longer claiming SMALL ENTITY status. See 37 CFR 1.27(g)(2).

NOTE: The Issue Fee and Publication Fee (if required) will not be accepted from anyone other than the applicant; a registered attorney or agent; or the assignee or other party in interest as shown by the records of the United States Patent and Trademark Office.

Authorized Signature /dexter chin/

Date 06/19/2012

Typed or printed name Dexter CHIN

Registration No. 38,842

This collection of information is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450.

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Horizon IP Pte Ltd.
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SINGAPORE

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Nannan CHEN	(Depositor's name)
/nannan chen/	(Signature)
06/19/2012	(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
12/500,620	07/10/2009	Guo Wei ZHANG	CSM P 2009 NAT 03 US 1	2701

TITLE OF INVENTION: HIGH VOLTAGE DEVICE

APPLN. TYPE	SMALL ENTITY	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	NO	\$1740	\$300	\$0	\$2040	06/19/2012

EXAMINER	ART UNIT	CLASS-SUBCLASS
HOANG, QUOC DINH	2892	257-335000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).
☐ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached.
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3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

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(A) NAME OF ASSIGNEE GLOBALFOUNDRIES Singapore Pte. Ltd.
(B) RESIDENCE: (CITY and STATE OR COUNTRY) Singapore

Please check the appropriate assignee category or categories (will not be printed on the patent): ☐ Individual ☒ Corporation or other private group entity ☐ Government

4a. The following fee(s) are submitted:

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☒ The Director is hereby authorized to charge the required fee(s), any deficiency, or credit any overpayment, to Deposit Account Number 502388 (enclose an extra copy of this form).

5. Change in Entity Status (from status indicated above)

☐ a. Applicant claims SMALL ENTITY status. See 37 CFR 1.27. ☐ b. Applicant is no longer claiming SMALL ENTITY status. See 37 CFR 1.27(g)(2).

NOTE: The Issue Fee and Publication Fee (if required) will not be accepted from anyone other than the applicant; a registered attorney or agent; or the assignee or other party in interest as shown by the records of the United States Patent and Trademark Office.

Authorized Signature /dexter chin/ Date 06/19/2012
Typed or printed name Dexter CHIN Registration No. 38,842

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TRANSMITTAL FORM <i>(to be used for all correspondence after initial filing)</i>	Application Number	12/500,620	
	Filing Date	07/10/2009	
	First Named Inventor	Guo Wei ZHANG	
	Art Unit	2892	
	Examiner Name	HOANG, QUOC DINH	
Total Number of Pages in This Submission	6	Attorney Docket Number	CSM P 2009 NAT 03 US 1

ENCLOSURES (Check all that apply)		
☒ Fee Transmittal Form ☒ Fee Attached ☐ Amendment/Reply ☐ After Final ☐ Affidavits/declaration(s) ☐ Extension of Time Request ☐ Express Abandonment Request ☐ Information Disclosure Statement ☐ Certified Copy of Priority Document(s) ☐ Reply to Missing Parts/ Incomplete Application ☐ Reply to Missing Parts under 37 CFR 1.52 or 1.53	☐ Drawing(s) ☐ Licensing-related Papers ☐ Petition ☐ Petition to Convert to a Provisional Application ☐ Power of Attorney, Revocation Change of Correspondence Address ☐ Terminal Disclaimer ☐ Request for Refund ☐ CD, Number of CD(s) _____ ☐ Landscape Table on CD	☐ After Allowance Communication to TC ☐ Appeal Communication to Board of Appeals and Interferences ☐ Appeal Communication to TC (Appeal Notice, Brief, Reply Brief) ☐ Proprietary Information ☐ Status Letter ☒ Other Enclosure(s) (please identify below):
Remarks Part B - Issue Fee(s) Transmittal (1 original & 1 copy)		

SIGNATURE OF APPLICANT, ATTORNEY, OR AGENT			
Firm Name	Horizon IP Pte. Ltd.		
Signature	/dexter chin/		
Printed name	Dexter CHIN		
Date	06/19/2012	Reg. No.	38,842

CERTIFICATE OF TRANSMISSION/MAILING			
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Signature	/nannan chen/		
Typed or printed name	Nannan CHEN	Date	06/19/2012

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3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
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8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
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Electronic Patent Application Fee Transmittal

Application Number:	12500620			
Filing Date:	10-Jul-2009			
Title of Invention:	HIGH VOLTAGE DEVICE			
First Named Inventor/Applicant Name:	Guo Wei ZHANG			
Filer:	Dexter Kam Chin			
Attorney Docket Number:	CSM P 2009 NAT 03 US 1			
Filed as Large Entity				
Utility under 35 USC 111(a) Filing Fees				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Utility Appl issue fee	1501	1	1740	1740
Publ. Fee- early, voluntary, or normal	1504	1	300	300

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				2040

Electronic Acknowledgement Receipt

EFS ID:	13045696
Application Number:	12500620
International Application Number:	
Confirmation Number:	2701
Title of Invention:	HIGH VOLTAGE DEVICE
First Named Inventor/Applicant Name:	Guo Wei ZHANG
Customer Number:	16162
Filer:	Dexter Kam Chin
Filer Authorized By:	
Attorney Docket Number:	CSM P 2009 NAT 03 US 1
Receipt Date:	19-JUN-2012
Filing Date:	10-JUL-2009
Time Stamp:	02:32:59
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	Deposit Account
Payment was successfully received in RAM	\$ 2040
RAM confirmation Number	7568
Deposit Account	502388
Authorized User	

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

Charge any Additional Fees required under 37 C.F.R. Section 1.20 (Post Issuance fees)

Charge any Additional Fees required under 37 C.F.R. Section 1.21 (Miscellaneous fees and charges)

File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Issue Fee Payment (PTO-85B)	CSMP2009NAT03US1-partbtransmittal_copy.pdf	90397 1b999e5deb952f0897078c82bb3d6acdbee68acf	no	1
Warnings:					
Information:					
2	Issue Fee Payment (PTO-85B)	CSMP2009NAT03US1-partbtransmittal_org.pdf	90397 1b999e5deb952f0897078c82bb3d6acdbee68acf	no	1
Warnings:					
Information:					
3	Miscellaneous Incoming Letter	CSMP2009NAT03US1-TF.pdf	278113 5bfadf97d12b6e5264bb97731f5bb9d9d7861c68	no	2
Warnings:					
Information:					
4	Fee Worksheet (SB06)	fee-info.pdf	31949 43534709a494163d2c3d181c911a78c0365e49cf	no	2
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Total Files Size (in bytes):			490856		
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APPLICATION NO.	ISSUE DATE	PATENT NO.	ATTORNEY DOCKET NO.	CONFIRMATION NO.
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Horizon IP Pte Ltd.
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ISSUE NOTIFICATION

The projected patent number and issue date are specified above.

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (application filed on or after May 29, 2000)

The Patent Term Adjustment is 496 day(s). Any patent to issue from the above-identified application will include an indication of the adjustment on the front page.

If a Continued Prosecution Application (CPA) was filed in the above-identified application, the filing date that determines Patent Term Adjustment is the filing date of the most recent CPA.

Applicant will be able to obtain more detailed information by accessing the Patent Application Information Retrieval (PAIR) WEB site (<http://pair.uspto.gov>).

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Application Assistance Unit (AAU) of the Office of Data Management (ODM) at (571)-272-4200.

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